

(12) **United States Patent**
Liu et al.

(10) **Patent No.:** **US 12,419,084 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **METHODS OF FORMING TRANSISTOR SOURCE/DRAIN REGIONS COMPRISING CARBON LINER LAYERS**

(58) **Field of Classification Search**
None
See application file for complete search history.

(71) Applicant: **Taiwan Semiconductor Manufacturing Co., Ltd.**, Hsinchu (TW)

(56) **References Cited**

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(72) Inventors: **Wei-Min Liu**, Hsinchu (TW); **Li-Li Su**, Chubei (TW); **Yee-Chia Yeo**, Hsinchu (TW)

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(73) Assignee: **Taiwan Semiconductor Manufacturing Co., Ltd.**, Hsinchu (TW)

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 580 days.

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(21) Appl. No.: **17/565,716**

Loubet, N. et al., "Stacked Nanosheet Gate-All-Around Transistor to Enable Scaling Beyond FinFET," 2017 Symposium on VLSI Technology Digest of Technical Papers, Jun. 5-8, 2017, pp. T230-T231.

(22) Filed: **Dec. 30, 2021**

Primary Examiner — Christine S. Kim

Assistant Examiner — Tyler J Wiegand

(65) **Prior Publication Data**

US 2022/0367625 A1 Nov. 17, 2022

(74) *Attorney, Agent, or Firm* — Slater Matsil, LLP

Related U.S. Application Data

(60) Provisional application No. 63/188,134, filed on May 13, 2021.

(51) **Int. Cl.**
H10D 62/10 (2025.01)
H10D 30/01 (2025.01)

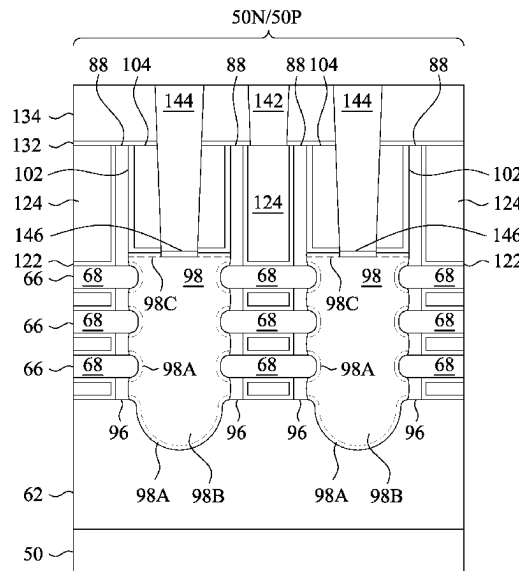
(Continued)

(52) **U.S. Cl.**
CPC **H10D 62/118** (2025.01); **H10D 30/031** (2025.01); **H10D 30/6713** (2025.01);
(Continued)

(57) **ABSTRACT**

In an embodiment, a device includes: a first nanostructure; a source/drain region adjoining a first channel region of the first nanostructure, the source/drain region including: a main layer; and a first liner layer between the main layer and the first nanostructure, a carbon concentration of the first liner layer being greater than a carbon concentration of the main layer; an inter-layer dielectric on the source/drain region; and a contact extending through the inter-layer dielectric, the contact connected to the main layer, the contact spaced apart from the first liner layer.

20 Claims, 45 Drawing Sheets



- (51) **Int. Cl.**
H10D 30/67 (2025.01)
H10D 84/01 (2025.01)
H10D 84/03 (2025.01)
H10D 30/69 (2025.01)
- (52) **U.S. Cl.**
 CPC *H10D 30/6735* (2025.01); *H10D 30/6757*
 (2025.01); *H10D 84/0128* (2025.01); *H10D*
84/013 (2025.01); *H10D 84/0147* (2025.01);
H10D 84/038 (2025.01); *H10D 30/797*
 (2025.01)
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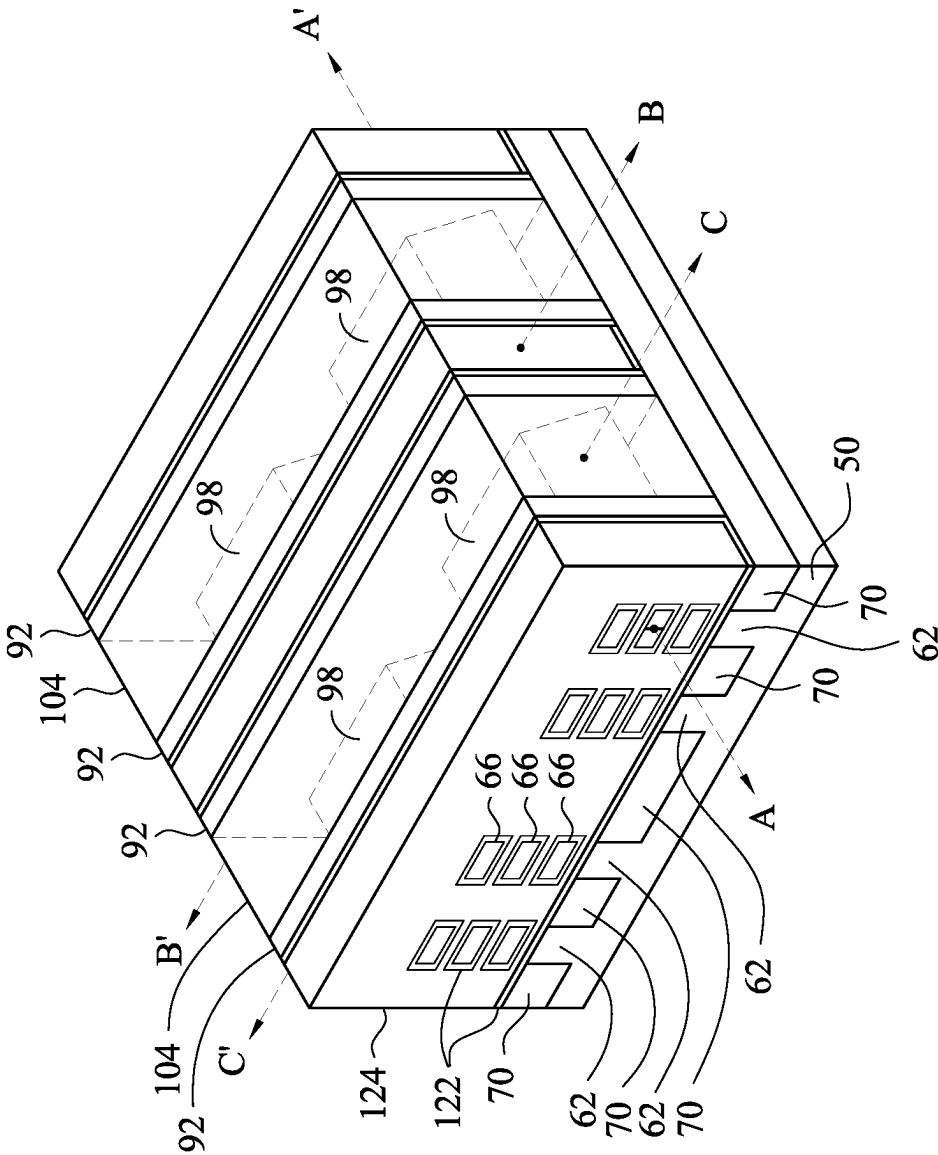


Figure 1

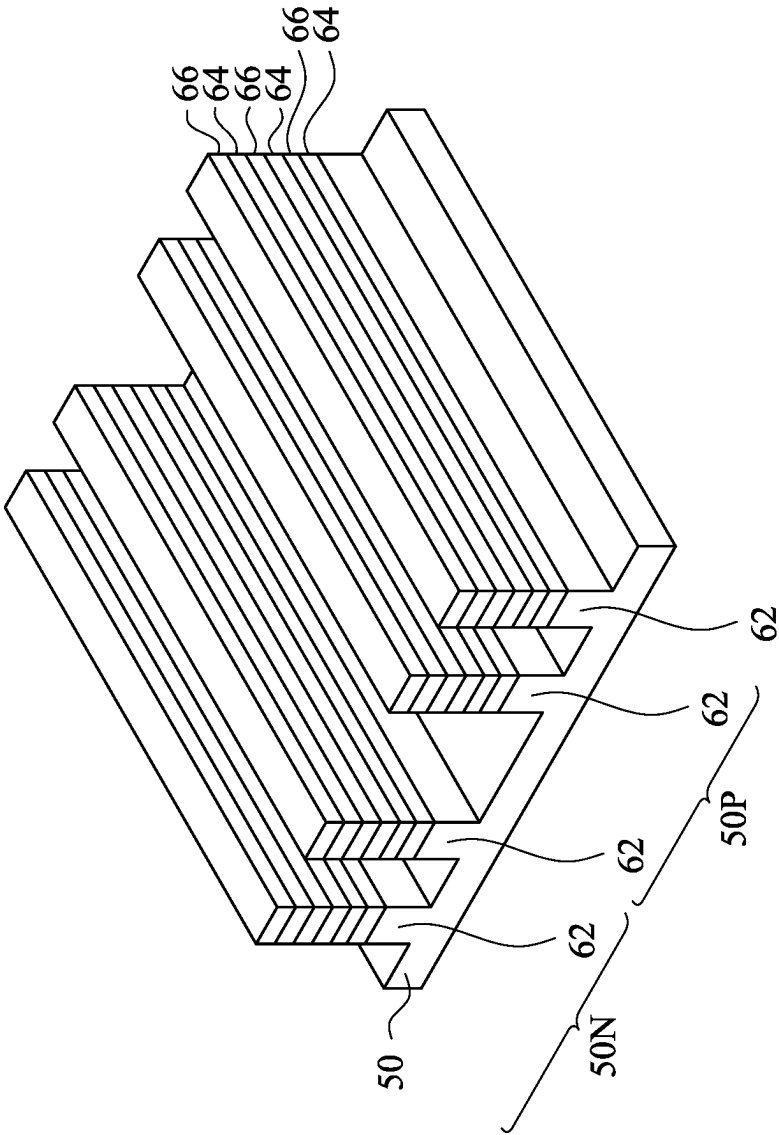


Figure 3

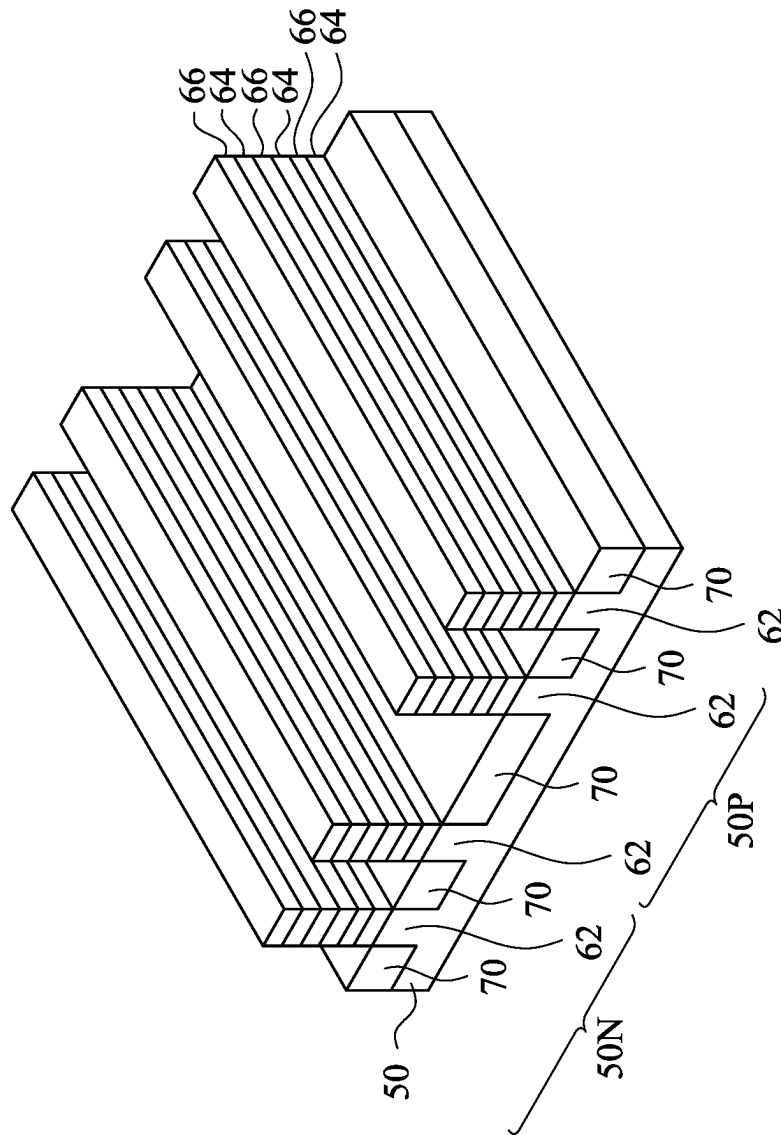


Figure 4

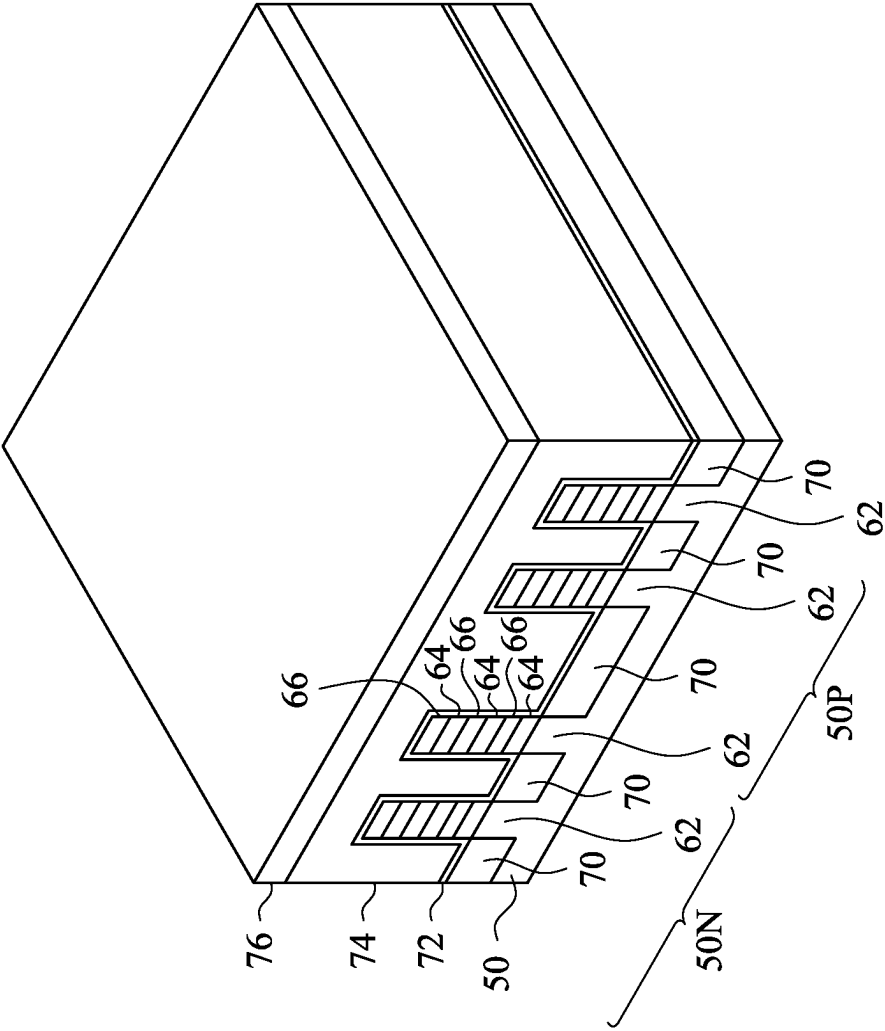


Figure 5

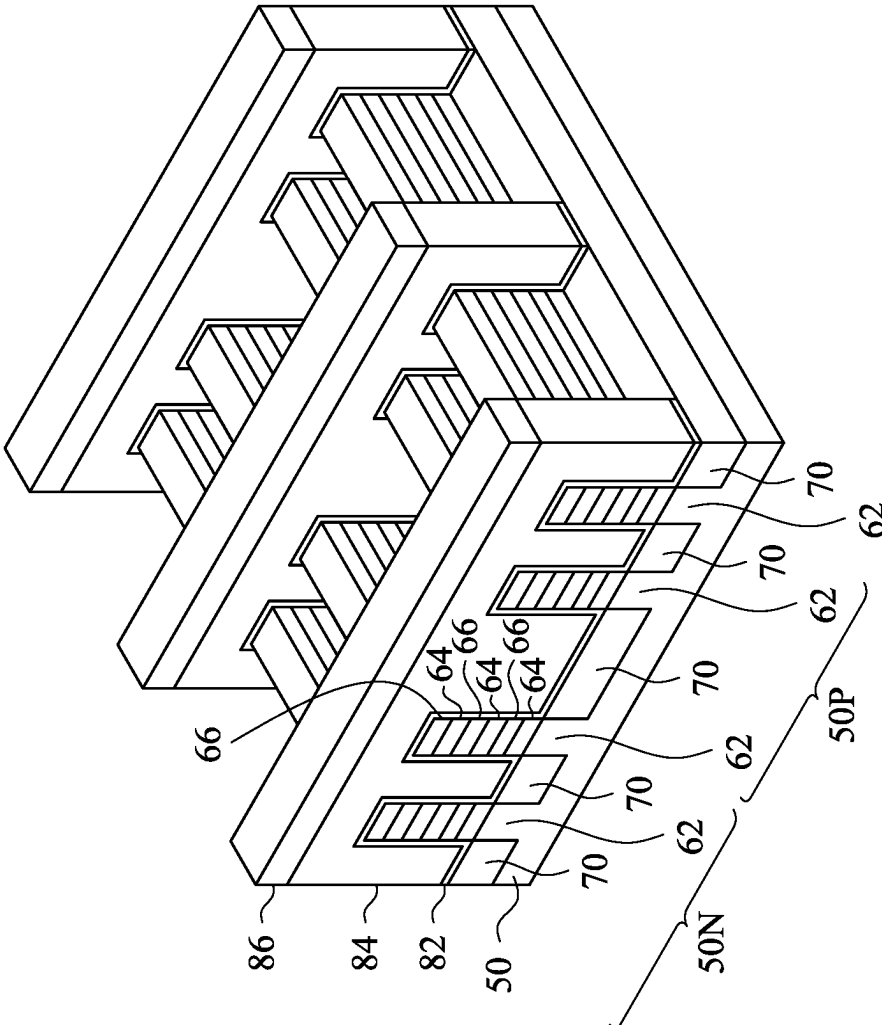


Figure 6

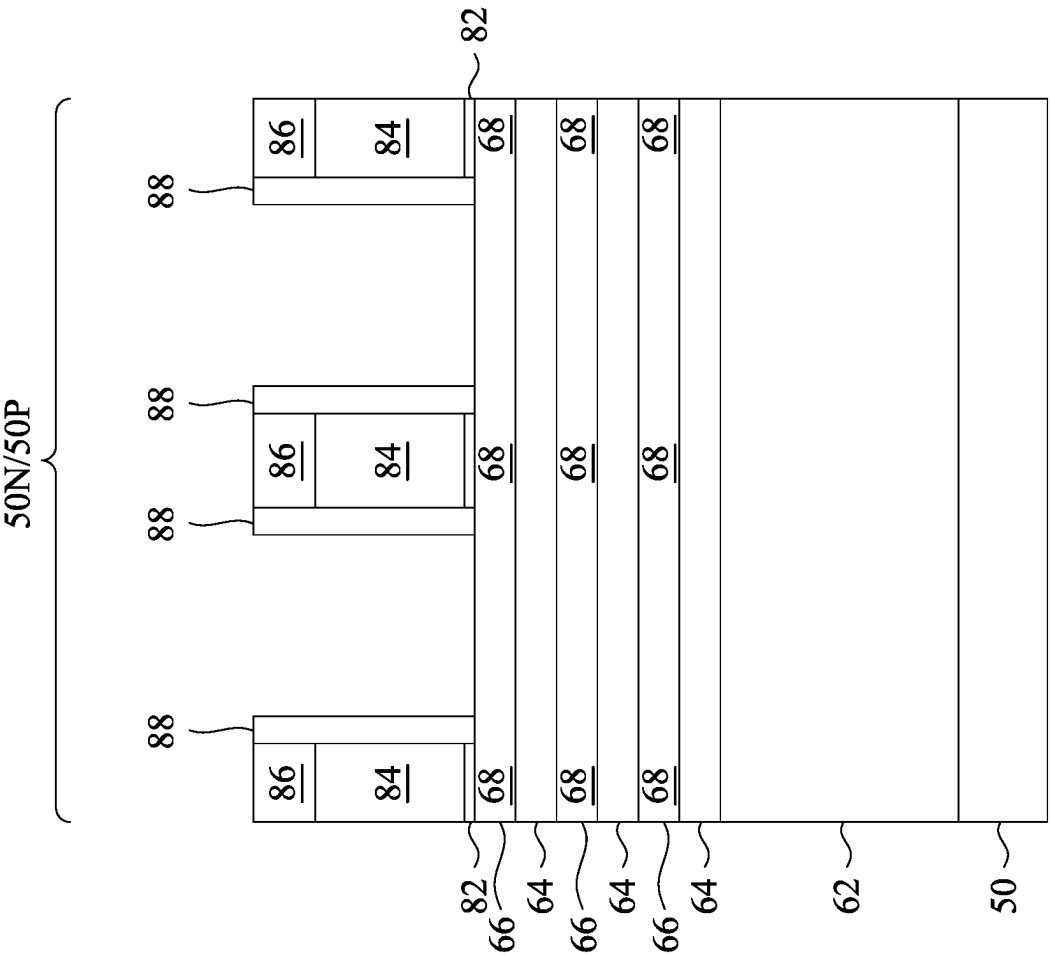


Figure 7A

50N/50P

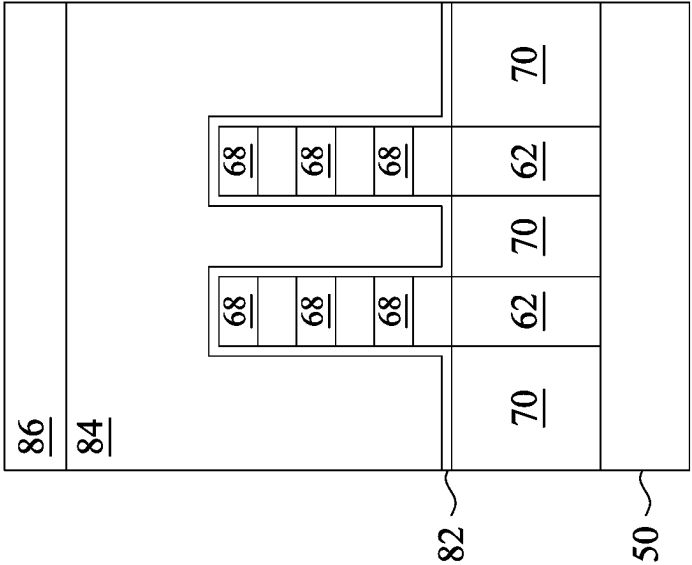


Figure 7B

50N/50P

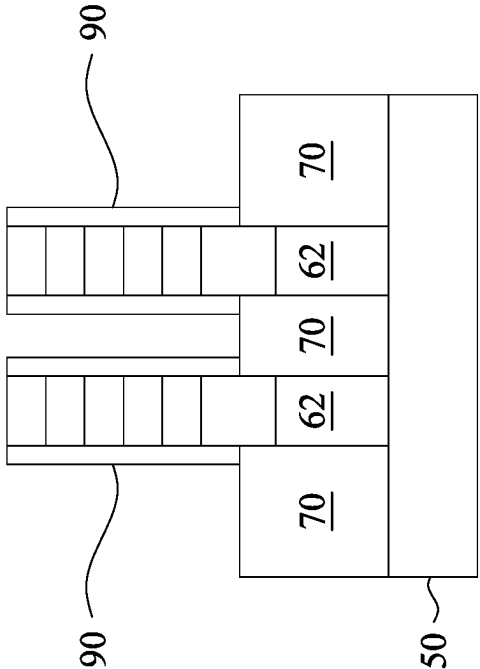


Figure 7C

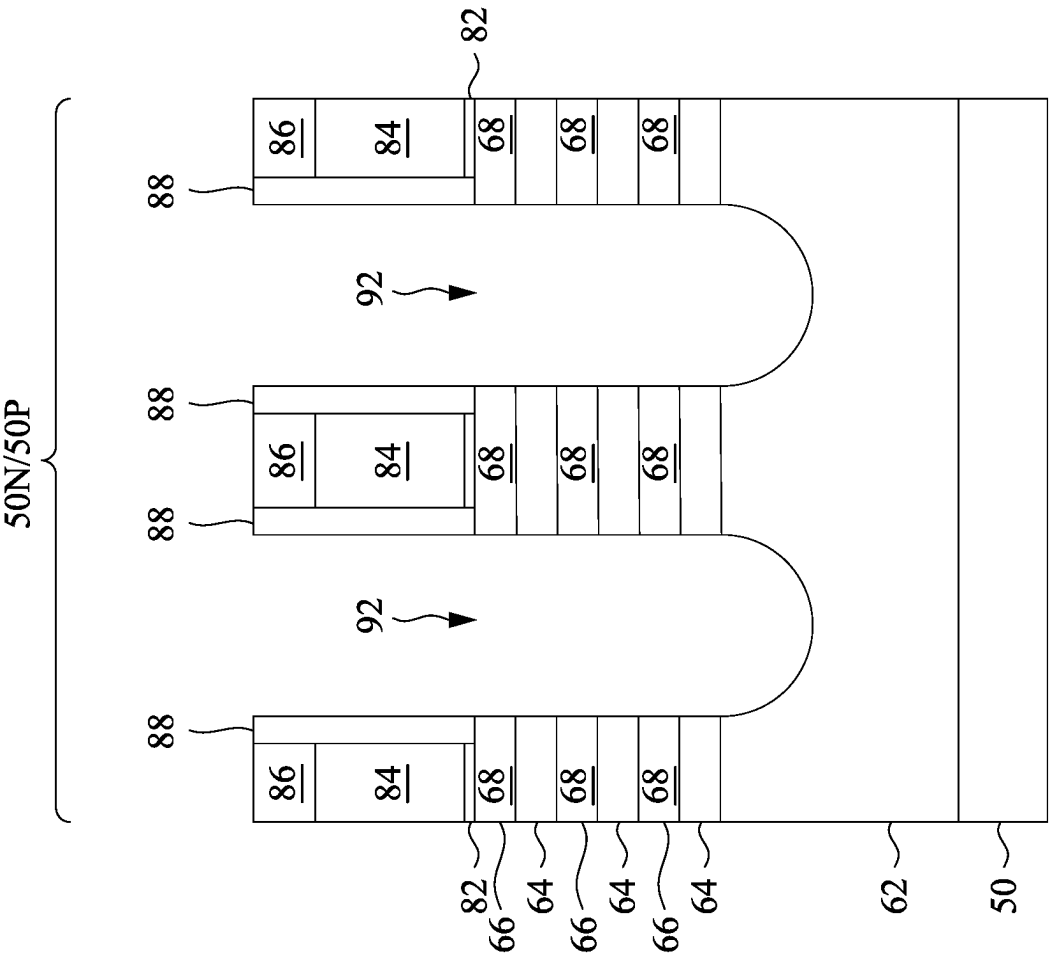


Figure 8A

50N/50P

50N/50P

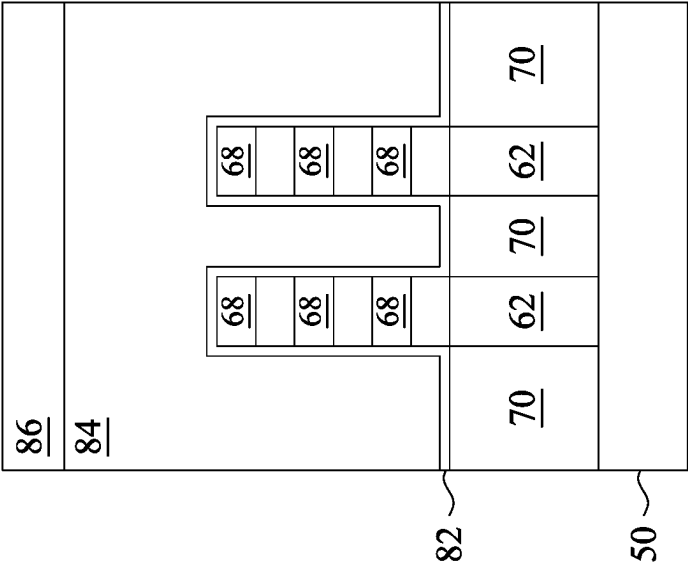


Figure 8B

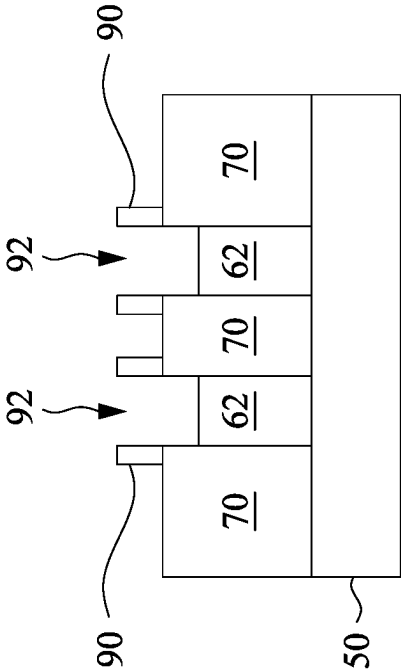


Figure 8C

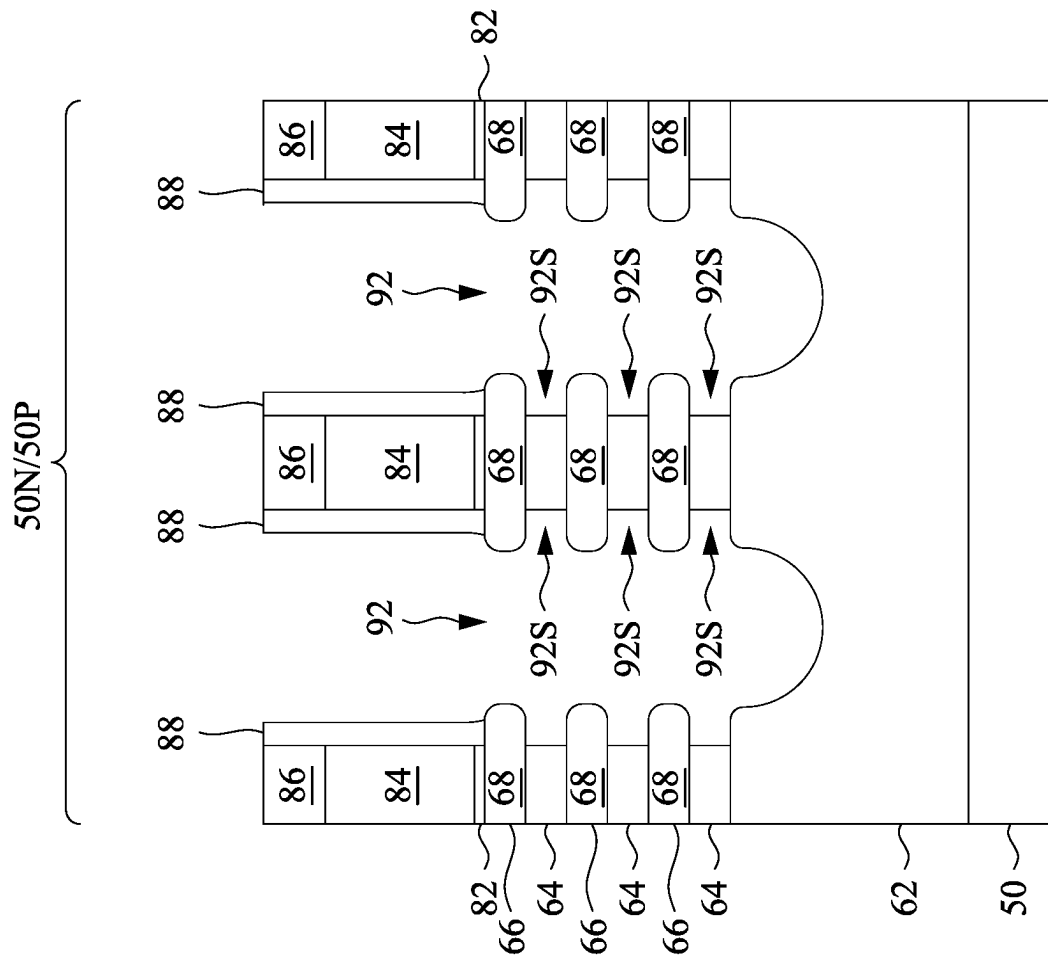


Figure 9A

50N/50P

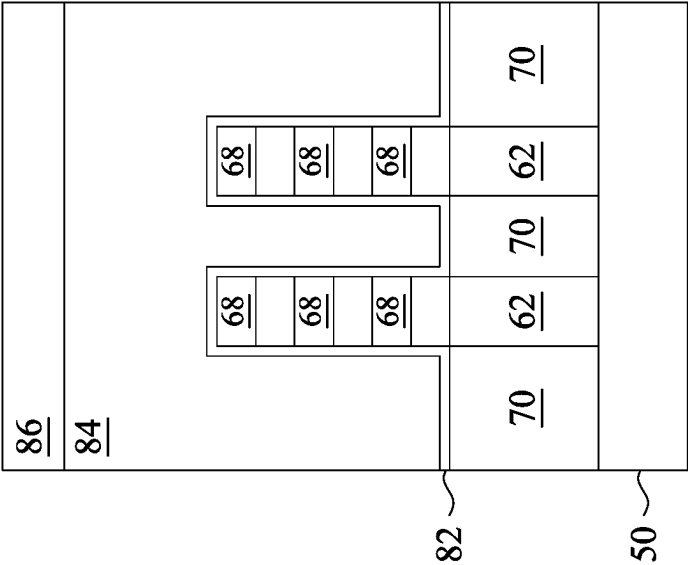


Figure 9B

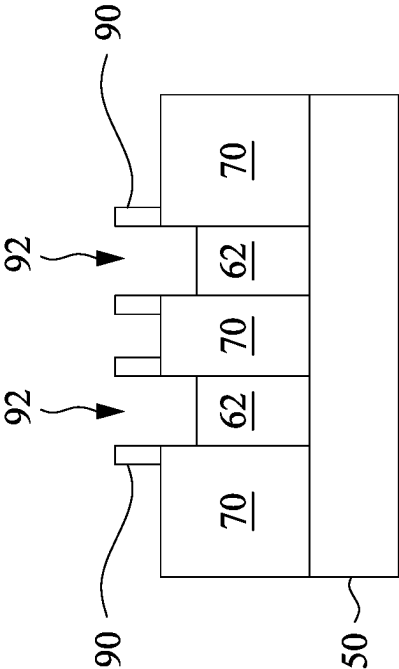


Figure 9C

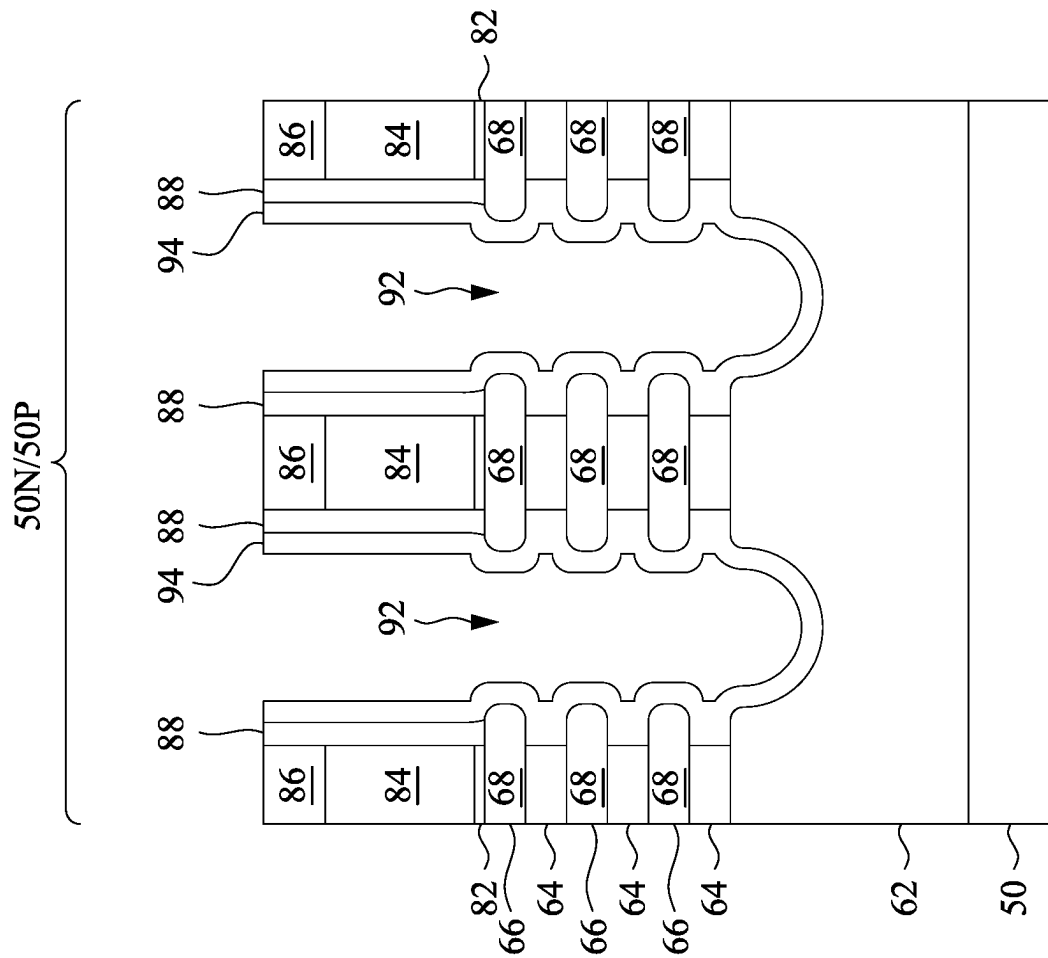


Figure 10A

50N/50P

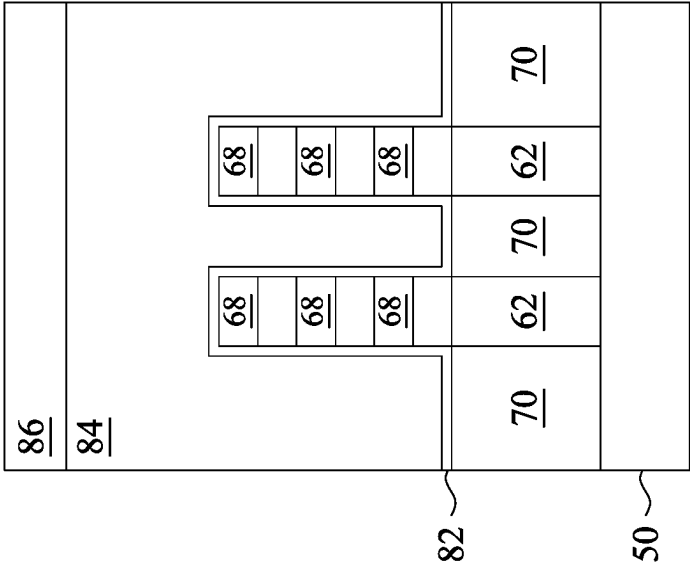


Figure 10B

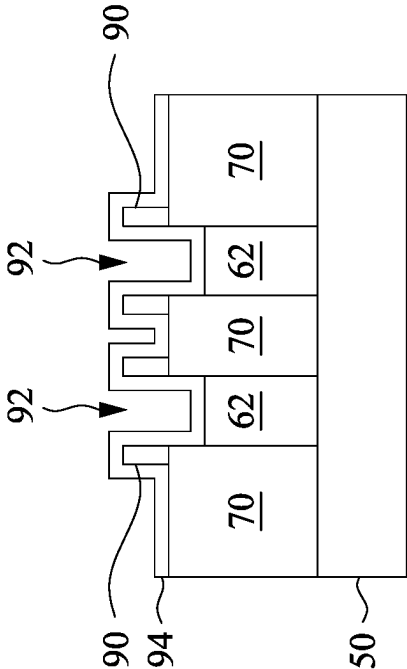


Figure 10C

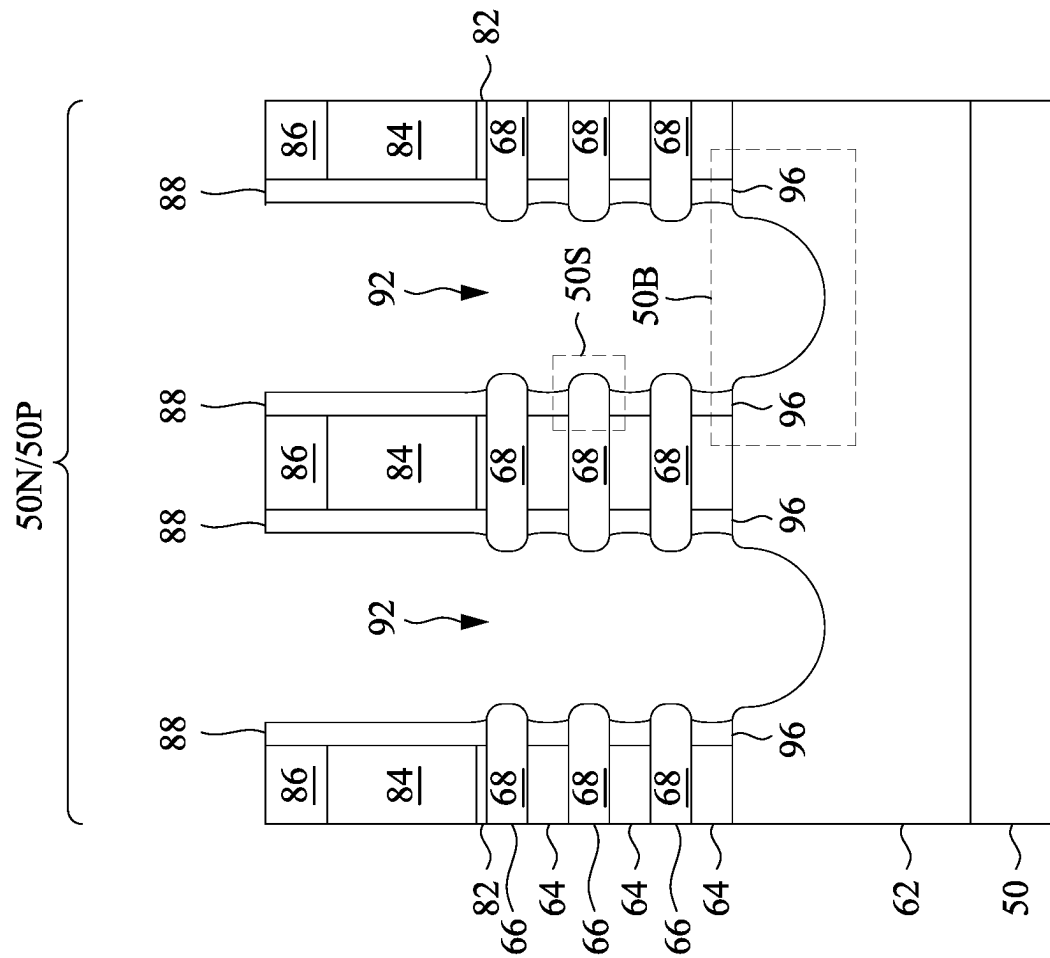


Figure 11A

50N/50P

50N/50P

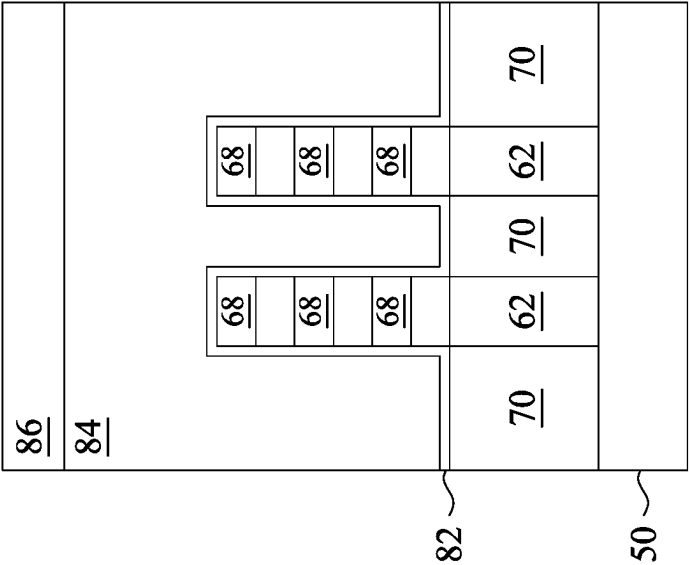


Figure 11B

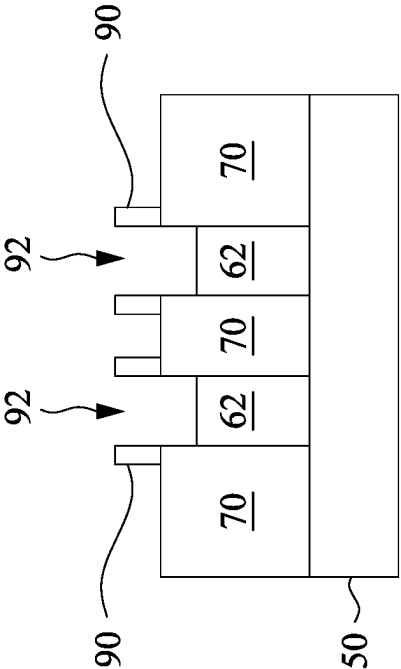


Figure 11C

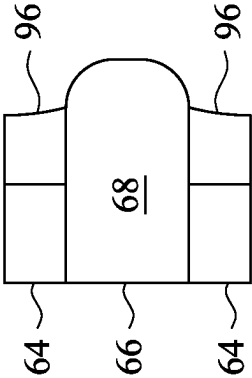


Figure 12A

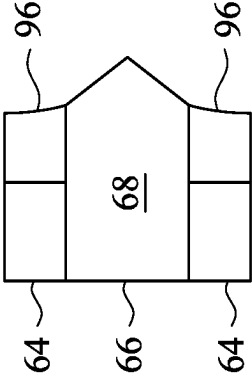


Figure 12B

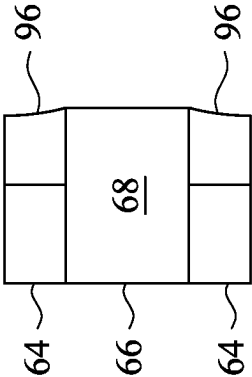


Figure 12C

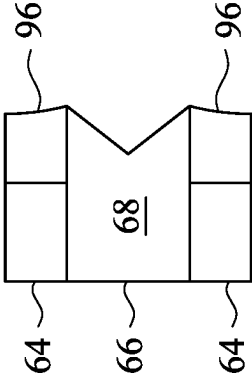


Figure 12D

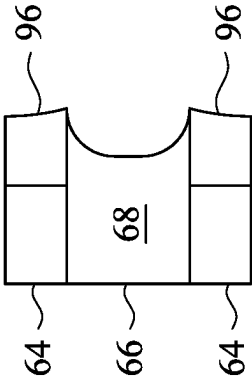


Figure 12E



Figure 13A

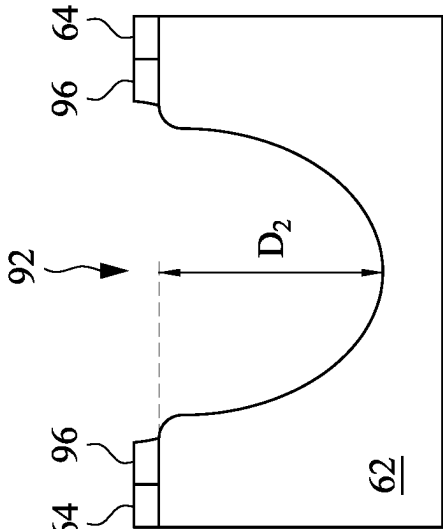


Figure 13B

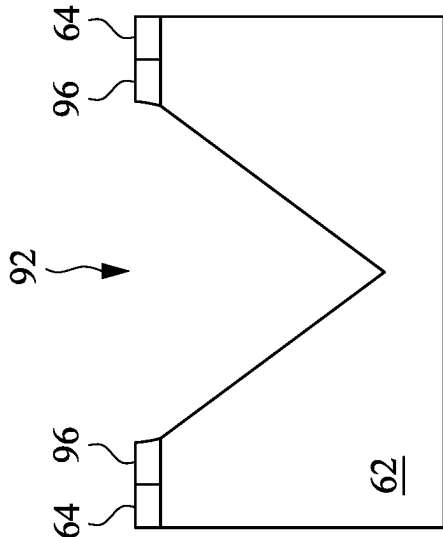


Figure 13C

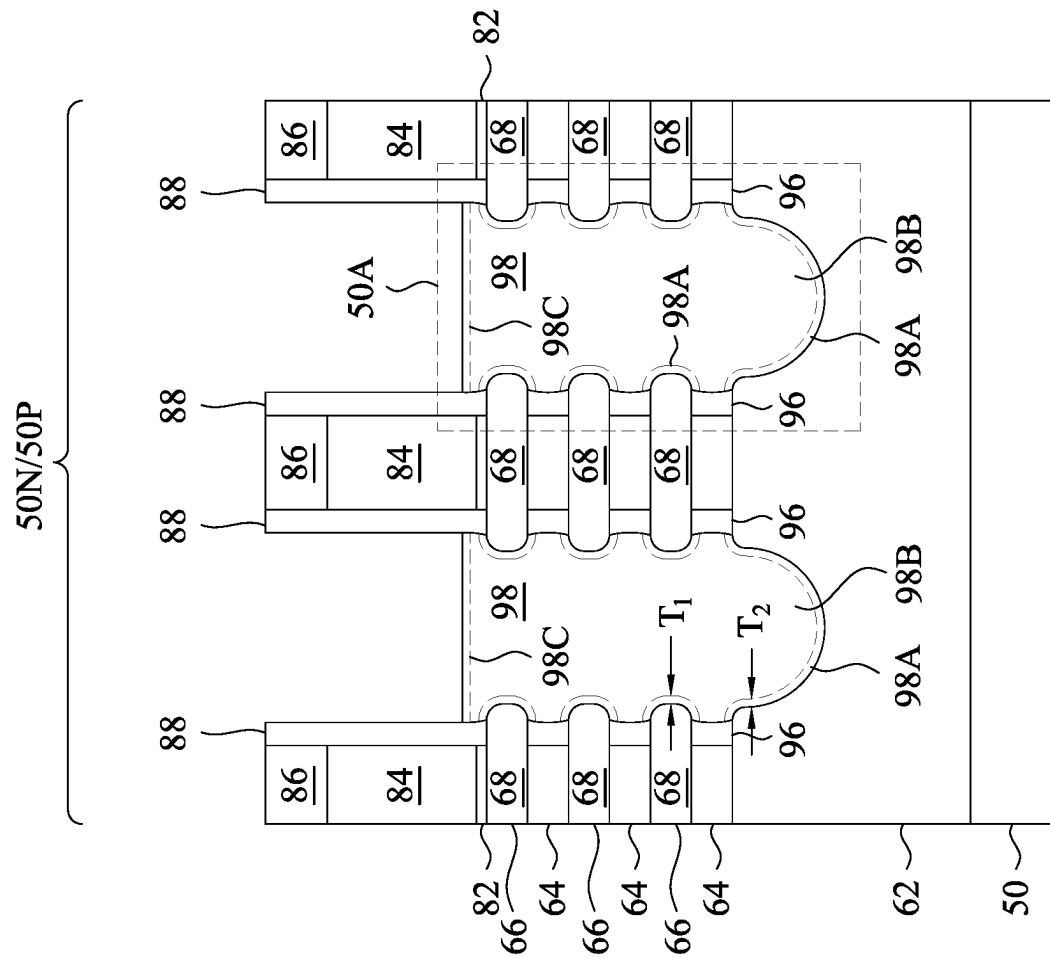


Figure 14A

50N/50P

50N/50P

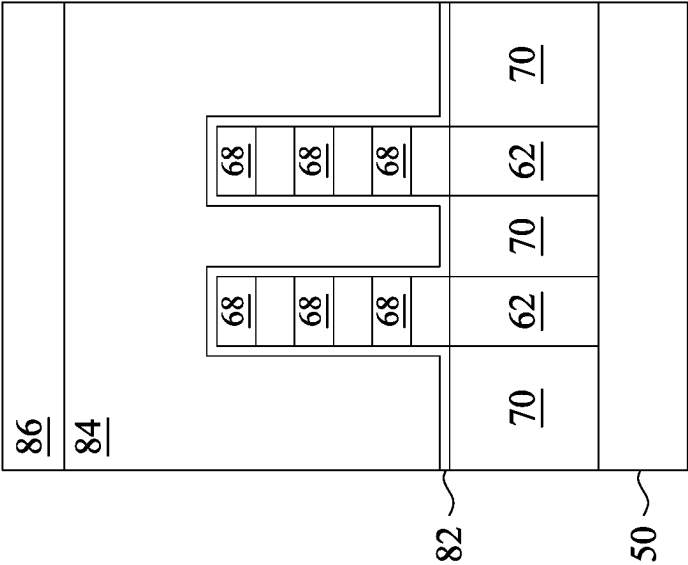


Figure 14B

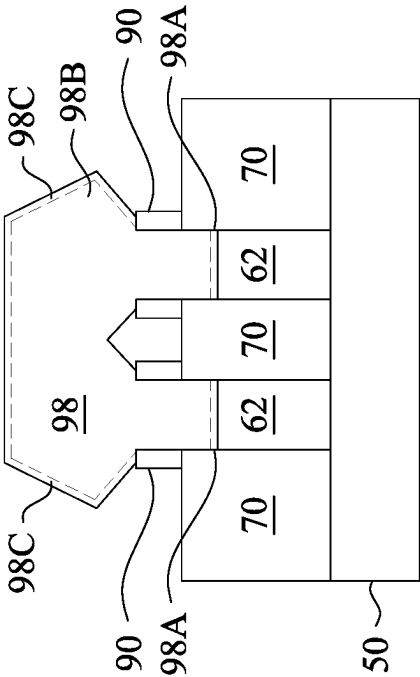


Figure 14C

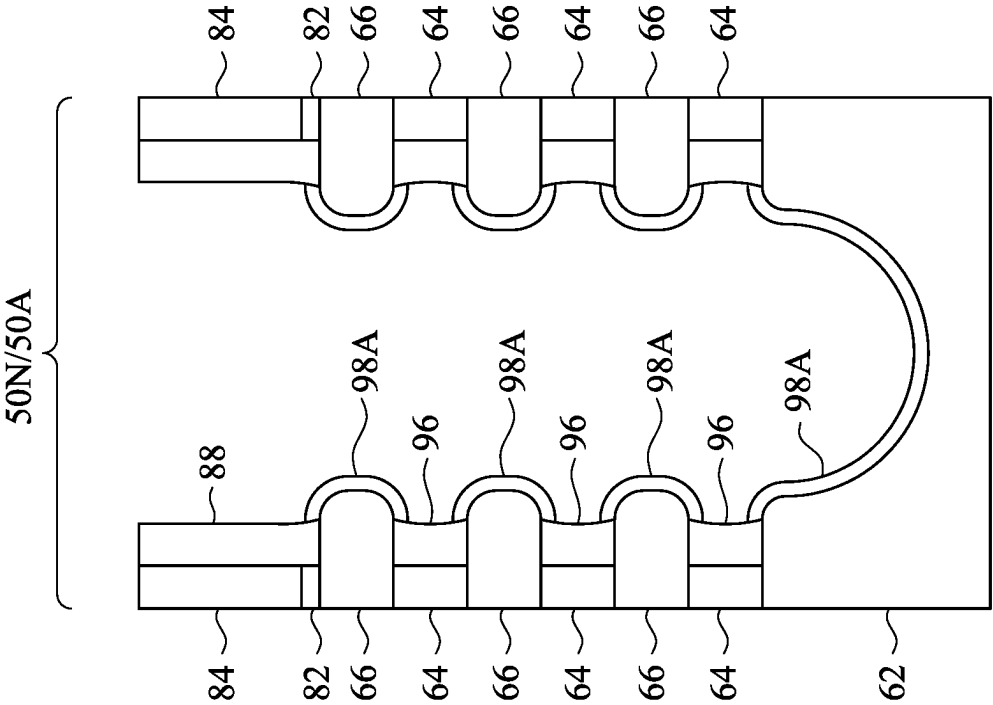


Figure 15

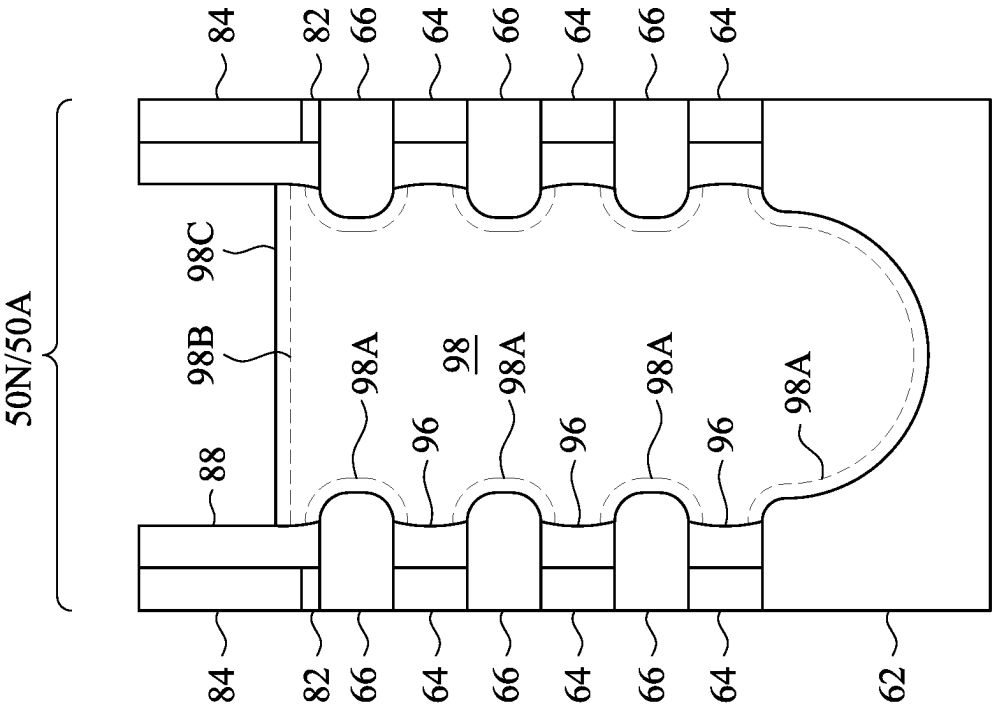


Figure 16

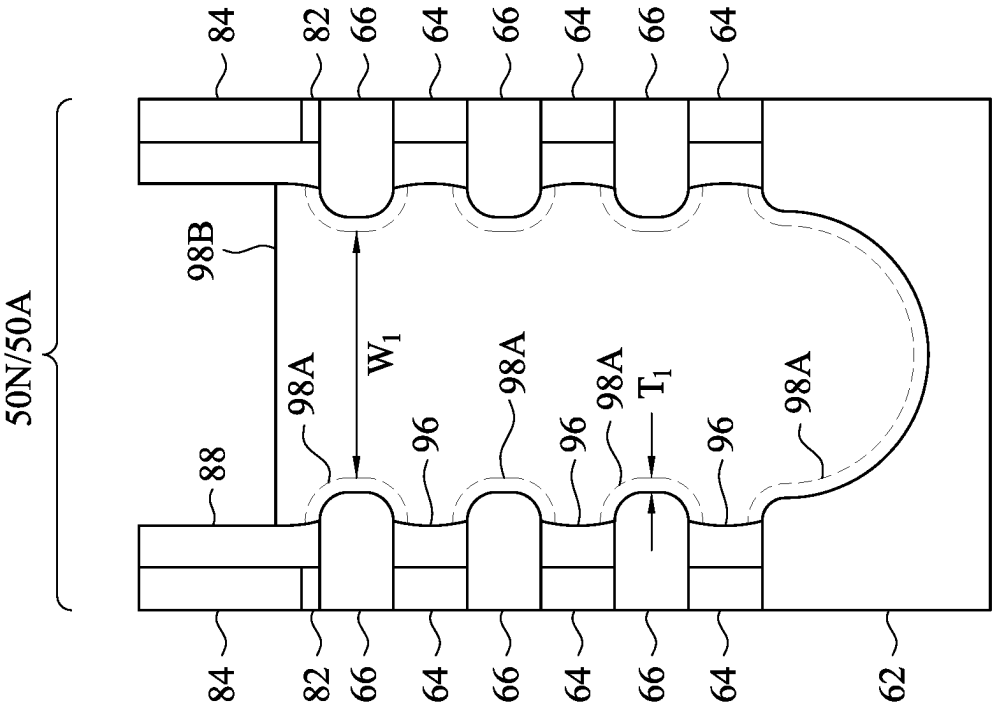


Figure 17

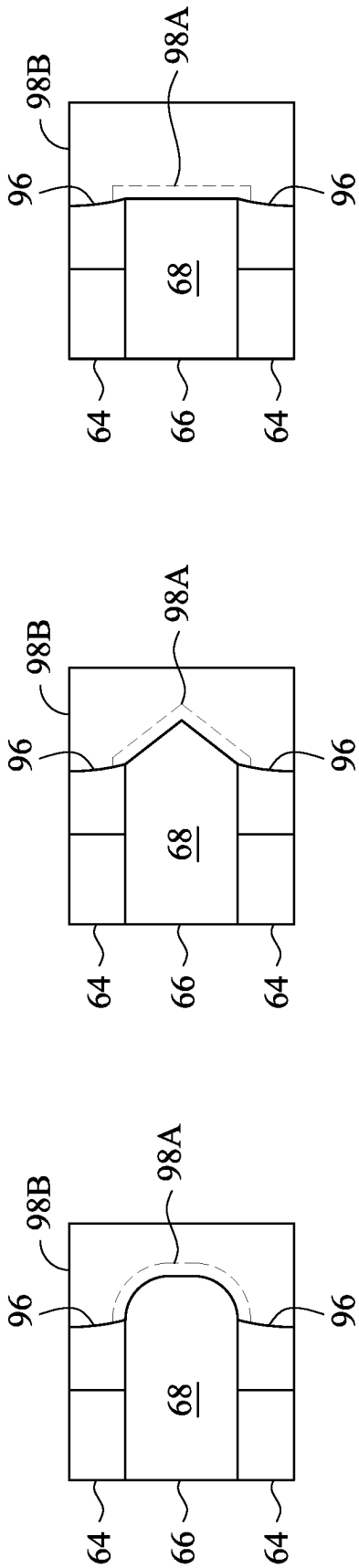


Figure 18A

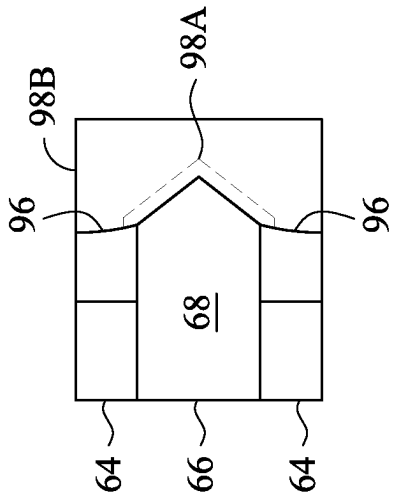


Figure 18B

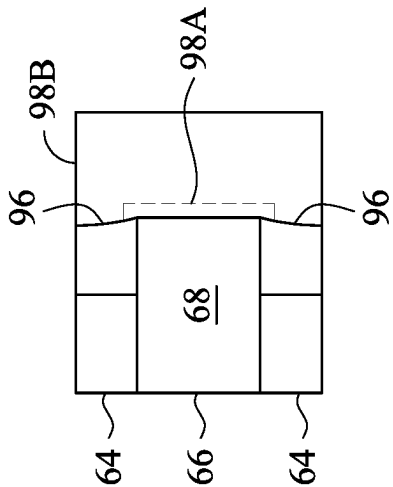


Figure 18C

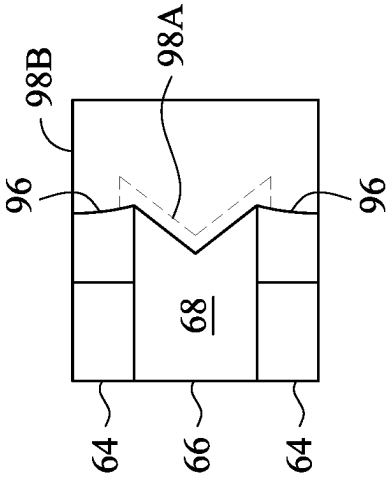


Figure 18D

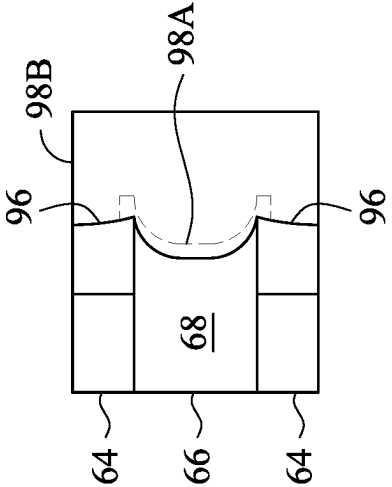


Figure 18E

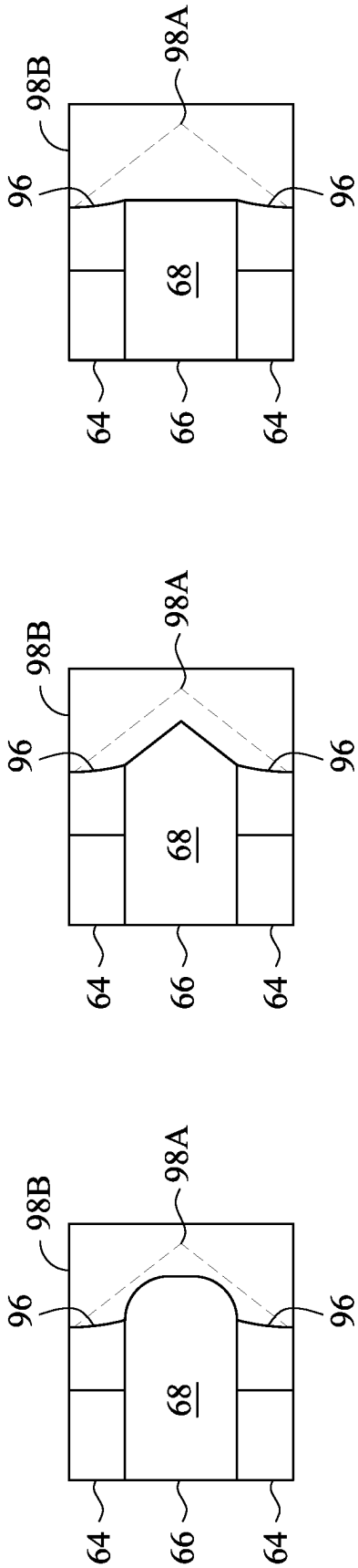


Figure 19A

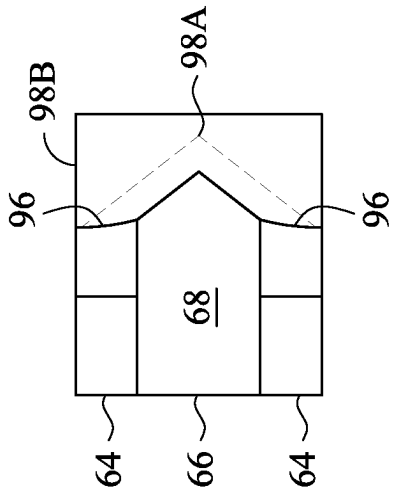


Figure 19B

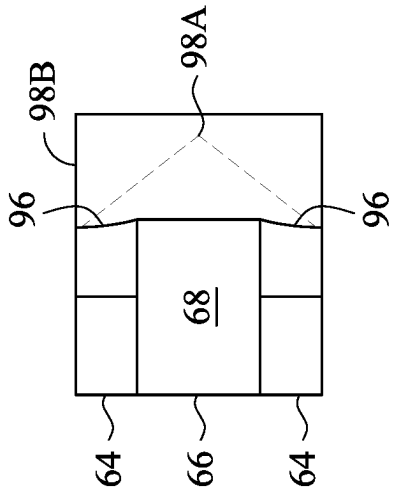


Figure 19C

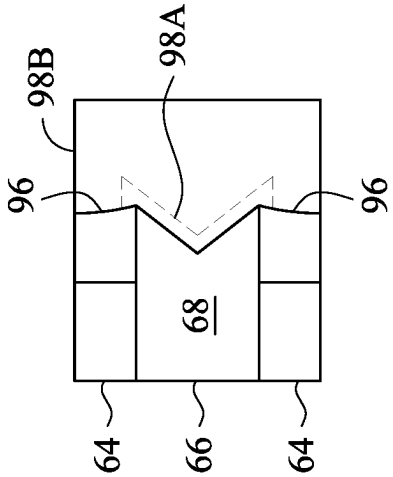


Figure 19D

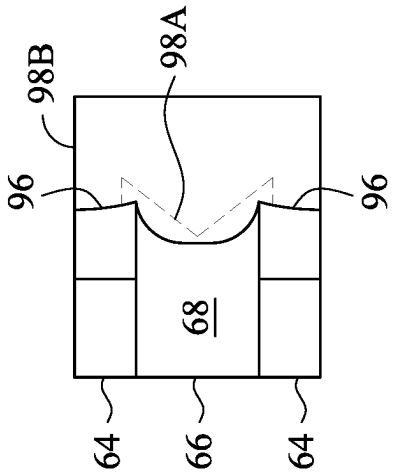


Figure 19E

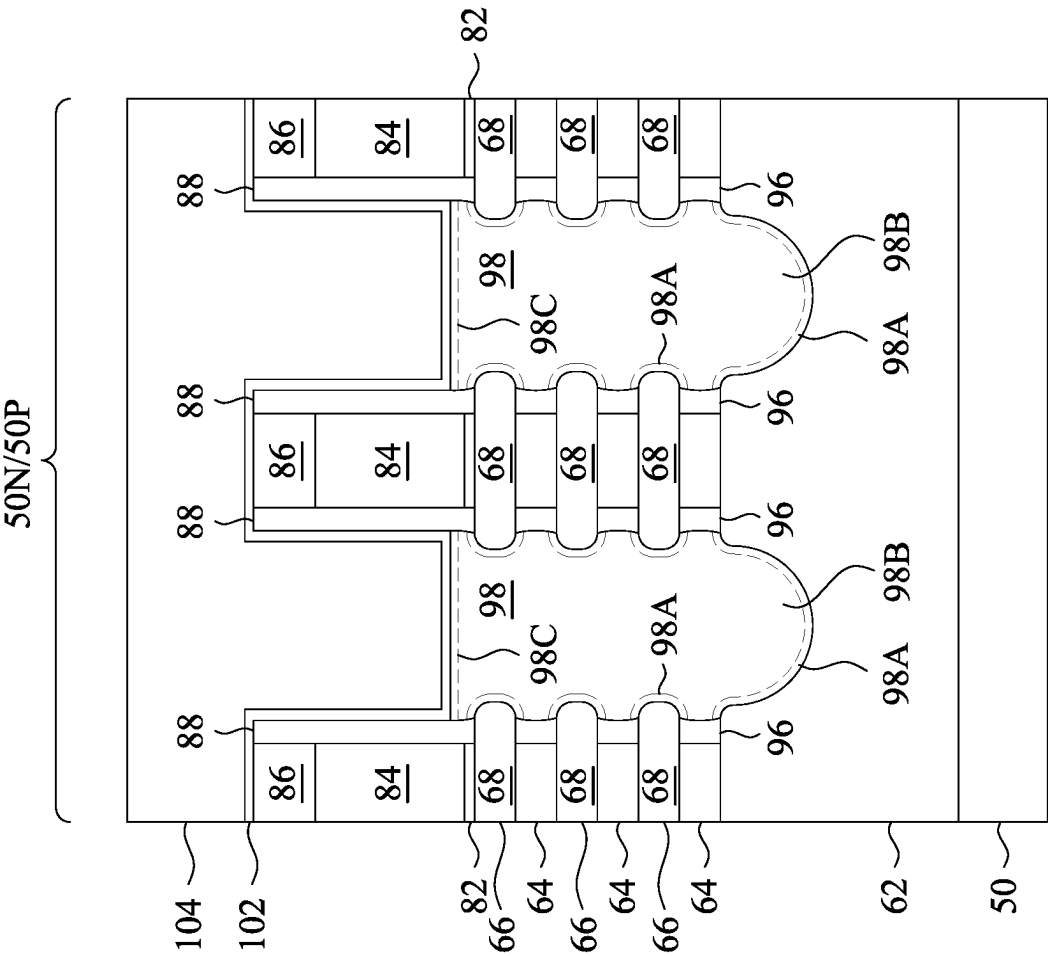


Figure 20A

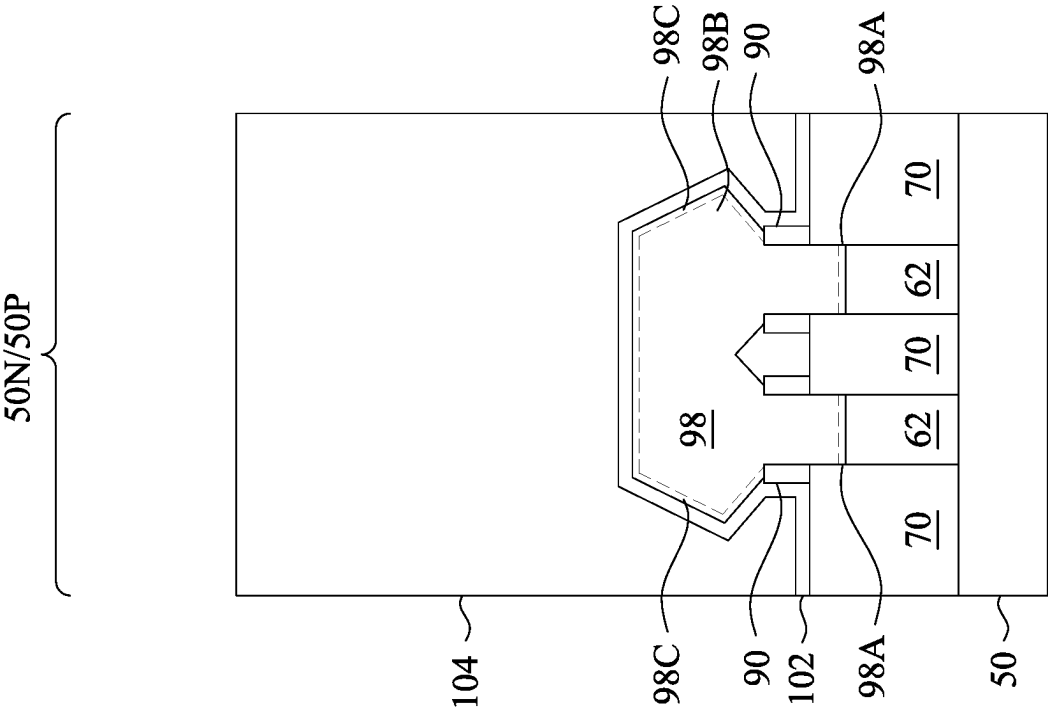


Figure 20C

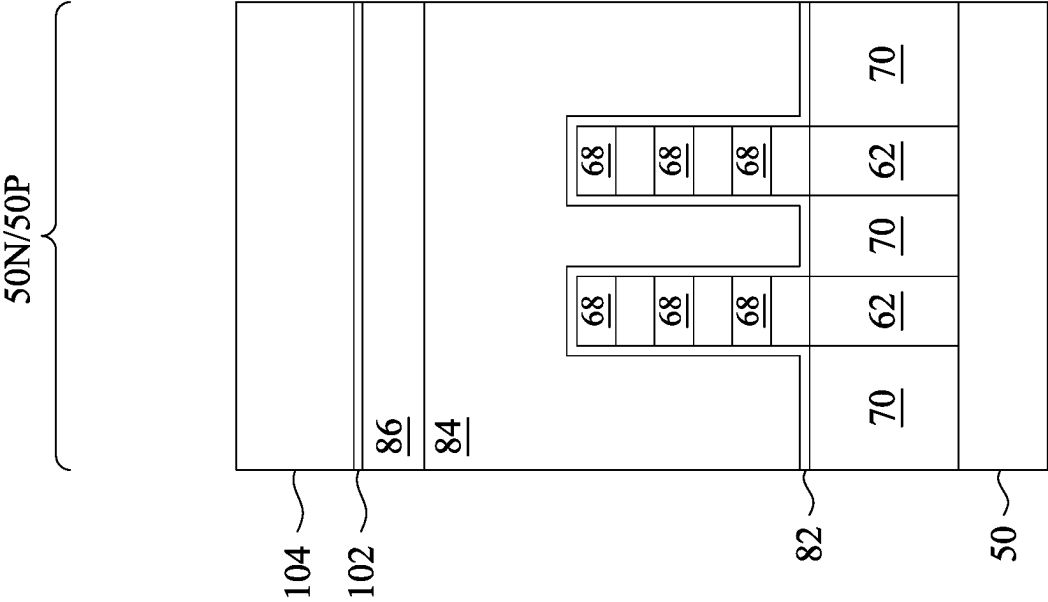


Figure 20B

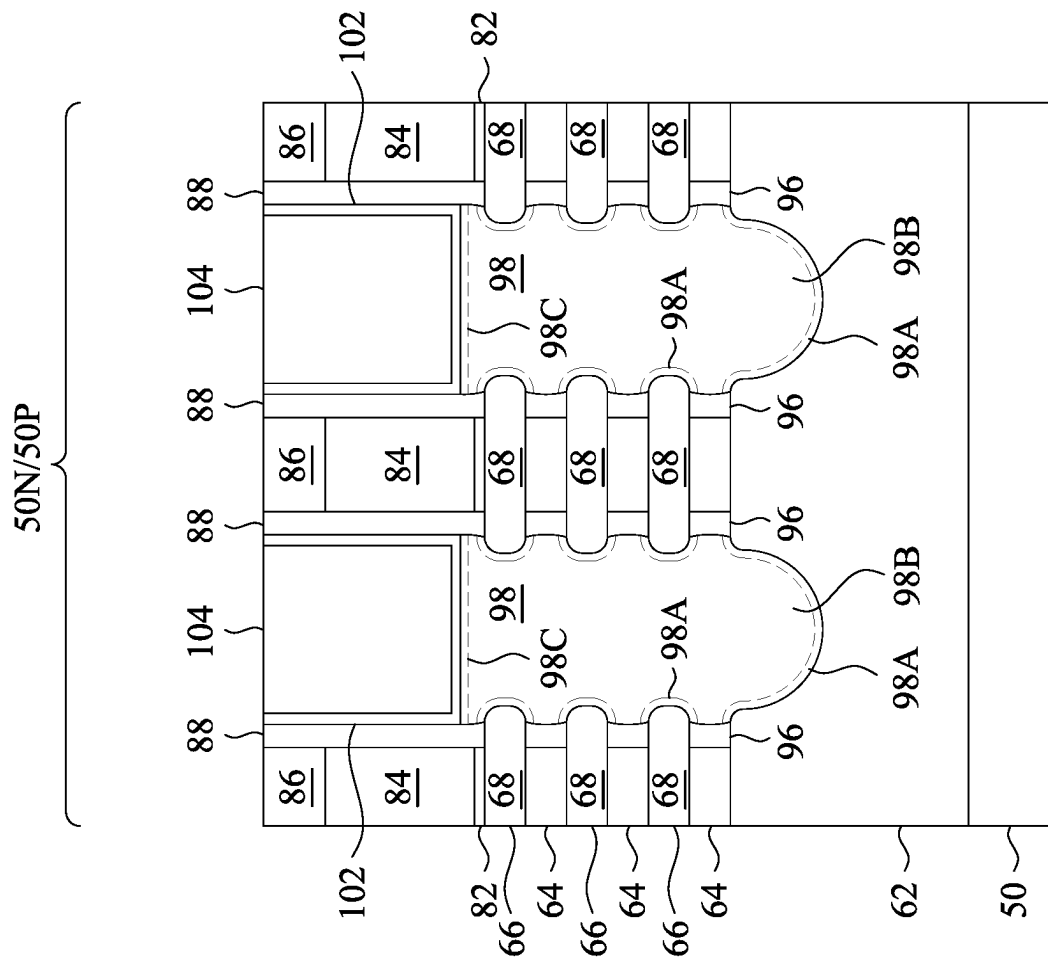


Figure 21A

50N/50P

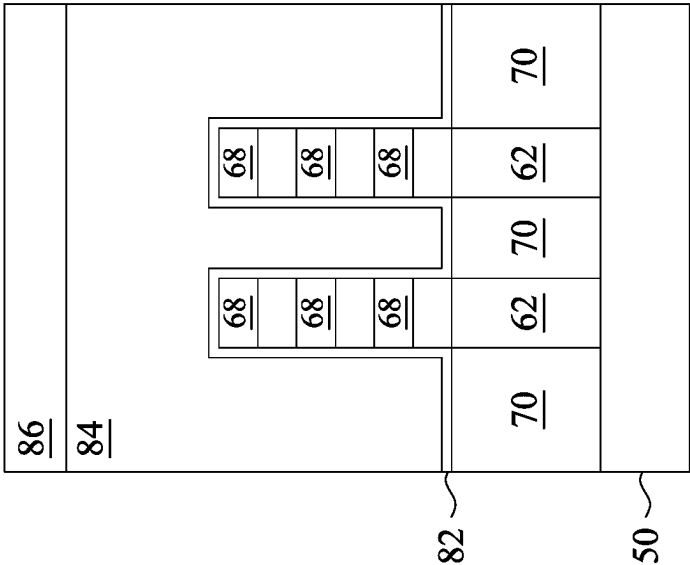


Figure 21B

50N/50P

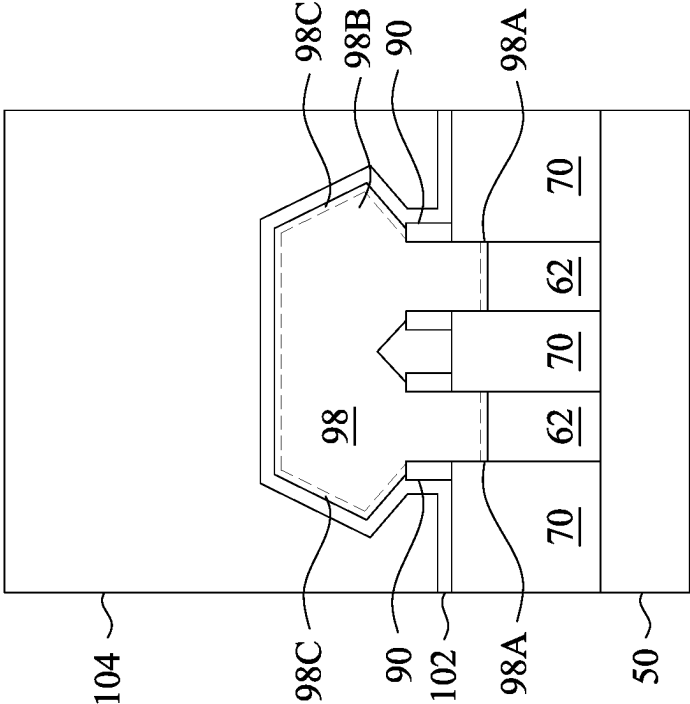


Figure 21C

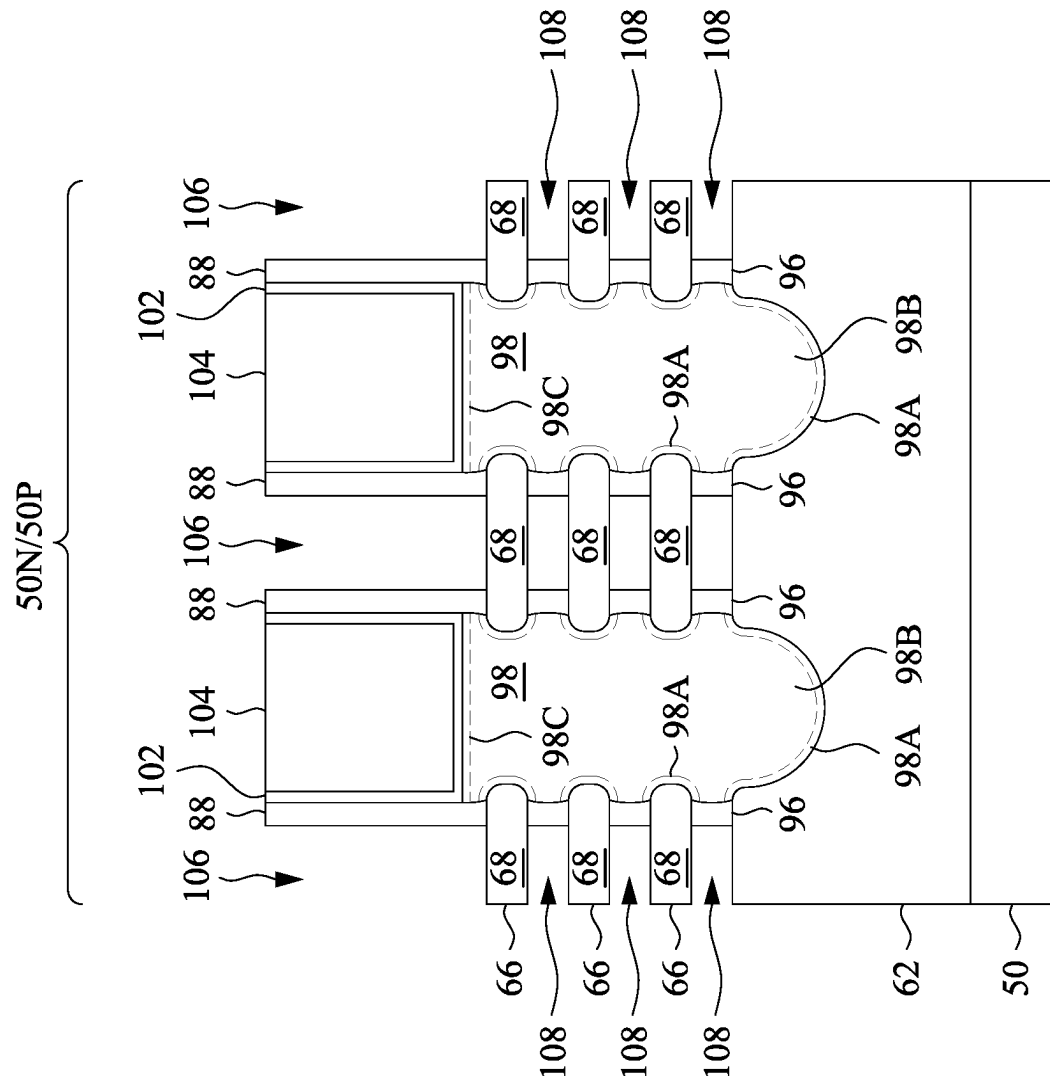


Figure 22A

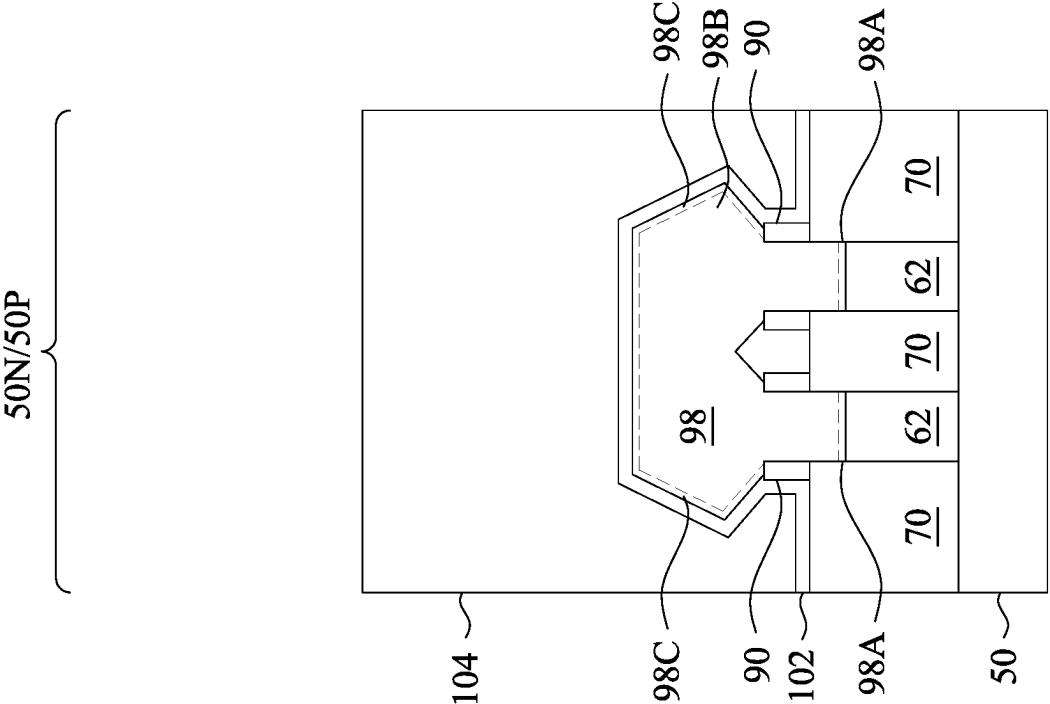


Figure 22B

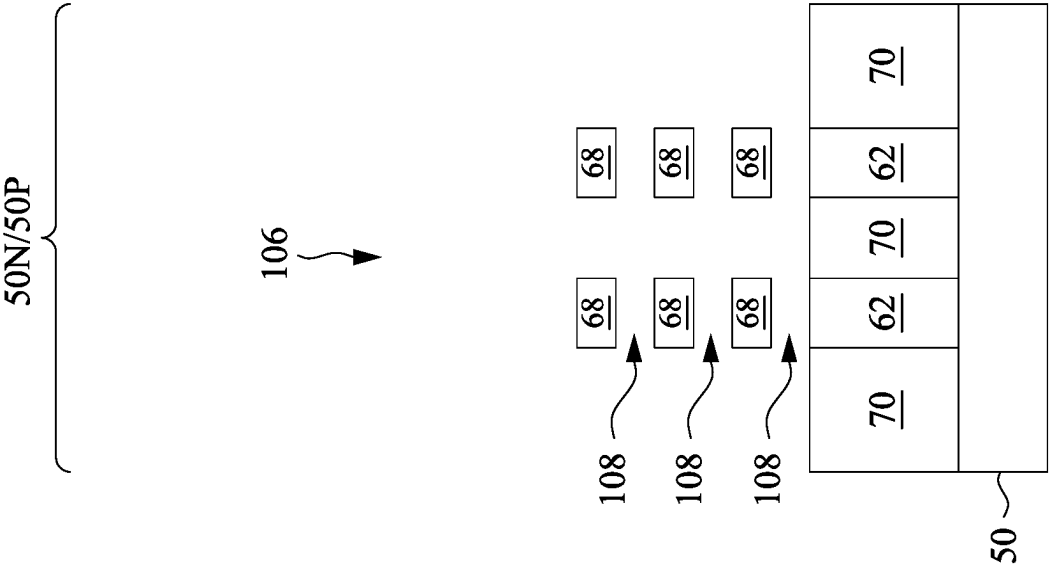


Figure 22C

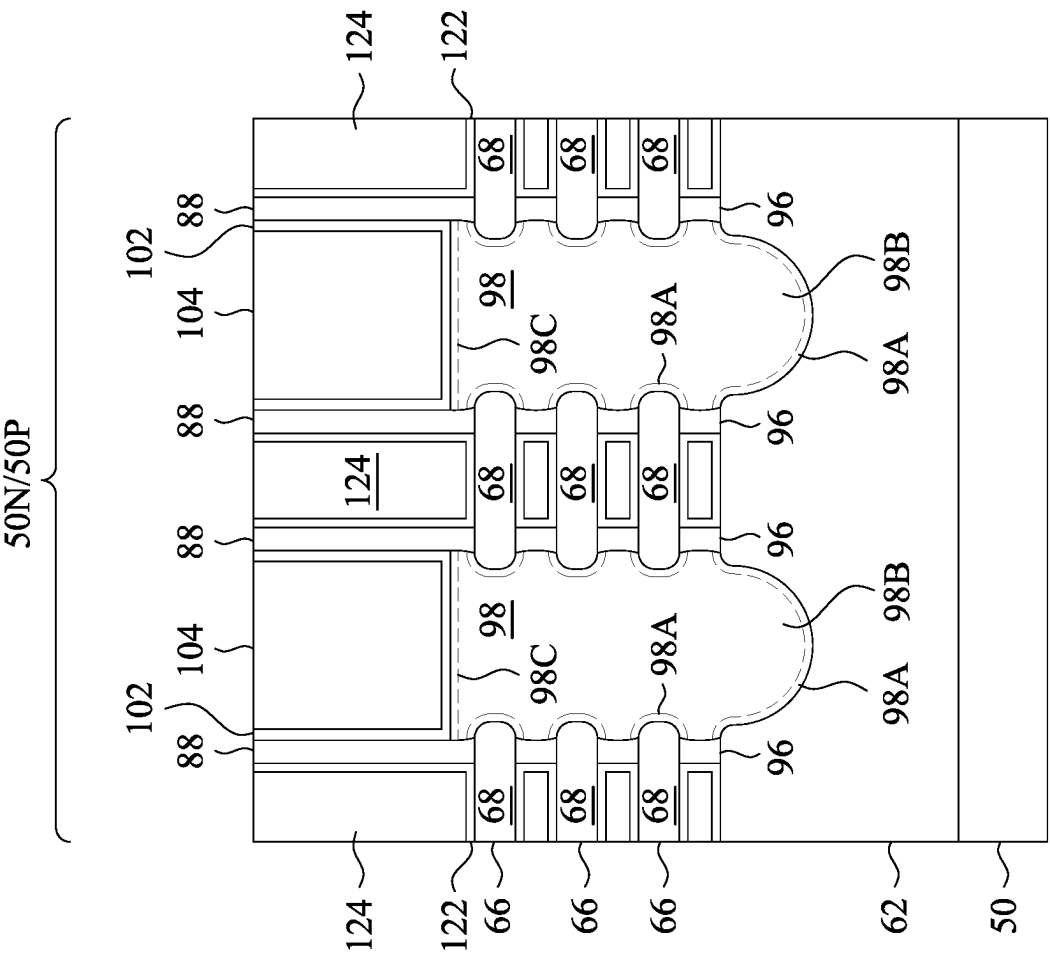


Figure 23A

50N/50P

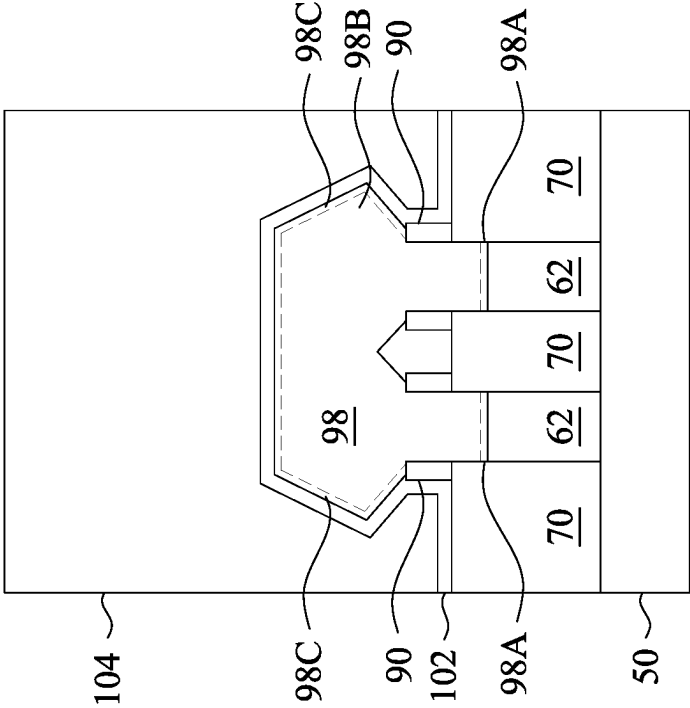


Figure 23B

50N/50P

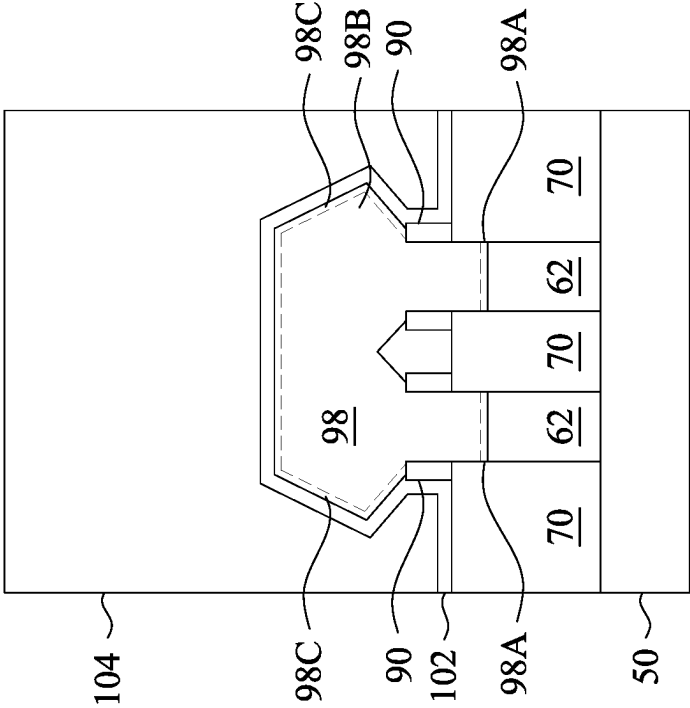


Figure 23C

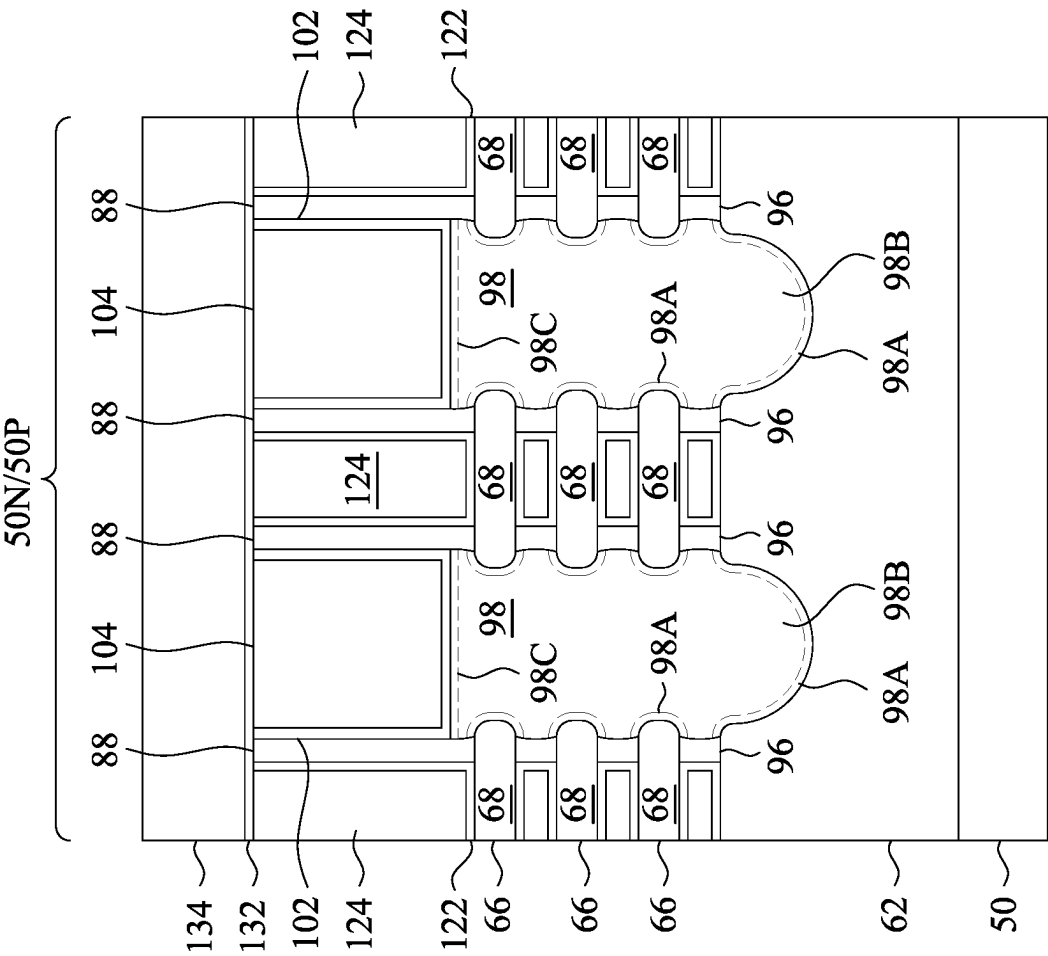


Figure 24A

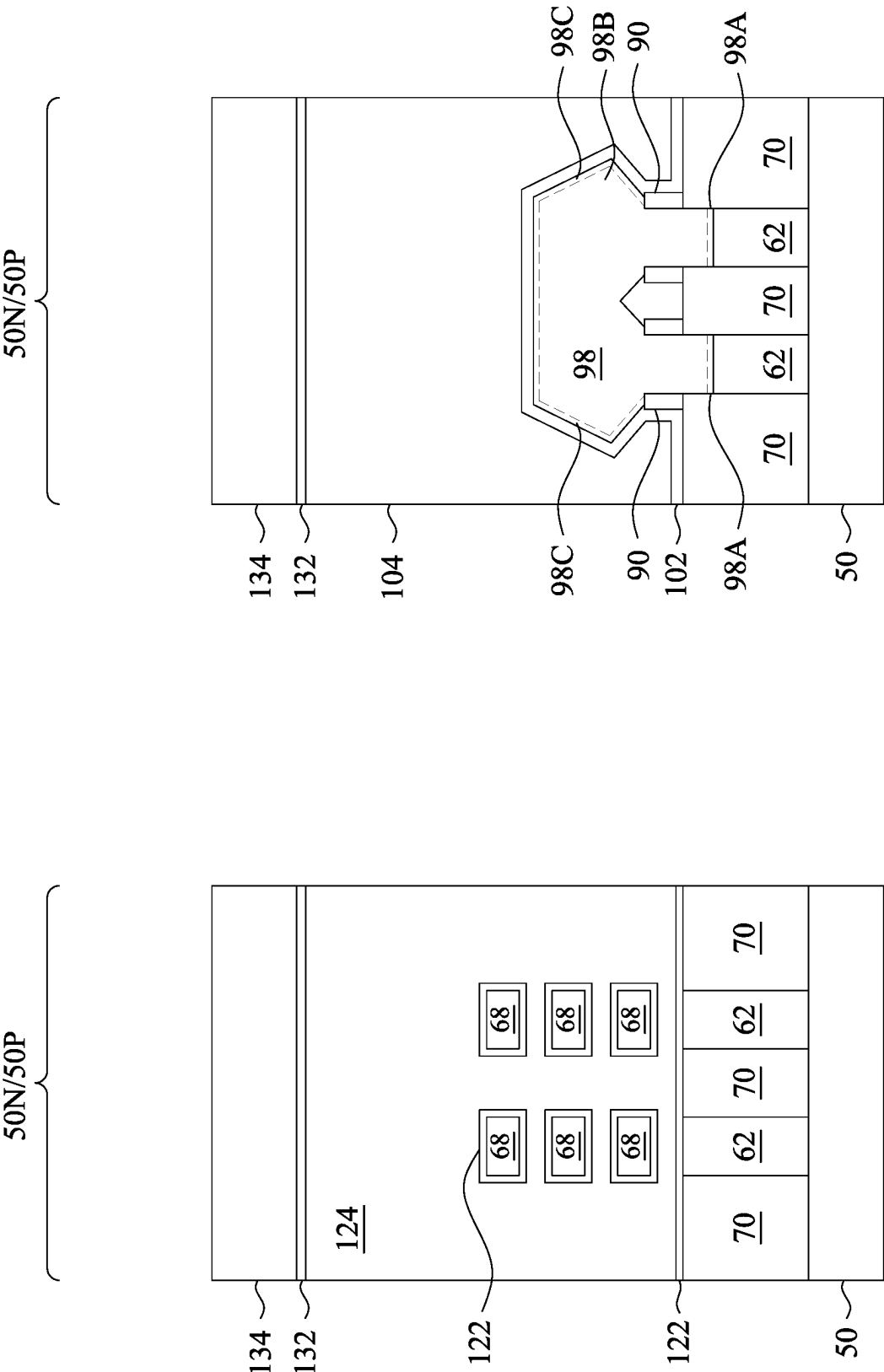


Figure 24C

Figure 24B

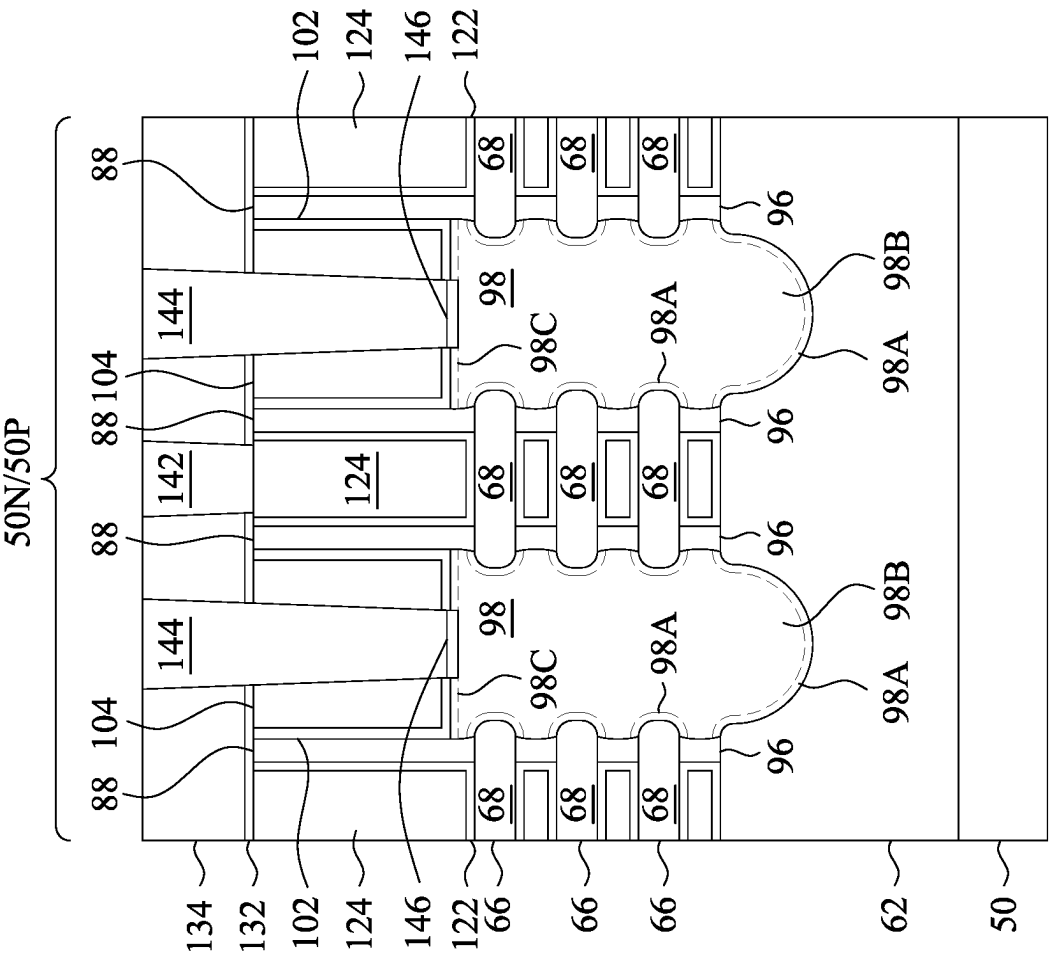


Figure 25A

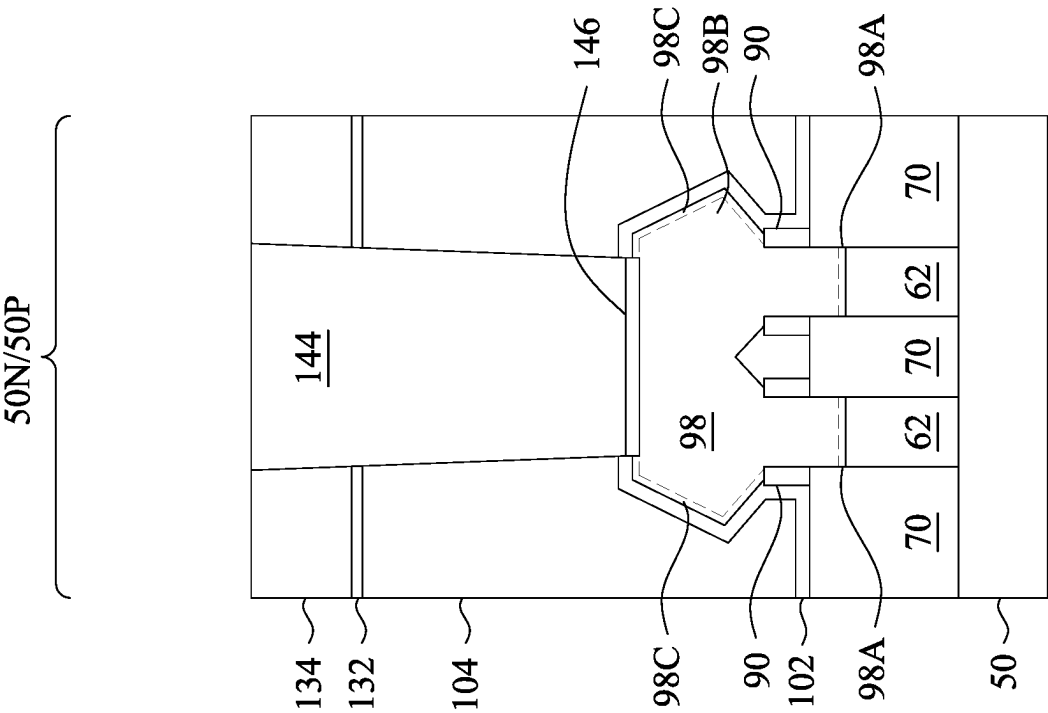


Figure 25C

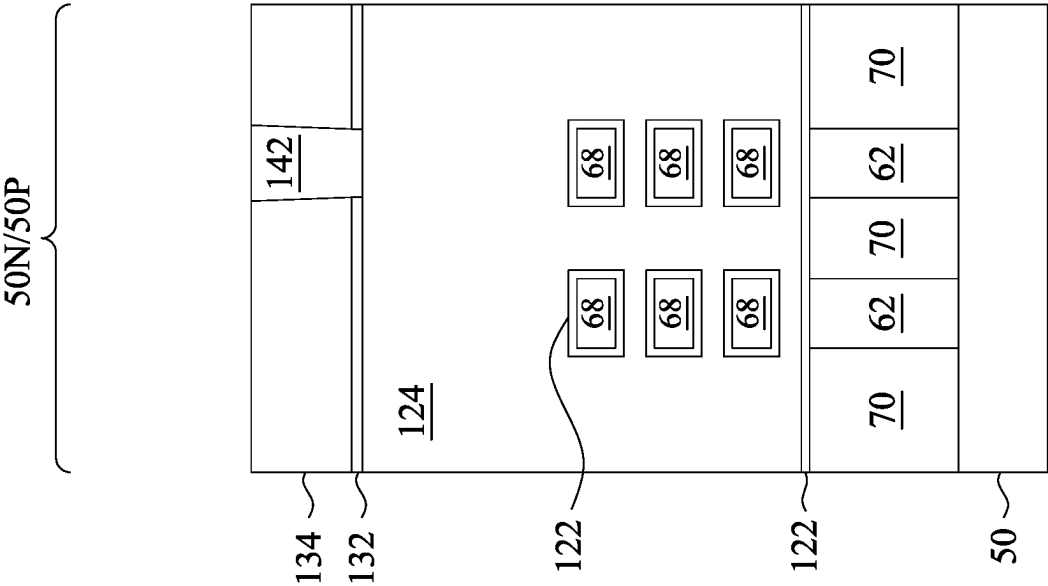


Figure 25B

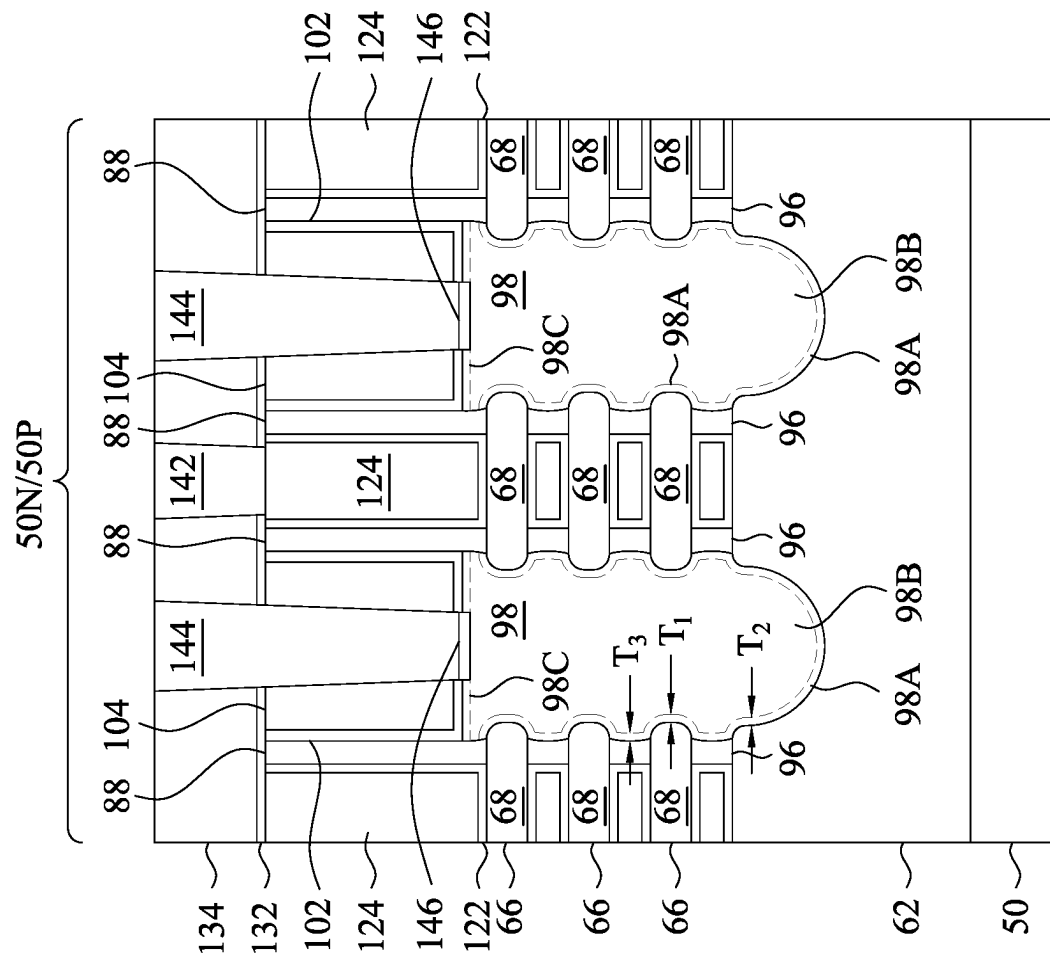


Figure 26

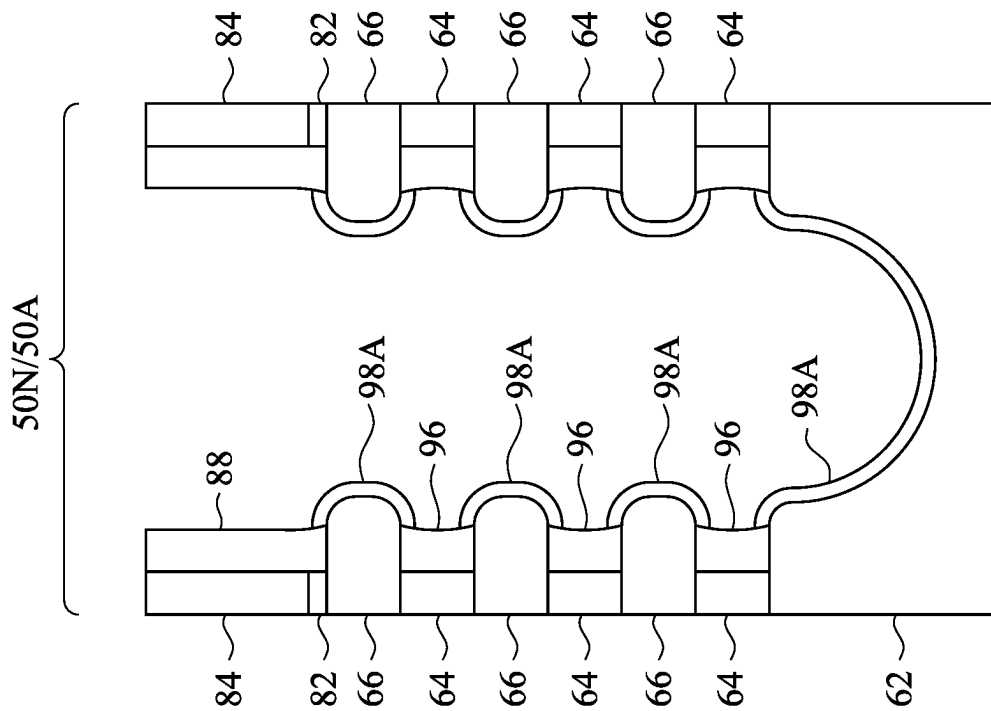


Figure 27

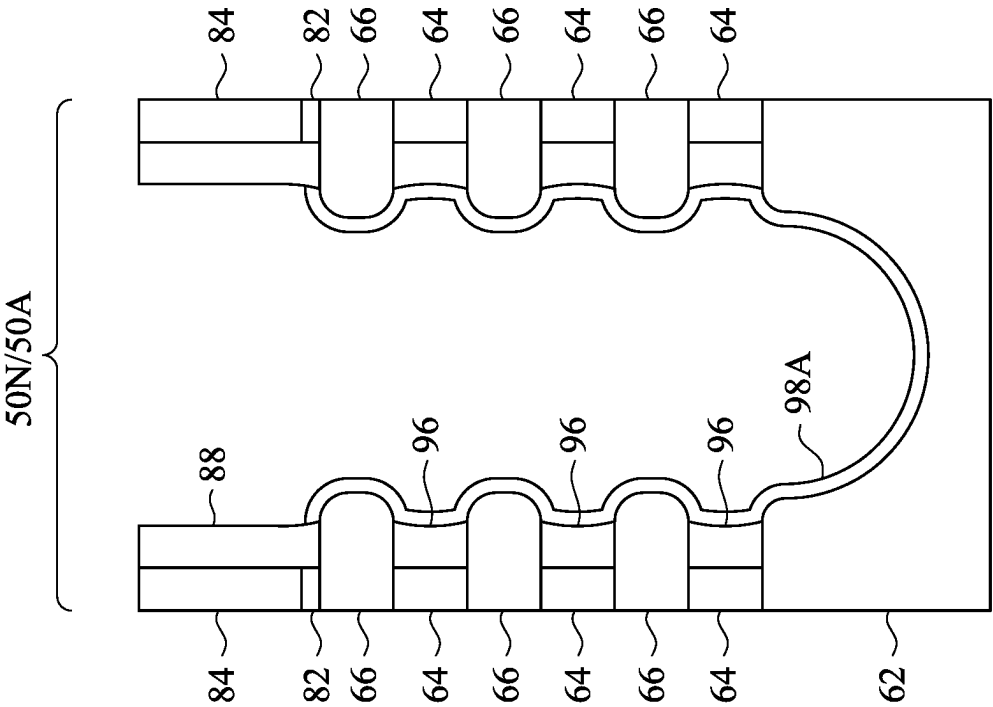


Figure 29

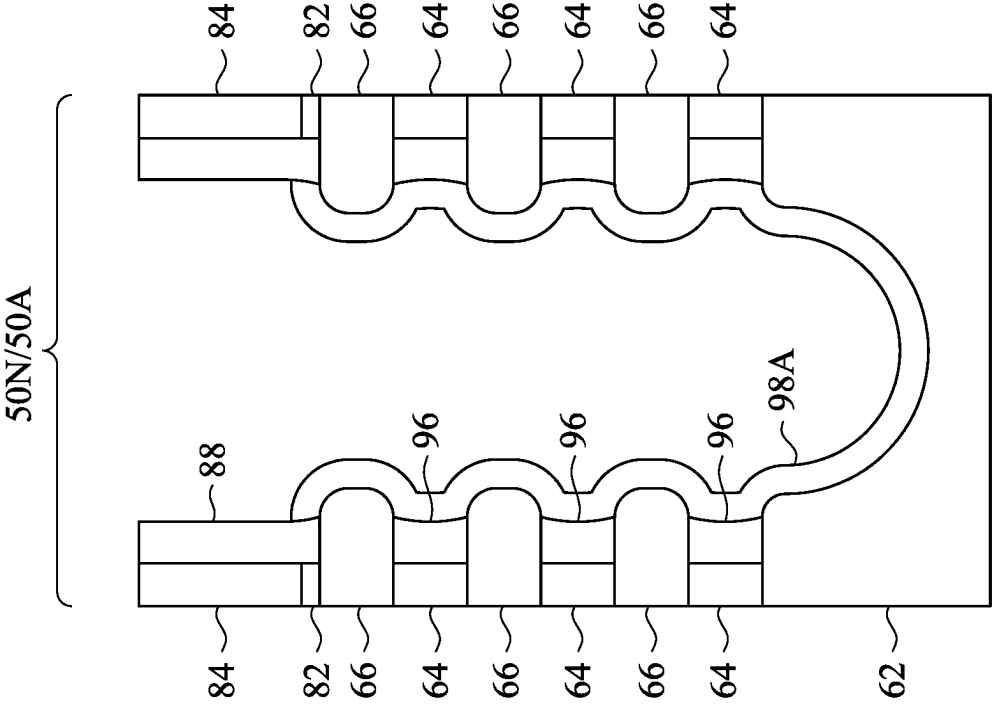


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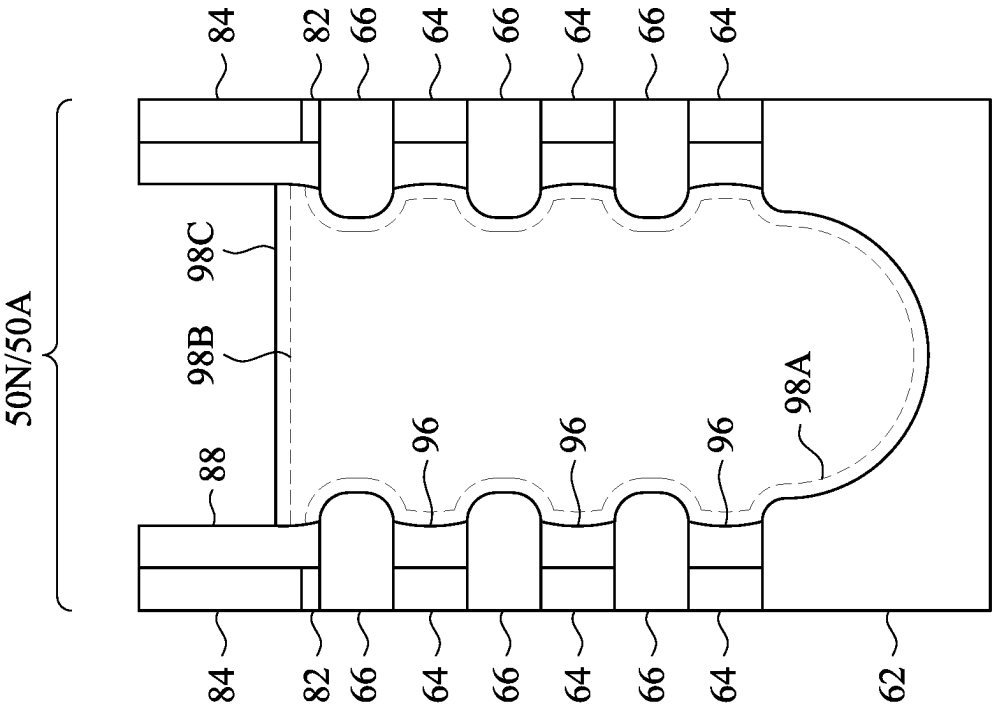


Figure 30

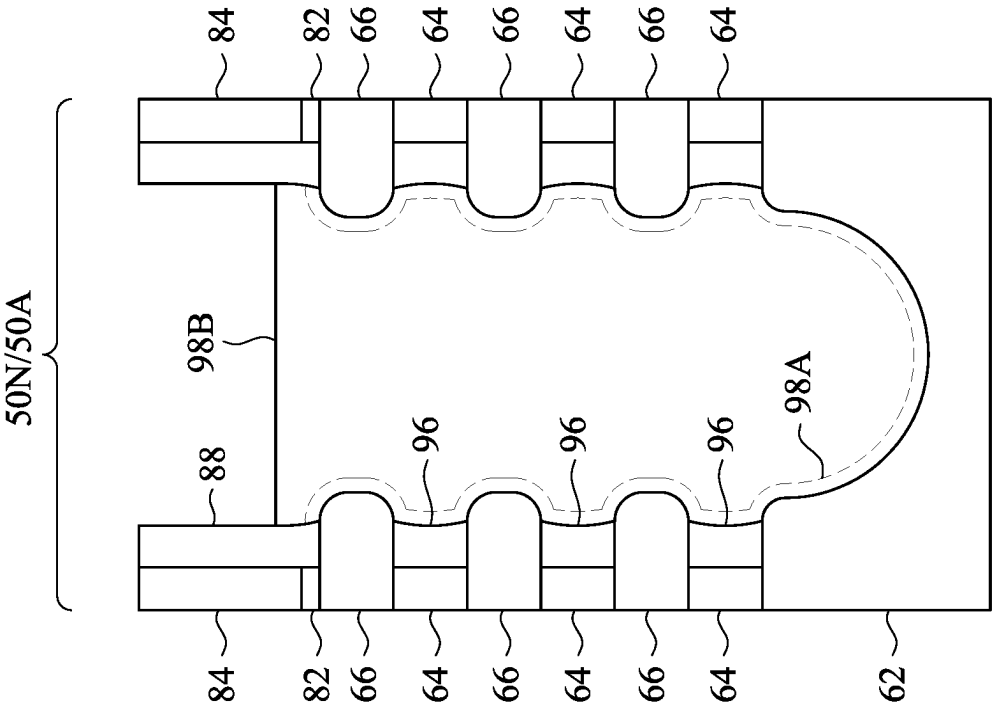


Figure 31

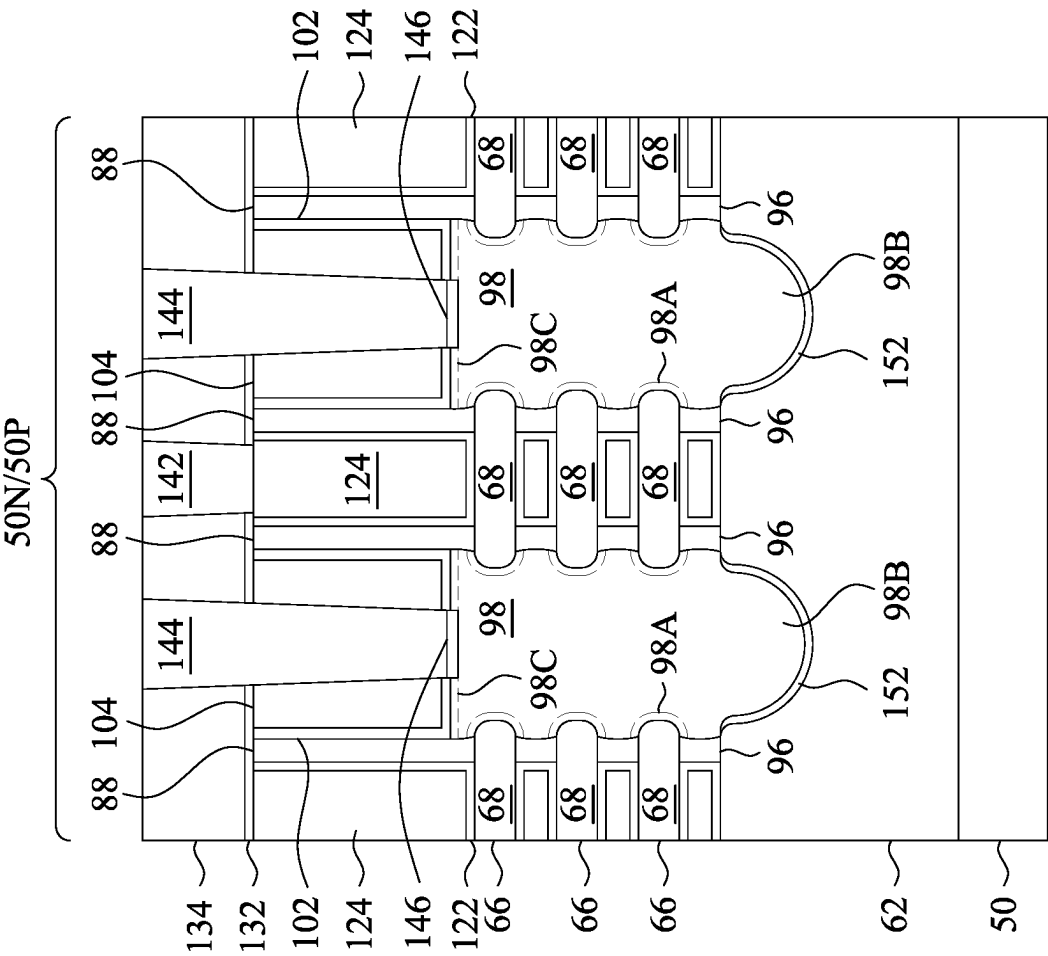


Figure 32A

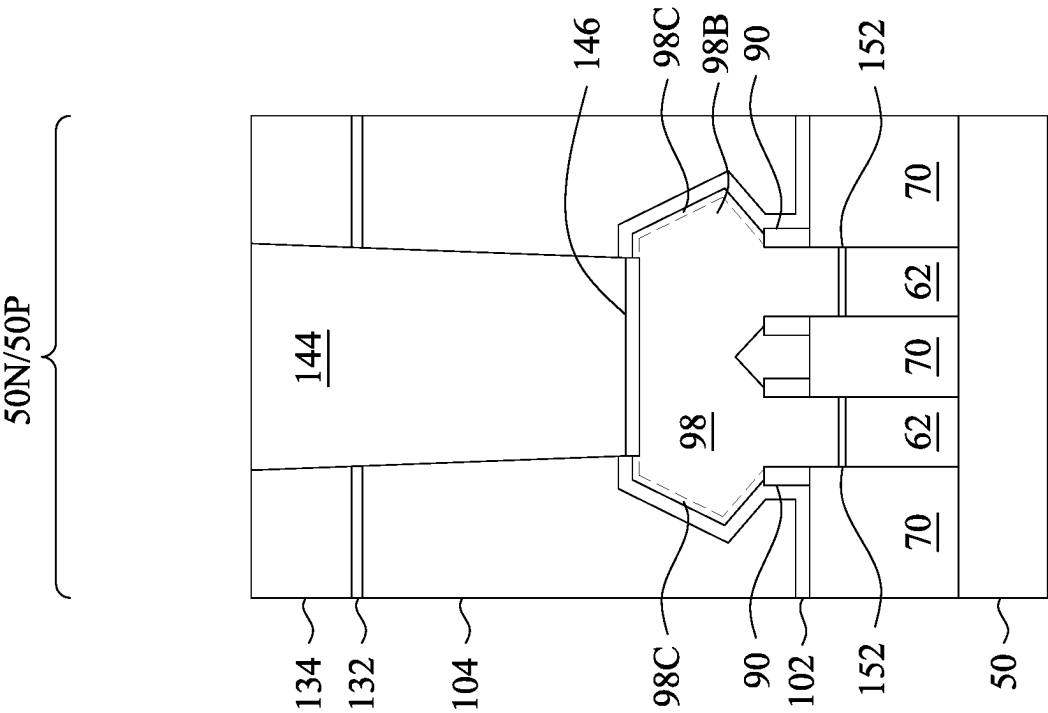


Figure 32C

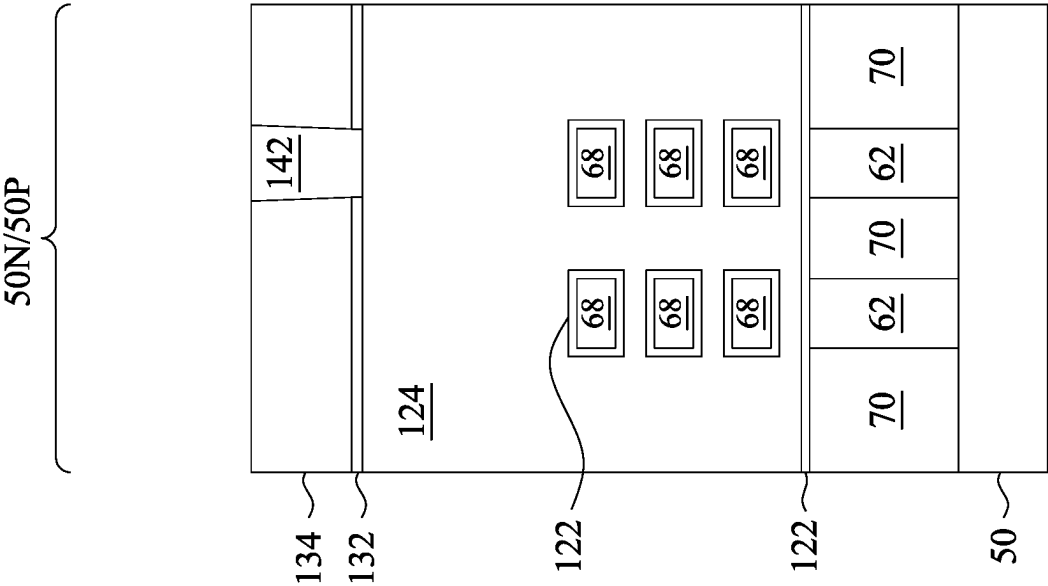


Figure 32B

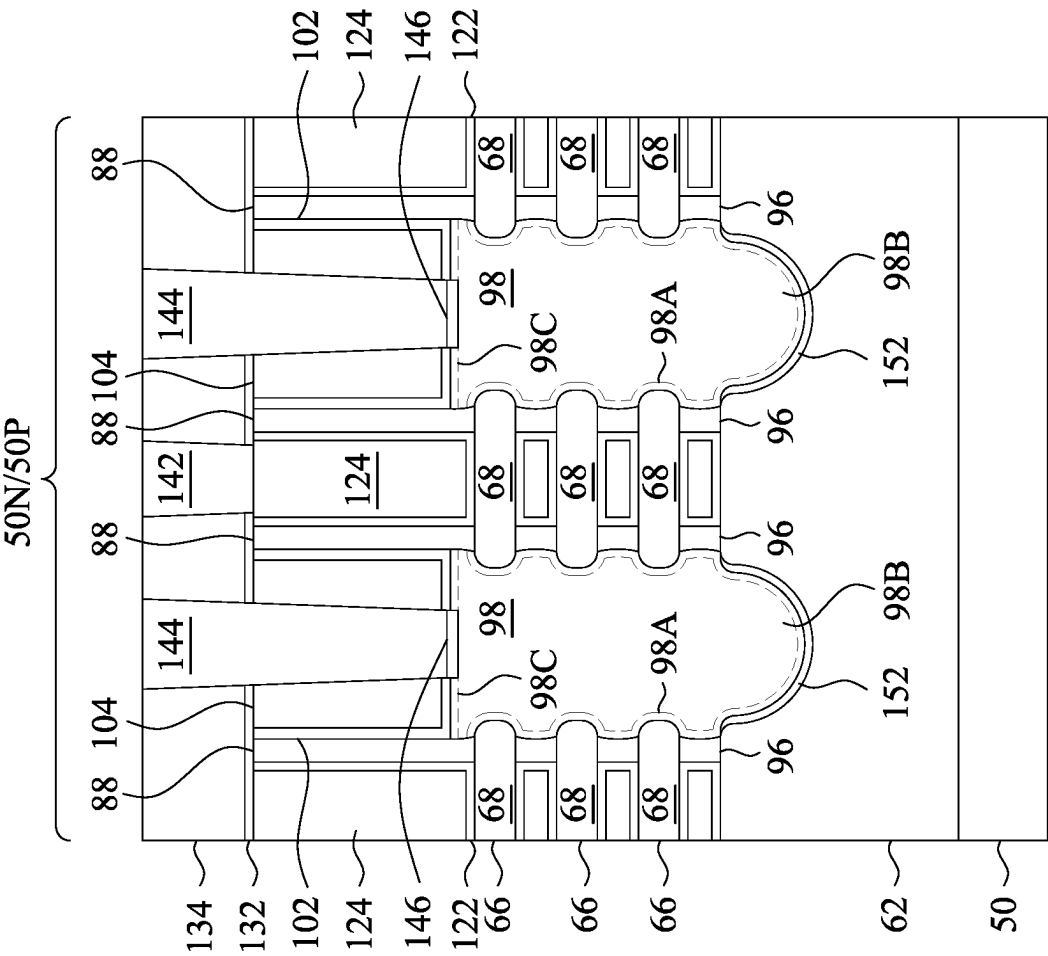


Figure 33

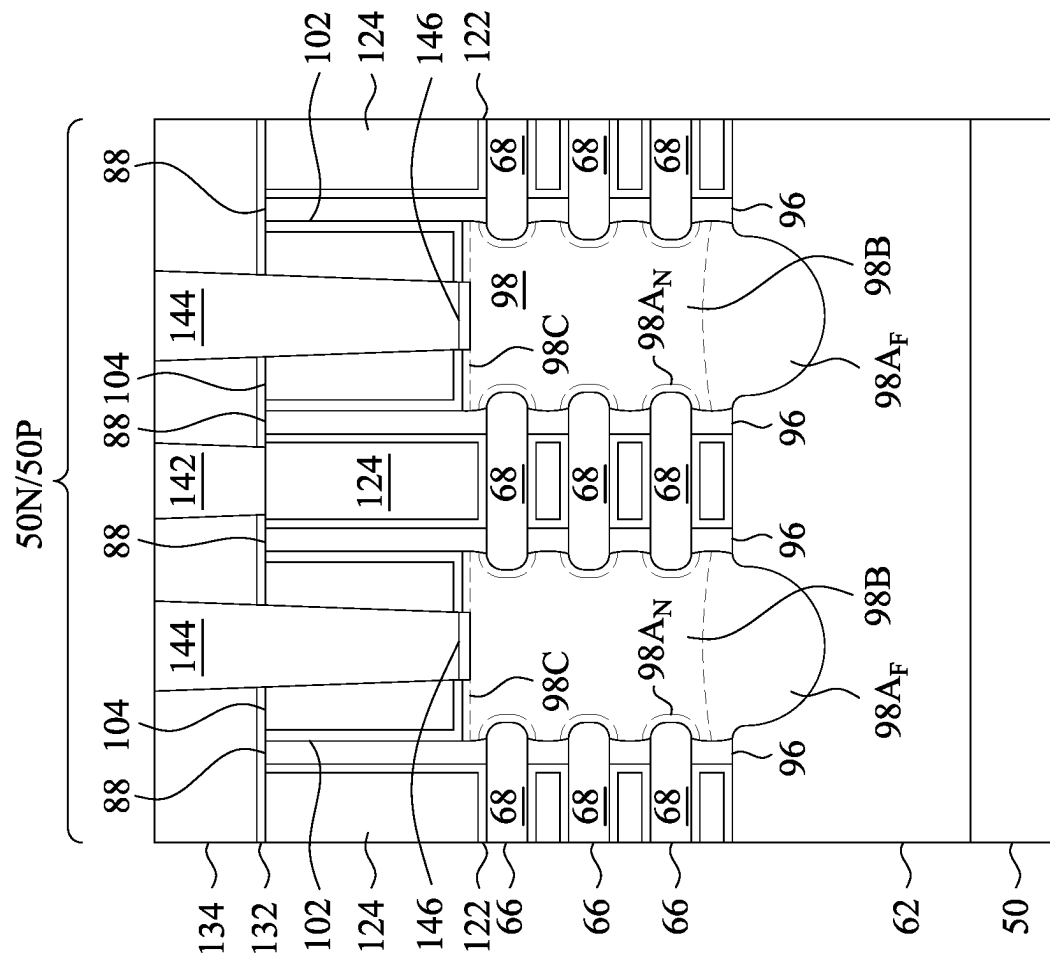


Figure 34A

1

METHODS OF FORMING TRANSISTOR SOURCE/RAIN REGIONS COMPRISING CARBON LINER LAYERS

PRIORITY CLAIM AND CROSS-REFERENCE

This application claims the benefit of U.S. Provisional Application No. 63/188,134, filed on May 13, 2021, which application is hereby incorporated herein by reference.

BACKGROUND

Semiconductor devices are used in a variety of electronic applications, such as, for example, personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers of material over a semiconductor substrate, and patterning the various material elements thereon.

The semiconductor industry continues to improve the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.) by continual reductions in minimum feature size, which allow more components to be integrated into a given area. However, as the minimum features sizes are reduced, additional problems arise that should be addressed.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 illustrates an example of a nanostructure field-effect transistor (nano-FET) in a three-dimensional view, in accordance with some embodiments.

FIGS. 2-25C are views of intermediate stages in the manufacturing of nano-FETs, in accordance with some embodiments.

FIG. 26 is a view of nano-FETs, in accordance with some other embodiments.

FIGS. 27-31 are views of intermediate stages in the manufacturing of nano-FETs, in accordance with some other embodiments.

FIGS. 32A-32C are views of nano-FETs, in accordance with some other embodiments.

FIG. 33 is a view of nano-FETs, in accordance with some other embodiments.

FIGS. 34A-34C are views of nano-FETs, in accordance with some other embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also

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include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

According to various embodiments, source/drain regions for n-type devices are epitaxially grown with liner layers. The liner layers include a semiconductor material that is composed of a blocker element, which helps reduce out-diffusion of dopants from the source/drain regions into adjoining channel regions. By reducing the diffusion of dopants into the channel regions, channel mobility may be increased. Device performance may thus be increased.

Embodiments are described in a particular context, a die including nano-FETs. Various embodiments may be applied, however, to dies including other types of transistors (e.g., fin field-effect transistors (finFETs), planar transistors, or the like) in lieu of or in combination with the nano-FETs.

FIG. 1 illustrates an example of nano-FETs (e.g., nanowire FETs, nanosheet FETs, or the like), in accordance with some embodiments. FIG. 1 is a three-dimensional view, where some features of the nano-FETs are omitted for illustration clarity. The nano-FETs may be nanosheet field-effect transistors (NSFETs), nanowire field-effect transistors (NWFETs), gate-all-around field-effect transistors (GAAFETs), or the like.

The nano-FETs include nanostructures 66 (e.g., nanosheets, nanowires, or the like) over fins 62 on a substrate 50 (e.g., a semiconductor substrate), with the nanostructures 66 acting as channel regions for the nano-FETs. The nanostructures 66 may include p-type nanostructures, n-type nanostructures, or a combination thereof. Isolation regions 70, such as shallow trench isolation (STI) regions, are disposed between adjacent fins 62, which may protrude above and from between adjacent isolation regions 70. Although the isolation regions 70 are described/illustrated as being separate from the substrate 50, as used herein, the term “substrate” may refer to the semiconductor substrate alone or a combination of the semiconductor substrate and the isolation regions. Additionally, although a bottom portion of the fins 62 are illustrated as being single, continuous materials with the substrate 50, the bottom portion of the fins 62 and/or the substrate 50 may include a single material or a plurality of materials. In this context, the fins 62 refer to the portion extending above and from between the adjacent isolation regions 70.

Gate dielectrics 122 are over top surfaces of the fins 62 and wrapped around the top surfaces, sidewalls, and bottom surfaces of the nanostructures 66. Gate electrodes 124 are over and wrapped around the gate dielectrics 122. Epitaxial source/drain regions 98 are disposed on the fins 62 at opposing sides of the gate dielectrics 122 and the gate electrodes 124. An inter-layer dielectric (ILD) 104 is formed over the epitaxial source/drain regions 98. Source/drain contact (subsequently described) to the epitaxial source/drain regions 98 are formed through the ILD 104. The epitaxial source/drain regions 98 may be shared between various nanostructures 66 and fins 62. For example, adjacent epitaxial source/drain regions 98 may be electrically connected, such as through coalescing the epitaxial source/drain regions 98 by epitaxial growth, or through coupling the epitaxial source/drain regions 98 with a same source/drain contact.

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FIG. 1 further illustrates reference cross-sections that are used in later figures. Cross-section A-A' is along a longitudinal axis of a fin 62 and in a direction of, for example, a current flow between the epitaxial source/drain regions 98 of a nano-FET. Cross-section B-B' is along a longitudinal axis of a gate electrode 124 and in a direction, for example, perpendicular to a direction of current flow between the epitaxial source/drain regions 98 of a nano-FET. Cross-section C-C' is parallel to cross-section B-B' and extends through epitaxial source/drain regions 98 of the nano-FETs. Subsequent figures refer to these reference cross-sections for clarity.

Some embodiments discussed herein are discussed in the context of nano-FETs formed using a gate-last process. In other embodiments, a gate-first process may be used. Also, some embodiments contemplate aspects used in planar devices, such as planar FETs, or in fin field-effect transistors (finFETs). For example, finFETs may include fins on a substrate, with the fins acting as channel regions for the finFETs. Similarly, planar FETs may include a substrate, with portions of the substrate acting as channel regions for the planar FETs.

FIGS. 2-25C are views of intermediate stages in the manufacturing of nano-FETs, in accordance with some embodiments. FIGS. 2-6 are three-dimensional views showing a similar three-dimensional view as FIG. 1. FIGS. 7A, 8A, 9A, 10A, 11A, 14A, 15, 16, 17, 20A, 21A, 22A, 23A, 24A, and 25A illustrate reference cross-section A-A' illustrated in FIG. 1. FIGS. 7B, 8B, 9B, 10B, 11B, 14B, 20B, 21B, 22B, 23B, 24B, and 25B illustrate reference cross-section B-B' illustrated in FIG. 1. FIGS. 7C, 8C, 9C, 10C, 11C, 14C, 20C, 21C, 22C, 23C, 24C, and 25C illustrate reference cross-section C-C' illustrated in FIG. 1.

In FIG. 2, a substrate 50 is provided for forming nano-FETs. The substrate 50 may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped (e.g., with a p-type or an n-type impurity) or undoped. The substrate 50 may be a wafer, such as a silicon wafer. Generally, a SOI substrate is a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, typically a silicon or glass substrate. Other substrates, such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate 50 may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; combinations thereof; or the like.

The substrate 50 has an n-type region 50N and a p-type region 50P. The n-type region 50N can be for forming n-type devices, such as NMOS transistors, e.g., n-type nano-FETs, and the p-type region 50P can be for forming p-type devices, such as PMOS transistors, e.g., p-type nano-FETs. The n-type region 50N may be physically separated from the p-type region 50P (not separately illustrated), and any number of device features (e.g., other active devices, doped regions, isolation structures, etc.) may be disposed between the n-type region 50N and the p-type region 50P. Although one n-type region 50N and one p-type region 50P are

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illustrated, any number of n-type regions 50N and p-type regions 50P may be provided.

The substrate 50 may be lightly doped with a p-type or an n-type impurity. An anti-punch-through (APT) implantation may be performed on an upper portion of the substrate 50 to form an APT region. During the APT implantation, impurities may be implanted in the substrate 50. The impurities may have a conductivity type opposite from a conductivity type of source/drain regions that will be subsequently formed in each of the n-type region 50N and the p-type region 50P. The APT region may extend under the source/drain regions in the nano-FETs. The APT region may be used to reduce the leakage from the source/drain regions to the substrate 50. In some embodiments, the impurity concentration in the APT region may be in the range of 10^{18} cm^{-3} to 10^{19} cm^{-3} .

A multi-layer stack 52 is formed over the substrate 50. The multi-layer stack 52 includes alternating first semiconductor layers 54 and second semiconductor layers 56. The first semiconductor layers 54 are formed of a first semiconductor material, and the second semiconductor layers 56 are formed of a second semiconductor material. The semiconductor materials may each be selected from the candidate semiconductor materials of the substrate 50. In the illustrated embodiment, the multi-layer stack 52 includes three layers of each of the first semiconductor layers 54 and the second semiconductor layers 56. It should be appreciated that the multi-layer stack 52 may include any number of the first semiconductor layers 54 and the second semiconductor layers 56.

In the illustrated embodiment, and as will be subsequently described in greater detail, the first semiconductor layers 54 will be removed and the second semiconductor layers 56 will be patterned to form channel regions for the nano-FETs in both the n-type region 50N and the p-type region 50P. The first semiconductor layers 54 are sacrificial layers (or dummy layers), which will be removed in subsequent processing to expose the top surfaces and the bottom surfaces of the second semiconductor layers 56. The first semiconductor material of the first semiconductor layers 54 is a material that has a high etching selectivity from the etching of the second semiconductor layers 56, such as silicon germanium. The second semiconductor material of the second semiconductor layers 56 is a material suitable for both n-type and p-type devices, such as silicon.

In another embodiment (not separately illustrated), the first semiconductor layers 54 will be patterned to form channel regions for nano-FETs in one region (e.g., the p-type region 50P), and the second semiconductor layers 56 will be patterned to form channel regions for nano-FETs in another region (e.g., the n-type region 50N). The first semiconductor material of the first semiconductor layers 54 may be a material suitable for p-type devices, such as silicon germanium (e.g., $\text{Si}_x\text{Ge}_{1-x}$, where x can be in the range of 0 to 1), pure germanium, a III-V compound semiconductor, a II-VI compound semiconductor, or the like. The second semiconductor material of the second semiconductor layers 56 may be a material suitable for n-type devices, such as silicon, silicon carbide, a III-V compound semiconductor, a II-VI compound semiconductor, or the like. The first semiconductor material and the second semiconductor material may have a high etching selectivity from the etching of one another, so that the first semiconductor layers 54 may be removed without removing the second semiconductor layers 56 in the n-type region 50N, and the second semiconductor layers 56 may be removed without removing the first semiconductor layers 54 in the p-type region 50P.

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Each of the layers of the multi-layer stack **52** may be grown by a process such as vapor phase epitaxy (VPE) or molecular beam epitaxy (MBE), deposited by a process such as chemical vapor deposition (CVD) or atomic layer deposition (ALD), or the like. Each of the layers may have a small thickness, such as a thickness in a range of 5 nm to 30 nm. In some embodiments, some layers of the multi-layer stack **52** (e.g., the second semiconductor layers **56**) are formed to be thinner than other layers of the multi-layer stack **52** (e.g., the first semiconductor layers **54**).

In FIG. 3, trenches are patterned in the substrate **50** and the multi-layer stack **52** to form fins **62**, first nanostructures **64**, and second nanostructures **66**. The fins **62** are semiconductor strips patterned in the substrate **50**. The first nanostructures **64** and the second nanostructures **66** include the remaining portions of the first semiconductor layers **54** and the second semiconductor layers **56**, respectively. The trenches may be patterned by any acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etching may be anisotropic.

The fins **62** and the nanostructures **64**, **66** may be patterned by any suitable method. For example, the fins **62** and the nanostructures **64**, **66** may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used as masks to pattern the fins **62** and the nanostructures **64**, **66**. In some embodiments, the mask (or other layer) may remain on the nanostructures **64**, **66**.

The fins **62** and the nanostructures **64**, **66** may each have widths in a range of 8 nm to 40 nm. In the illustrated embodiment, the fins **62** and the nanostructures **64**, **66** have substantially equal widths in the n-type region **50N** and the p-type region **50P**. In another embodiment, the fins **62** and the nanostructures **64**, **66** in one region (e.g., the n-type region **50N**) are wider or narrower than the fins **62** and the nanostructures **64**, **66** in another region (e.g., the p-type region **50P**).

In FIG. 4, STI regions **70** are formed over the substrate **50** and between adjacent fins **62**. The STI regions **70** are disposed around at least a portion of the fins **62** such that at least a portion of the nanostructures **64**, **66** protrude from between adjacent STI regions **70**. In the illustrated embodiment, the top surfaces of the STI regions **70** are coplanar (within process variations) with the top surfaces of the fins **62**. In some embodiments, the top surfaces of the STI regions **70** are above or below the top surfaces of the fins **62**. The STI regions **70** separate the features of adjacent devices.

The STI regions **70** may be formed by any suitable method. For example, an insulation material can be formed over the substrate **50** and the nanostructures **64**, **66**, and between adjacent fins **62**. The insulation material may be an oxide, such as silicon oxide, a nitride, such as silicon nitride, the like, or a combination thereof, which may be formed by a chemical vapor deposition (CVD) process, such as high density plasma CVD (HDP-CVD), flowable chemical vapor deposition (FCVD), the like, or a combination thereof. Other insulation materials formed by any acceptable process may

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be used. In some embodiments, the insulation material is silicon oxide formed by FCVD. An anneal process may be performed once the insulation material is formed. In an embodiment, the insulation material is formed such that excess insulation material covers the nanostructures **64**, **66**. Although the STI regions **70** are each illustrated as a single layer, some embodiments may utilize multiple layers. For example, in some embodiments a liner (not separately illustrated) may first be formed along surfaces of the substrate **50**, the fins **62**, and the nanostructures **64**, **66**. Thereafter, a fill material, such as those previously described may be formed over the liner.

A removal process is then applied to the insulation material to remove excess insulation material over the nanostructures **64**, **66**. In some embodiments, a planarization process such as a chemical mechanical polish (CMP), an etch-back process, combinations thereof, or the like may be utilized. In embodiments in which a mask remains on the nanostructures **64**, **66**, the planarization process may expose the mask or remove the mask. After the planarization process, the top surfaces of the insulation material and the mask (if present) or the nanostructures **64**, **66** are coplanar (within process variations). Accordingly, the top surfaces of the mask (if present) or the nanostructures **64**, **66** are exposed through the insulation material. In the illustrated embodiment, no mask remains on the nanostructures **64**, **66**. The insulation material is then recessed to form the STI regions **70**. The insulation material is recessed such that at least a portion of the nanostructures **64**, **66** protrude from between adjacent portions of the insulation material. Further, the top surfaces of the STI regions **70** may have a flat surface as illustrated, a convex surface, a concave surface (such as dishing), or a combination thereof. The top surfaces of the STI regions **70** may be formed flat, convex, and/or concave by an appropriate etch. The insulation material may be recessed using any acceptable etching process, such as one that is selective to the material of the insulation material (e.g., selectively etches the insulation material of the STI regions **70** at a faster rate than the materials of the fins **62** and the nanostructures **64**, **66**). For example, an oxide removal may be performed using dilute hydrofluoric (DHF) acid.

The process previously described is just one example of how the fins **62** and the nanostructures **64**, **66** may be formed. In some embodiments, the fins **62** and/or the nanostructures **64**, **66** may be formed using a mask and an epitaxial growth process. For example, a dielectric layer can be formed over a top surface of the substrate **50**, and trenches can be etched through the dielectric layer to expose the underlying substrate **50**. Epitaxial structures can be epitaxially grown in the trenches, and the dielectric layer can be recessed such that the epitaxial structures protrude from the dielectric layer to form the fins **62** and/or the nanostructures **64**, **66**. The epitaxial structures may include the alternating semiconductor materials previously described, such as the first semiconductor material and the second semiconductor material. In some embodiments where epitaxial structures are epitaxially grown, the epitaxially grown materials may be in situ doped during growth, which may obviate prior and/or subsequent implantations, although in situ and implantation doping may be used together.

Further, appropriate wells (not separately illustrated) may be formed in the nanostructures **64**, **66**, the fins **62**, and/or the substrate **50** by doping (e.g., with a p-type or an n-type impurity). The wells may have a conductivity type opposite from a conductivity type of source/drain regions that will be subsequently formed in each of the n-type region **50N** and the p-type region **50P**. In some embodiments, a p-type well

is formed in the n-type region **50N**, and an n-type well is formed in the p-type region **50P**. In some embodiments, a p-type well or an n-type well is formed in both the n-type region **50N** and the p-type region **50P**.

In embodiments with different well types, different implant steps for the n-type region **50N** and the p-type region **50P** may be achieved using mask (not separately illustrated) such as a photoresist. For example, a photoresist may be formed over the fins **62**, the nanostructures **64**, **66**, and the STI regions **70** in the n-type region **50N**. The photoresist is patterned to expose the p-type region **50P**. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable photolithography techniques. Once the photoresist is patterned, an n-type impurity implant is performed in the p-type region **50P**, and the photoresist may act as a mask to substantially prevent n-type impurities from being implanted into the n-type region **50N**. The n-type impurities may be phosphorus, arsenic, antimony, or the like implanted in the region to a concentration in the range of 10^{13} cm^{-3} to 10^{14} cm^{-3} . After the implant, the photoresist may be removed, such as by any acceptable ashing process.

Following or prior to the implanting of the p-type region **50P**, a mask (not separately illustrated) such as a photoresist is formed over the fins **62**, the nanostructures **64**, **66**, and the STI regions **70** in the p-type region **50P**. The photoresist is patterned to expose the n-type region **50N**. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable photolithography techniques. Once the photoresist is patterned, a p-type impurity implant may be performed in the n-type region **50N**, and the photoresist may act as a mask to substantially prevent p-type impurities from being implanted into the p-type region **50P**. The p-type impurities may be boron, boron fluoride, indium, or the like implanted in the region to a concentration in the range of 10^{13} cm^{-3} to 10^{14} cm^{-3} . After the implant, the photoresist may be removed, such as by any acceptable ashing process.

After the implants of the n-type region **50N** and the p-type region **50P**, an anneal may be performed to repair implant damage and to activate the p-type and/or n-type impurities that were implanted. In some embodiments where epitaxial structures are epitaxially grown for the fins **62** and/or the nanostructures **64**, **66**, the grown materials may be in situ doped during growth, which may obviate the implantations, although in situ and implantation doping may be used together.

In FIG. **5**, a dummy dielectric layer **72** is formed on the fins **62** and the nanostructures **64**, **66**. The dummy dielectric layer **72** may be formed of a dielectric material such as silicon oxide, silicon nitride, a combination thereof, or the like, which may be deposited or thermally grown according to acceptable techniques. A dummy gate layer **74** is formed over the dummy dielectric layer **72**, and a mask layer **76** is formed over the dummy gate layer **74**. The dummy gate layer **74** may be deposited over the dummy dielectric layer **72** and then planarized, such as by a CMP. The dummy gate layer **74** may be formed of a conductive or non-conductive material, such as amorphous silicon, polycrystalline-silicon (polysilicon), poly-crystalline silicon germanium (poly-SiGe), a metal, a metallic nitride, a metallic silicide, a metallic oxide, or the like, which may be deposited by physical vapor deposition (PVD), CVD, or the like. The dummy gate layer **74** may be formed of material(s) that have a high etching selectivity from the etching of insulation materials, e.g., the STI regions **70** and/or the dummy dielectric layer **72**. The mask layer **76** may be deposited over the

dummy gate layer **74**. The mask layer **76** may be formed of a dielectric material such as silicon nitride, silicon oxynitride, or the like. In this example, a single dummy gate layer **74** and a single mask layer **76** are formed across the n-type region **50N** and the p-type region **50P**. In the illustrated embodiment, the dummy dielectric layer **72** covers the fins **62**, the nanostructures **64**, **66**, and the STI regions **70**, such that the dummy dielectric layer **72** extends over the STI regions **70** and between the dummy gate layer **74** and the STI regions **70**. In another embodiment, the dummy dielectric layer **72** covers only the fins **62** and the nanostructures **64**, **66**.

In FIG. **6**, the mask layer **76** is patterned using acceptable photolithography and etching techniques to form masks **86**. The pattern of the masks **86** is then transferred to the dummy gate layer **74** by any acceptable etching technique to form dummy gates **84**. The pattern of the masks **86** may optionally be further transferred to the dummy dielectric layer **72** by any acceptable etching technique to form dummy dielectrics **82**. The dummy gates **84** cover portions of the nanostructures **64**, **66** that will be exposed in subsequent processing to form channel regions. Specifically, the dummy gates **84** extend along the portions of the second nanostructures **66** that will be patterned to form channel regions **68**. The pattern of the masks **86** may be used to physically separate adjacent dummy gates **84**. The dummy gates **84** may also have lengthwise directions substantially perpendicular (within process variations) to the lengthwise directions of the fins **62**. The masks **86** can optionally be removed after patterning, such as by any acceptable etching technique.

FIGS. **7A-11C**, **14A-17**, and **20A-25C** illustrate various additional steps in the manufacturing of embodiment devices. FIGS. **7A-11C**, **14A-17**, and **20A-25C** illustrate features in either of the n-type region **50N** and the p-type region **50P**. For example, the structures illustrated may be applicable to both the n-type region **50N** and the p-type region **50P**. Differences (if any) in the structures of the n-type region **50N** and the p-type region **50P** are explained in the description accompanying each figure.

In FIGS. **7A-7C**, gate spacers **88** are formed over the nanostructures **64**, **66**, on exposed sidewalls of the masks **86** (if present), the dummy gates **84**, and the dummy dielectrics **82**. The gate spacers **88** may be formed by conformally depositing one or more dielectric material(s) and subsequently etching the dielectric material(s). Acceptable dielectric materials may include silicon oxide, silicon nitride, silicon oxynitride, silicon oxycarbonitride, or the like, which may be formed by a conformal deposition process such as chemical vapor deposition (CVD), plasma-enhanced chemical vapor deposition (PECVD), atomic layer deposition (ALD), plasma-enhanced atomic layer deposition (PEALD), or the like. Other insulation materials formed by any acceptable process may be used. Any acceptable etch process, such as a dry etch, a wet etch, the like, or a combination thereof, may be performed to pattern the dielectric material(s). The etching may be anisotropic. The dielectric material(s), when etched, have portions left on the sidewalls of the dummy gates **84** (thus forming the gate spacers **88**). The dielectric material(s), when etched, may also have portions left on the sidewalls of the fins **62** and/or the nanostructures **64**, **66** (thus forming fin spacers **90**). After etching, the fin spacers **90** (if present) and the gate spacers **88** can have straight sidewalls (as illustrated) or can have curved sidewalls (not separately illustrated).

Further, implants may be performed to form lightly doped source/drain (LDD) regions (not separately illustrated). In the embodiments with different device types, similar to the

implants for the wells previously described, a mask (not separately illustrated) such as a photoresist may be formed over the n-type region 50N, while exposing the p-type region 50P, and appropriate type (e.g., p-type) impurities may be implanted into the fins 62 and/or the nanostructures 64, 66 exposed in the p-type region 50P. The mask may then be removed. Subsequently, a mask (not separately illustrated) such as a photoresist may be formed over the p-type region 50P while exposing the n-type region 50N, and appropriate type impurities (e.g., n-type) may be implanted into the fins 62 and/or the nanostructures 64, 66 exposed in the n-type region 50N. The mask may then be removed. The n-type impurities may be any of the n-type impurities previously described, and the p-type impurities may be any of the p-type impurities previously described. During the implanting, the channel regions 68 remain covered by the dummy gates 84, so that the channel regions 68 remain substantially free of the impurity implanted to form the LDD regions. The LDD regions may have a concentration of impurities in the range of 10^{15} cm^{-3} to 10^{19} cm^{-3} . An anneal may be used to repair implant damage and to activate the implanted impurities.

It is noted that the previous disclosure generally describes a process of forming spacers and LDD regions. Other processes and sequences may be used. For example, fewer or additional spacers may be utilized, different sequence of steps may be utilized, additional spacers may be formed and removed, and/or the like. Furthermore, the n-type devices and the p-type devices may be formed using different structures and steps.

In FIGS. 8A-8C, source/drain recesses 92 are formed in the nanostructures 64, 66. In the illustrated embodiment, the source/drain recesses 92 extend through the nanostructures 64, 66 and into the fins 62. The source/drain recesses 92 may also extend into the substrate 50. In various embodiments, the source/drain recesses 92 may extend to a top surface of the substrate 50 without etching the substrate 50; the fins 62 may be etched such that bottom surfaces of the source/drain recesses 92 are disposed below the top surfaces of the STI regions 70; or the like. The source/drain recesses 92 may be formed by etching the nanostructures 64, 66 using an anisotropic etching processes, such as a RIE, a NBE, or the like. The gate spacers 88 and the dummy gates 84 collectively mask portions of the fins 62 and/or the nanostructures 64, 66 during the etching processes used to form the source/drain recesses 92. A single etch process may be used to etch each of the nanostructures 64, 66, or multiple etch processes may be used to etch the nanostructures 64, 66. Timed etch processes may be used to stop the etching of the source/drain recesses 92 after the source/drain recesses 92 reach a desired depth. The fin spacers 90 (if present) may be etched during or after the etching of the source/drain recesses 92, so that the height of the fin spacers 90 is reduced. The dimensions of the source/drain regions that will be subsequently formed in the source/drain recesses 92 may be controlled by adjusting the height of the fin spacers 90.

In FIGS. 9A-9C, the source/drain recesses 92 are laterally expanded to form sidewall recesses 92S in the source/drain recesses 92. Specifically, portions of the sidewalls of the first nanostructures 64 exposed by the source/drain recesses 92 are recessed. Although sidewalls of the first nanostructures 64 are illustrated as being straight, the sidewalls may be concave or convex. The sidewalls may be recessed by any acceptable etching process, such as one that is selective to the material of the first nanostructures 64 (e.g., selectively etches the material of the first nanostructures 64 at a faster

rate than the material of the second nanostructures 66). The etching may be isotropic. For example, when the second nanostructures 66 are formed of silicon and the first nanostructures 64 are formed of silicon germanium, the etching process may be a wet etch using tetramethylammonium hydroxide (TMAH), ammonium hydroxide (NH_4OH), or the like. In another embodiment, the etching process may be a dry etch using a fluorine-based gas such as hydrogen fluoride (HF) gas. In some embodiments, the same etching process may be continually performed to both form the source/drain recesses 92 and recess the sidewalls of the first nanostructures 64.

In some embodiments, the widths of the gate spacers 88 are reduced, such as by the etching process used to form and/or expand the source/drain recesses 92. Further, the sidewalls of the second nanostructures 66 may be etched by the etching process used to form and/or expand the source/drain recesses 92. In some embodiments, the sidewalls of the second nanostructures 66 are rounded convex sidewalls at this stage of processing.

In FIGS. 10A-10C, a spacer layer 94 is deposited in the source/drain recesses 92, including in the sidewall recesses 92S (see FIGS. 9A-9C). The spacer layer 94 may also be deposited on the STI regions 70. The spacer layer 94 may be formed of a dielectric material such as silicon nitride, silicon oxynitride, silicon oxycarbonitride, or the like, which may be deposited by a conformal deposition process, such as ALD, CVD, or the like. A low-k dielectric material (e.g., dielectric materials having a k-value less than about 3.5) may be utilized. Other insulation materials formed by any acceptable process may be used.

In FIGS. 11A-11C, the spacer layer 94 is patterned to form inner spacers 96 in the sidewall recesses 92S (see FIGS. 9A-9C) of the source/drain recesses 92. The spacer layer 94 may be patterned by any acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etching may be anisotropic. Although outer sidewalls of the inner spacers 96 are illustrated as being flush with respect to the sidewalls of the gate spacers 88, the outer sidewalls of the inner spacers 96 may extend beyond or be recessed from the sidewalls of the gate spacers 88. In other words, the inner spacers 96 may partially fill, completely fill, or overfill the sidewall recesses 92S. Moreover, although the sidewalls of the inner spacers 96 are illustrated as being concave, the sidewalls of the inner spacers 96 may be straight or convex.

After formation, the inner spacers 96 are disposed on the sidewalls of the remaining portions of the first nanostructures 64, e.g., those sidewalls exposed by the source/drain recesses 92. As will be subsequently described in greater detail, source/drain regions will be subsequently formed in the source/drain recesses 92, and the first nanostructures 64 will be subsequently replaced with corresponding gate structures. The inner spacers 96 act as isolation features between the subsequently formed source/drain regions and the subsequently formed gate structures. Further, the inner spacers 96 may be used to substantially prevent damage to the subsequently formed source/drain regions by subsequent etching processes, such as etching processes used to subsequently remove the first nanostructures 64.

In this embodiment, the spacer layer 94 is removed from the bottoms of the source/drain recesses 92 so that no spacers are formed on the bottoms of the source/drain recesses 92. In other embodiments (subsequently described for FIG. 32A-33), the spacer etch used to form the inner

spacers **96** is adjusted to also form spacers on the bottoms of the source/drain recesses **92**, such as on the surfaces of the fins **62**.

After the source/drain recesses **92** and the inner spacers **96** are formed, the second nanostructures **66** can have a variety of sidewall types. FIGS. **12A-12E** illustrate different sidewall types in a region **50S** in FIG. **11A**. The sidewalls of the second nanostructures **66** may be rounded convex sidewalls, as shown by FIG. **12A**. The sidewalls of the second nanostructures **66** may be polygonal convex sidewalls, as shown by FIG. **12B**. The sidewalls of the second nanostructures **66** may be straight sidewalls, as shown by FIG. **12C**. The sidewalls of the second nanostructures **66** may be polygonal concave sidewalls, as shown by FIG. **12D**. The sidewalls of the second nanostructures **66** may be rounded concave sidewalls, as shown by FIG. **12E**. Subsequent processing steps are shown for the embodiment of FIG. **12A**, although those processing steps may be performed for any of the embodiments.

Further, after the source/drain recesses **92** and the inner spacers **96** are formed, the source/drain recesses **92** which extend into the fins **62** can have a variety of bottom types. FIGS. **13A-13C** illustrate different bottom types in a region **50B** in FIG. **11A**. The bottoms of the source/drain recesses **92** which extend into the fins **62** may be shallow rounded concave bottoms, as shown by FIG. **13A**. The bottoms of the source/drain recesses **92** which extend into the fins **62** may be deep rounded concave bottoms, as shown by FIG. **13B**. The deep rounded concave bottoms shown in FIG. **13B** extend further into the fins **62** than the shallow rounded concave bottoms shown in FIG. **13A**. In some embodiments, the shallow rounded concave bottoms have a depth D_1 in the range of 3 nm to 20 nm, and the deep rounded concave bottoms have a depth D_2 in the range of 20 nm to 50 nm. The bottoms of the source/drain recesses **92** which extend into the fins **62** may be polygonal concave bottoms, as shown by FIG. **13C**. Subsequent processing steps are shown for the embodiment of FIG. **13A**, although those processing steps may be performed for any of the embodiments.

The second nanostructures **66** and/or the source/drain recesses **92** may be shaped to have the sidewall types and/or bottom types described for FIGS. **12A-13C** by an appropriate etch. The shaping of the second nanostructures **66** and/or the source/drain recesses **92** may occur as a result of the etch used to initially form the source/drain recesses **92** (see FIGS. **8A-8C**), as a result of the etch used to expand the source/drain recesses **92** (see FIGS. **9A-9C**), and/or as a result of the etch used to pattern the spacer layer **94** (see FIGS. **11A-11C**). The second nanostructures **66** and/or the source/drain recesses **92** may also be shaped by a separate etch performed after the inner spacers **96** are formed. In some embodiments, the etch is an anisotropic wet etch performed with potassium hydroxide (KOH), tetramethylammonium hydroxide (TMAH), and ethylenediamine pyrocatechol (EDP). In some embodiments, the etch is an anisotropic dry etch performed with carbon tetrafluoride (CF_4), fluoromethane (CH_3F), hydrogen bromide (HBr), and oxygen (O_2) gas in helium (He) and/or argon (Ar) while generating a plasma with a bias voltage or a bias power. In some embodiments, the etch is an isotropic dry etch performed with nitrogen trifluoride (NF_3), chlorine (Cl_2) gas, and hydrogen (H_2) gas in helium (He) and/or argon (Ar). In some embodiments, the etch is a combination of these processes.

In FIGS. **14A-14C**, epitaxial source/drain regions **98** are formed in the source/drain recesses **92**. The epitaxial source/drain regions **98** are formed in the source/drain recesses **92** such that each dummy gate **84** (and corresponding channel

region **68**) is disposed between respective adjacent pairs of the epitaxial source/drain regions **98**. In some embodiments, the gate spacers **88** and the inner spacers **96** are used to separate the epitaxial source/drain regions **98** from, respectively, the dummy gates **84** and the first nanostructures **64** by an appropriate lateral distance so that the epitaxial source/drain regions **98** do not short out with subsequently formed gates of the resulting nano-FETs. A material of the epitaxial source/drain regions **98** may be selected to exert stress in the respective channel regions **68**, thereby improving performance.

The epitaxial source/drain regions **98** in the n-type region **50N** may be formed by masking the p-type region **50P**. Then, the epitaxial source/drain regions **98** in the n-type region **50N** are epitaxially grown in the source/drain recesses **92** in the n-type region **50N**. The epitaxial source/drain regions **98** may include any acceptable material appropriate for n-type devices. For example, if the second nanostructures **66** are silicon, the epitaxial source/drain regions **98** in the n-type region **50N** may include materials exerting a tensile strain on the channel regions **68**, such as silicon, silicon carbide, phosphorus-doped silicon carbide, silicon phosphide, or the like. The epitaxial source/drain regions **98** in the n-type region **50N** may be referred to as "n-type source/drain regions." The epitaxial source/drain regions **98** in the n-type region **50N** may have surfaces raised from respective surfaces of the fins **62** and the nanostructures **64**, **66**, and may have facets.

The epitaxial source/drain regions **98** in the p-type region **50P** may be formed by masking the n-type region **50N**. Then, the epitaxial source/drain regions **98** in the p-type region **50P** are epitaxially grown in the source/drain recesses **92** in the p-type region **50P**. The epitaxial source/drain regions **98** may include any acceptable material appropriate for p-type devices. For example, if the second nanostructures **66** are silicon, the epitaxial source/drain regions **98** in the p-type region **50P** may include materials exerting a compressive strain on the channel regions **68**, such as silicon germanium, boron-doped silicon germanium, germanium, germanium tin, or the like. The epitaxial source/drain regions **98** in the p-type region **50P** may be referred to as "p-type source/drain regions." The epitaxial source/drain regions **98** in the p-type region **50P** may have surfaces raised from respective surfaces of the fins **62** and the nanostructures **64**, **66**, and may have facets.

The epitaxial source/drain regions **98**, the nanostructures **64**, **66**, and/or the fins **62** may be implanted with impurities to form source/drain regions, similar to the process previously described for forming LDD regions, followed by an anneal. The source/drain regions may have an impurity concentration in the range of 10^{19} cm^{-3} to 10^{21} cm^{-3} . The n-type and/or p-type impurities for source/drain regions may be any of the impurities previously described. In some embodiments, the epitaxial source/drain regions **98** may be in situ doped during growth.

As a result of the epitaxy processes used to form the epitaxial source/drain regions **98**, upper surfaces of the epitaxial source/drain regions have facets which expand laterally outward beyond sidewalls of the fins **62** and the nanostructures **64**, **66**. In some embodiments, these facets cause adjacent epitaxial source/drain regions **98** to merge as illustrated by FIG. **14C**. In some embodiments, adjacent epitaxial source/drain regions **98** remain separated after the epitaxy process is completed (not separately illustrated). In the illustrated embodiments, the spacer etch used to form the gate spacers **88** is adjusted to also form the fin spacers **90** on sidewalls of the fins **62** and/or the nanostructures **64**, **66**. The

fin spacers 90 are formed to cover a portion of the sidewalls of the fins 62 and/or the nanostructures 64, 66 that extend above the STI regions 70, thereby blocking the epitaxial growth. In another embodiment, the spacer etch used to form the gate spacers 88 is adjusted to not form fin spacers, so as to allow the epitaxial source/drain regions 98 to extend to the surface of the STI regions 70.

The epitaxial source/drain regions 98 include multiple semiconductor material layers. Specifically, the epitaxial source/drain regions 98 each include a liner layer 98A, a main layer 98B, and (optionally) a finishing layer 98C (or more generally, first, second, and third epitaxial layers). Other quantities of semiconductor material layers may be used for the epitaxial source/drain regions 98. The liner layers 98A are grown on the surfaces of semiconductor features (e.g., surfaces of the fins 62 and the second nanostructures 66) in the source/drain recesses 92. The main layers 98B are grown on the liner layers 98A. The finishing layers 98C (if present) are grown on the main layers 98B. Each of the liner layer 98A, the main layer 98B, and the finishing layer 98C may be formed of different semiconductor materials and may be doped (e.g., with a p-type or an n-type impurity) to different impurity concentrations. In some embodiments, the main layers 98B have a greater impurity concentration than the finishing layers 98C, and the finishing layers 98C have a greater impurity concentration than the liner layers 98A. Forming the liner layers 98A with a lesser impurity concentration than the main layers 98B may increase adhesion in the source/drain recesses 92, and forming the finishing layers 98C with a lesser impurity concentration than the main layers 98B may reduce out-diffusion of dopants from the main layers 98B in subsequent processing.

When the second nanostructures 66 are formed of silicon and the first nanostructures 64 are formed of silicon germanium, a tensile strain may be exerted on the channel regions 68 in the n-type region 50N, increasing the propensity of dopant diffusion into the channel regions 68 during growth of the epitaxial source/drain regions 98 in the n-type region 50N. For example, a tensile strain of up to 0.5% may be exerted on the channel regions 68 in the n-type region 50N after the source/drain recesses 92 are formed in the nanostructures 64, 66. According to various embodiments, the liner layers 98A in the n-type region 50N are formed of a material which helps reduce out-diffusion of dopants from the main layers 98B into the second nanostructures 66 or the fins 62. Diffusion of dopants into the channel regions 68 may thus be reduced, thereby increasing the channel mobility and thus increasing device performance.

The semiconductor material of the liner layers 98A in the n-type region 50N is a compound semiconductor which is composed of a semiconductor element and a blocker element. The blocker element helps the semiconductor material of the liner layers 98A suppress diffusion of n-type dopants (e.g., phosphorous, arsenic, etc.). In some embodiments, the semiconductor element is silicon and the blocker element is carbon. The semiconductor material(s) of the main layers 98B and/or the finishing layers 98C (if present) may not be composed of the blocker element. In some embodiments, the main layers 98B and/or the finishing layers 98C are formed of the same semiconductor material, and that semiconductor material is an elemental semiconductor which is composed of the semiconductor element but is not composed of the blocker element. In other words, the semiconductor material of the liner layers 98A may be a carbon-containing semiconductor material (e.g., silicon carbide), but the semiconductor material(s) of the main layers 98B and/or the finish-

ing layers 98C may be carbon-free semiconductor material (s) (e.g., silicon). The concentration of the blocker element in the semiconductor material of the liner layers 98A determines how well the liner layers 98A reduce out-diffusion of dopants from the main layers 98B. In some embodiments, the semiconductor material of the liner layers 98A has a concentration of the blocker element (e.g., a carbon concentration) in the range of 0.1 atomic percent (at %) to 2 at %. Forming the semiconductor material of the liner layers 98A with a concentration of the blocker element that is less than 0.1 at % may not sufficiently reduce out-diffusion of dopants from the main layers 98B, decreasing device performance. Forming the semiconductor material of the liner layers 98A with a concentration of the blocker element that is greater than 2 at % may cause the resistance of the epitaxial source/drain regions 98 to be excessively high, decreasing device performance. The liner layers 98A have a greater concentration of the blocker element than the main layers 98B and the finishing layers 98C, e.g., the liner layers 98A have a greater carbon concentration than the main layers 98B and the finishing layers 98C.

One or more of the layers 98A, 98B, 98C in the n-type region 50N further include appropriate type impurities (e.g., n-type) to form source/drain regions, which are different from the blocker element. The n-type impurities may be any of the n-type impurities previously described. In some embodiments, the layers 98A, 98B, 98C each include the impurities. For example, when the n-type impurities are phosphorous, the finishing layers 98C may be formed of phosphorous-doped silicon, the main layers 98B may be formed of phosphorous-doped silicon, and the liner layers 98A may be formed of phosphorous-doped silicon carbide or undoped silicon carbide. Similarly, when the n-type impurities are arsenic, the finishing layers 98C may be formed of arsenic-doped silicon, the main layers 98B may be formed of arsenic-doped silicon, and the liner layers 98A may be formed of arsenic-doped silicon carbide or undoped silicon carbide. The concentration of the impurities in the liner layers 98A and the main layers 98B determines how well the liner layers 98A reduce out-diffusion of dopants from the main layers 98B. In some embodiments, the liner layers 98A have an impurity concentration in the range of $2 \times 10^{20} \text{ cm}^{-3}$ to $2 \times 10^{21} \text{ cm}^{-3}$. In some embodiments, the main layers 98B have an impurity concentration in the range of $1 \times 10^{20} \text{ cm}^{-3}$ to $5 \times 10^{21} \text{ cm}^{-3}$. In some embodiments, the main layers 98B and the finishing layers 98C include the impurities, but the liner layers 98A are free of the impurities.

The thickness of the liner layers 98A in the n-type region 50N determines how well the liner layers 98A reduce out-diffusion of dopants from the main layers 98B. The thickness T_1 of the portions of the liner layers 98A along the second nanostructures 66 may be smaller than the thickness T_2 of the portions of the liner layers 98A along the fins 62. In some embodiments, the portions of the liner layers 98A along the second nanostructures 66 have a thickness T_1 in the range of 1 nm to 6 nm, and the portions of the liner layers 98A along the fins 62 have a thickness T_2 in the range of 1 nm to 30 nm. Forming the liner layers 98A with a thickness T_1 that is less than 1 nm or with a thickness T_2 that is less than 1 nm may not sufficiently reduce out-diffusion of dopants from the main layers 98B, decreasing device performance. Forming the liner layers 98A with a thickness T_1 that is greater than 6 nm or with a thickness T_2 that is greater than 30 nm may cause the resistance of the epitaxial source/drain regions 98 to be excessively high, decreasing device performance.

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FIGS. 15-17 illustrate a process for forming the epitaxial source/drain regions 98 in the n-type region 50N. FIGS. 15-17 illustrate features in a region 50A in FIG. 14A. As previously described, the semiconductor material of the liner layers 98A in the n-type region 50N is composed of a semiconductor element and a blocker element, which helps reduce out-diffusion of dopants from the main layers 98B in subsequent processing.

In FIG. 15, the liner layers 98A in the n-type region 50N are epitaxially grown on the surfaces of semiconductor features (e.g., surfaces of the fins 62 and the second nanostructures 66) in the source/drain recesses 92. The liner layers 98A may be grown from the second nanostructures 66 and the fins 62 by exposing the second nanostructures 66 and the fins 62 to a semiconductor-containing precursor, a blocker-containing precursor, and (optionally) a dopant-containing precursor. When the semiconductor element is silicon, the semiconductor-containing precursor may be a silicon-containing precursor such as a silane, such as monosilane (SiH_4), disilane (Si_2H_6), trisilane (Si_3H_8), trichlorosilane (HCl_3Si), dichlorosilane (H_2SiCl_2), or the like. When the blocker element is carbon, the blocker-containing precursor may be a carbon-containing precursor such as trimethylsilane ($(\text{CH}_3)_3\text{SiH}$) or the like. When the dopant is arsenic, the dopant-containing precursor may be an arsenic-containing precursor such as arsine (AsH_3) or the like. When the dopant is phosphorous, the dopant-containing precursor may be a phosphorous-containing precursor such as diphosphine (P_2H_6), phosphorus trichloride (PCl_3), or the like. In some embodiments, the second nanostructures 66 and the fins 62 are exposed to the semiconductor-containing precursor, the blocker-containing precursor, and the dopant-containing precursor (if present) at a temperature in the range of 500° C. to 800° C., at a pressure in the range of 1 Torr to 760 Torr, and for a duration in the range of 5 seconds to 20 minutes. Growing the liner layers 98A at a temperature and at a pressure in these ranges allows the liner layers 98A to have a desired thickness and blocker element concentration (previously described). Growing the liner layers 98A at a temperature or at a pressure outside of these ranges may not allow the liner layers 98A to have the desired thickness or blocker element concentration.

In this embodiment, the portions of the liner layers 98A in each source/drain recess 92 are grown so that they remain separated and do not merge after growth. Specifically growth of the liner layers 98A is stopped before the liner layers 98A merge in the source/drain recess 92. Portions of the sidewalls of the inner spacers 96 are thus not covered by the liner layers 98A, and remain exposed after growth of the liner layers 98A is completed. In another embodiment (subsequently described for FIGS. 26-32C), the liner layers 98A are grown until they merge in each source/drain recess 92, so that the sidewalls of the inner spacers 96 are covered by the liner layers 98A.

In FIG. 16, the main layers 98B in the n-type region 50N are epitaxially grown on the liner layers 98A. The main layers 98B may be grown from the liner layers 98A by exposing the liner layers 98A to a semiconductor-containing precursor and a dopant-containing precursor. When the semiconductor element is silicon, the semiconductor-containing precursor may be a silicon-containing precursor such as a silane, such as monosilane (SiH_4), disilane (Si_2H_6), trisilane (Si_3H_8), trichlorosilane (HCl_3Si), dichlorosilane (H_2SiCl_2), or the like. When the dopant is arsenic, the dopant-containing precursor may be an arsenic-containing precursor such as arsine (AsH_3) or the like. When the dopant is phosphorous, the dopant-containing precursor may be a

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phosphorous-containing precursor such as diphosphine (P_2H_6), phosphorus trichloride (PCl_3), or the like. The semiconductor-containing precursor and/or the dopant-containing precursor used to grow the main layers 98B may be the same as those used to grow the liner layers 98A, or may include different precursors. In some embodiments, the liner layers 98A are exposed to the semiconductor-containing precursor and the dopant-containing precursor at a temperature in the range of 400° C. to 800° C. and at a pressure in the range of 1 Torr to 760 Torr.

The main layers 98B in the n-type region 50N are grown from the liner layers 98A until the portions of the main layers 98B in each source/drain recess 92 merge. In this embodiment, the main layers 98B cover the portions of the sidewalls of the inner spacers 96 that are not covered by the liner layers 98A. In another embodiment (subsequently described for FIGS. 26-32C) where the liner layers 98A cover all of the sidewalls of the inner spacers 96, the main layers 98B are separated from the inner spacers 96 by the liner layers 98A.

The width of the source/drain recesses 92 and the thickness T_1 (previously described) of the liner layers 98A is controlled so that the liner layers 98A at opposing sides of the epitaxial source/drain regions 98 are separated by a desired distance. As a result, the width W_1 of the main layers 98B between opposing liner layers 98A may be controlled. In some embodiments, the width W_1 of the main layers 98B between opposing liner layers 98A is in the range of 3 nm to 50 nm.

In FIG. 17, the finishing layers 98C in the n-type region 50N are epitaxially grown on the main layers 98B. The finishing layers 98C may be grown from the main layers 98B by exposing the main layers 98B to a semiconductor-containing precursor and a dopant-containing precursor. When the semiconductor element is silicon, the semiconductor-containing precursor may be a silicon-containing precursor such as a silane, such as monosilane (SiH_4), disilane (Si_2H_6), trisilane (Si_3H_8), trichlorosilane (HCl_3Si), dichlorosilane (H_2SiCl_2), or the like. When the dopant is arsenic, the dopant-containing precursor may be an arsenic-containing precursor such as arsine (AsH_3) or the like. When the dopant is phosphorous, the dopant-containing precursor may be a phosphorous-containing precursor such as diphosphine (P_2H_6), phosphorus trichloride (PCl_3), or the like. The semiconductor-containing precursor and/or the dopant-containing precursor used to grow the finishing layers 98C may be the same as those used to grow the main layers 98B and/or the liner layers 98A, or may include different precursors. In some embodiments, the main layers 98B are exposed to the semiconductor-containing precursor and the dopant-containing precursor at a temperature in the range of 500° C. to 800° C. and at a pressure in the range of 1 Torr to 760 Torr.

Some embodiments contemplate other layers of the epitaxial source/drain regions 98 comprising a semiconductor material that is composed of the blocker element. For example, the semiconductor material of the main layers 98B may also be a compound semiconductor which is composed of the semiconductor element and the blocker element. In some embodiments, the semiconductor material of the main layers 98B has a concentration of the blocker element (e.g., a carbon concentration) in the range of 0 at % to 2 at %. In such embodiments, a carbon-containing precursor such as that previously described may also be used to grow the main layers 98B. In other words, the semiconductor material of the main layers 98B may have a zero or non-zero carbon

concentration. The liner layers **98A** have a greater concentration of the blocker element than the main layers **98B**.

After the epitaxial source/drain regions **98** are formed, the liner layers **98A** can have a variety of profile types. The liner layers **98A** may be grown conformally so that the liner layers **98A** have conformal profiles, as shown by FIGS. **18A-18E**, which correspond to FIGS. **12A-12E**, respectively. The liner layers **98A** may be grown along the $\langle 111 \rangle$ direction so that the liner layers **98A** have $\langle 111 \rangle$ faceted profiles, as shown by FIGS. **19A-19E**, which correspond to FIGS. **12A-12E**, respectively. The facet types of the liner layers **98A** may be controlled by controlling the flow rates of the various precursors used to grow the liner layers **98A**.

In FIGS. **20A-20C**, a first ILD **104** is deposited over the epitaxial source/drain regions **98**, the gate spacers **88**, and the masks **86** (if present) or the dummy gates **84**. The first ILD **104** may be formed of a dielectric material, which may be deposited by any suitable method, such as CVD, plasma-enhanced CVD (PECVD), FCVD, or the like. Acceptable dielectric materials may include phospho-silicate glass (PSG), boro-silicate glass (BSG), boron-doped phospho-silicate glass (BPSG), undoped silicate glass (USG), or the like. Other insulation materials formed by any acceptable process may be used.

In some embodiments, a contact etch stop layer (CESL) **102** is formed between the first ILD **104** and the epitaxial source/drain regions **98**, the gate spacers **88**, and the masks **86** (if present) or the dummy gates **84**. The CESL **102** may be formed of a dielectric material, such as silicon nitride, silicon oxide, silicon oxynitride, or the like, having a high etching selectivity from the etching of the first ILD **104**. The CESL **102** may be formed by any suitable method, such as CVD, ALD, or the like.

In FIGS. **21A-21C**, a removal process is performed to level the top surfaces of the first ILD **104** with the top surfaces of the gate spacers **88** and the masks **86** (if present) or the dummy gates **84**. In some embodiments, a planarization process such as a chemical mechanical polish (CMP), an etch-back process, combinations thereof, or the like may be utilized. The planarization process may also remove the masks **86** on the dummy gates **84**, and portions of the gate spacers **88** along sidewalls of the masks **86**. After the planarization process, the top surfaces of the gate spacers **88**, the first ILD **104**, the CESL **102**, and the masks **86** (if present) or the dummy gates **84** are coplanar (within process variations). Accordingly, the top surfaces of the masks **86** (if present) or the dummy gates **84** are exposed through the first ILD **104**. In the illustrated embodiment, the masks **86** remain, and the planarization process levels the top surfaces of the first ILD **104** with the top surfaces of the masks **86**.

In FIGS. **22A-22C**, the masks **86** (if present) and the dummy gates **84** are removed in an etching process, so that recesses **106** are formed. Portions of the dummy dielectrics **82** in the recesses **106** are also removed. In some embodiments, the dummy gates **84** are removed by an anisotropic dry etch process. For example, the etching process may include a dry etch process using reaction gas(es) that selectively etch the dummy gates **84** at a faster rate than the first ILD **104** or the gate spacers **88**. During the removal, the dummy dielectrics **82** may be used as etch stop layers when the dummy gates **84** are etched. The dummy dielectrics **82** are then removed. Each recess **106** exposes and/or overlies portions of the channel regions **68**. Portions of the second nanostructures **66** which act as the channel regions **68** are disposed between and adjoin adjacent pairs of the epitaxial source/drain regions **98**.

The remaining portions of the first nanostructures **64** are then removed to expand the recesses **106**, such that openings **108** are formed between the second nanostructures **66**. The remaining portions of the first nanostructures **64** can be removed by any acceptable etching process that selectively etches the material of the first nanostructures **64** at a faster rate than the material of the second nanostructures **66**. The etching may be isotropic. For example, when the first nanostructures **64** are formed of silicon germanium and the second nanostructures **66** are formed of silicon, the etching process may be a wet etch using tetramethylammonium hydroxide (TMAH), ammonium hydroxide (NH_4OH), or the like. In some embodiments, a trim process (not separately illustrated) is performed to decrease the thicknesses of the exposed portions of the second nanostructures **66**.

In FIGS. **23A-23C**, gate dielectrics **122** and gate electrodes **124** are formed for replacement gates. Each respective pair of a gate dielectric **122** and a gate electrode **124** may be collectively referred to as a "gate structure." Each gate structure extends along sidewalls and a top surface of a channel region **68** of a second nanostructure **66**. Some of the gate structures also extend along sidewalls and/or a top surface of a fin **62**. The gate dielectrics **122** include one or more gate dielectric layer(s) disposed around the second nanostructure **66** and on the sidewalls of the gate spacers **88** and the inner spacers **96**. The gate dielectrics **122** may be formed of an oxide such as silicon oxide or a metal oxide, a silicate such as a metal silicate, combinations thereof, multi-layers thereof, or the like. Additionally or alternatively, the gate dielectrics **122** may be formed of a high-k dielectric material (e.g., dielectric materials having a k-value greater than about 7.0), such as a metal oxide or a silicate of hafnium, aluminum, zirconium, lanthanum, manganese, barium, titanium, lead, and combinations thereof. The dielectric material(s) of the gate dielectrics **122** may be formed by molecular-beam deposition (MBD), ALD, PECVD, or the like. Although a single-layered gate dielectrics **122** are illustrated, the gate dielectrics **122** may include any number of interfacial layers and any number of main layers. For example, the gate dielectrics **122** may include an interfacial layer and an overlying high-k dielectric layer.

The gate electrodes **124** include one or more gate electrode layer(s) disposed over the gate dielectrics **122**. The gate electrodes **124** may be formed of a metal-containing material such as titanium nitride, titanium oxide, tantalum nitride, tantalum carbide, tungsten, cobalt, ruthenium, aluminum, combinations thereof, multi-layers thereof, or the like. Although single-layered gate electrodes **124** are illustrated, the gate electrodes **124** may include any number of work function tuning layers, any number of barrier layers, any number of glue layers, and a fill material.

As an example to form the gate structures, one or more gate dielectric layer(s) may be deposited in the recesses **106** and the openings **108**. The gate dielectric layer(s) may also be deposited on the top surfaces of the first ILD **104**, the CESL **102**, and the gate spacers **88**. Subsequently, one or more gate electrode layer(s) may be deposited on the gate dielectric layer(s), and in the remaining portions of the recesses **106** and the openings **108**. A removal process may then be performed to remove the excess portions of the gate dielectric layer(s) and the gate electrode layer(s), which excess portions are over the top surfaces of the first ILD **104**, the CESL **102**, and the gate spacers **88**. The gate dielectric layer(s), after the removal process, have portions left in the recesses **106** and the openings **108** (thus forming the gate dielectrics **122**). The gate electrode layer(s), after the removal process, have portions left in the recesses **106** and

the openings **108** (thus forming the gate electrodes **124**). In some embodiments, a planarization process such as a chemical mechanical polish (CMP), an etch-back process, combinations thereof, or the like may be utilized. After the planarization process, the top surfaces of the gate spacers **88**, the CESL **102**, the first ILD **104**, the gate dielectrics **122**, and the gate electrodes **124** are coplanar (within process variations).

The formation of the gate dielectrics **122** in the n-type region **50N** and the p-type region **50P** may occur simultaneously such that the gate dielectrics **122** in each region are formed of the same material(s), and the formation of the gate electrodes **124** may occur simultaneously such that the gate electrodes **124** in each region are formed of the same material(s). In some embodiments, the gate dielectrics **122** in each region may be formed by distinct processes, such that the gate dielectrics **122** may include different materials and/or have a different number of layers, and/or the gate electrodes **124** in each region may be formed by distinct processes, such that the gate electrodes **124** may include different materials and/or have a different number of layers. Various masking steps may be used to mask and expose appropriate regions when using distinct processes.

In FIGS. **24A-24C**, a second ILD **134** is formed over the gate spacers **88**, the CESL **102**, the first ILD **104**, the gate dielectrics **122**, and the gate electrodes **124**. In some embodiments, the second ILD **134** is a flowable film formed by a flowable CVD method. In some embodiments, the second ILD **134** is formed of a dielectric material such as PSG, BSG, BPSG, USG, or the like, which may be deposited by any suitable method, such as CVD, PECVD, or the like.

In some embodiments, an etch stop layer (ESL) **132** is formed between the second ILD **134** and the gate spacers **88**, the CESL **102**, the first ILD **104**, the gate dielectrics **122**, and the gate electrodes **124**. The ESL **132** may include a dielectric material, such as, silicon nitride, silicon oxide, silicon oxynitride, or the like, having a high etching selectivity from the etching of the second ILD **134**.

In FIGS. **25A-25C**, gate contacts **142** and source/drain contacts **144** are formed to contact, respectively, the gate electrodes **124** and the epitaxial source/drain regions **98**. The gate contacts **142** are physically and electrically coupled to the gate electrodes **124**. The source/drain contacts **144** are physically and electrically coupled to the epitaxial source/drain regions **98**.

As an example to form the gate contacts **142** and the source/drain contacts **144**, openings for the gate contacts **142** are formed through the second ILD **134** and the ESL **132**, and openings for the source/drain contacts **144** are formed through the second ILD **134**, the ESL **132**, the first ILD **104**, and the CESL **102**. The openings may be formed using acceptable photolithography and etching techniques. A liner (not separately illustrated), such as a diffusion barrier layer, an adhesion layer, or the like, and a conductive material are formed in the openings. The liner may include titanium, titanium nitride, tantalum, tantalum nitride, or the like. The conductive material may be copper, a copper alloy, silver, gold, tungsten, cobalt, aluminum, nickel, or the like. A planarization process, such as a CMP, may be performed to remove excess material from a surface of the second ILD **134**. The remaining liner and conductive material form the gate contacts **142** and the source/drain contacts **144** in the openings. The gate contacts **142** and the source/drain contacts **144** may be formed in distinct processes, or may be formed in the same process. Although shown as being formed in the same cross-sections, it should be appreciated

that each of the gate contacts **142** and the source/drain contacts **144** may be formed in different cross-sections, which may avoid shorting of the contacts.

Optionally, metal-semiconductor alloy regions **146** are formed at the interfaces between the epitaxial source/drain regions **98** and the source/drain contacts **144**. The metal-semiconductor alloy regions **146** can be silicide regions formed of a metal silicide (e.g., titanium silicide, cobalt silicide, nickel silicide, etc.), germanide regions formed of a metal germanide (e.g. titanium germanide, cobalt germanide, nickel germanide, etc.), silicon germanide regions formed of both a metal silicide and a metal germanide, or the like. The metal-semiconductor alloy regions **146** can be formed before the material(s) of the source/drain contacts **144** by depositing a metal in the openings for the source/drain contacts **144** and then performing a thermal anneal process. In some embodiments, the metal reacts with the finishing layers **98C** of the epitaxial source/drain regions **98** such that the metal-semiconductor alloy regions **146** extend through the finishing layers **98C**. The metal can be any metal capable of reacting with the semiconductor materials (e.g., silicon, silicon carbide, silicon germanium, germanium, etc.) of the epitaxial source/drain regions **98** to form a low-resistance metal-semiconductor alloy, such as nickel, cobalt, titanium, tantalum, platinum, tungsten, other noble metals, other refractory metals, rare earth metals or their alloys. The metal may be deposited by a deposition process such as ALD, CVD, PVD, or the like. After the thermal anneal process, a cleaning process, such as a wet clean, may be performed to remove any residual metal from the openings for the source/drain contacts **144**, such as from surfaces of the metal-semiconductor alloy regions **146**. The material(s) of the source/drain contacts **144** can then be formed on the metal-semiconductor alloy regions **146**.

The width of the source/drain contacts **144** is controlled so that the width W_2 of the source/drain contacts **144** is less than the width W_1 (previously described) of the main layers **98B** between opposing liner layers **98A**. As a result, the source/drain contacts **144** are formed to contact the main layers **98B** of the epitaxial source/drain regions **98**, and are not formed to contact the liner layers **98A** of the epitaxial source/drain regions **98**. In other words, the source/drain contacts **144** are spaced apart from the liner layers **98A**. The contact resistance to the epitaxial source/drain regions **98** may thus be reduced, increasing device performance. In some embodiments, the width W_2 of the source/drain contacts **144** is in the range of 3 nm to 50 nm.

Subsequently, the nano-FET devices may be interconnected by metallization layers in an overlying interconnect structure to form integrated circuits. The overlying interconnect structure can be formed in a back end of line (BEOL) process, in which the metallization layers are connected to the gate contacts **142** and the source/drain contacts **144**. In some types of devices, such as memory devices, the source/drain contacts **144** to source regions of the devices may be connected to ground. Additional features, such as passive devices, memories (e.g., magnetoresistive random-access memory (MRAM), resistive random access memory (RRAM), phase-change random access memory (PCRAM), etc.), or the like may be integrated with the interconnect structure during the BEOL process.

FIG. **26** is a view of nano-FETs, in accordance with some other embodiments. This embodiment is similar to that described for FIG. **25A**, except the liner layers **98A** are grown until they merge in each source/drain recess **92** (see FIG. **28**), so that the sidewalls of the inner spacers **96** are covered by the liner layers **98A**. As such, the liner layers

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98A are extending continuously along the sidewalls of the second nanostructures 66 and the inner spacers 96, and the main layers 98B are separated from the inner spacers 96 by the liner layers 98A.

The thickness T_3 of the portions of the liner layers 98A along the inner spacers 96 may be smaller than the thickness T_1 of the portions of the liner layers 98A along the second nanostructures 66 and the thickness T_2 of the portions of the liner layers 98A along the fins 62. In some embodiments, the portions of the liner layers 98A along the inner spacers 96 have a thickness T_3 in the range of 0 nm to 8 nm. Forming the liner layers 98A with a thickness T_3 that is greater than 8 nm may cause the resistance of the epitaxial source/drain regions 98 to be excessively high, decreasing device performance.

FIGS. 27-31 are views of intermediate stages in the manufacturing of nano-FETs, in accordance with some other embodiments. Specifically, FIGS. 27-31 illustrate another process for forming the epitaxial source/drain regions 98 in the n-type region 50N. FIGS. 27-31 illustrate features in a region 50A in FIG. 14A. As previously described, the semiconductor material of the liner layers 98A in the n-type region 50N is composed of a semiconductor element and a blocker element, which helps reduce out-diffusion of dopants from the main layers 98B in subsequent processing.

In FIGS. 27-28, the liner layers 98A in the n-type region 50N are epitaxially grown on the surfaces of semiconductor features (e.g., surfaces of the fins 62 and the second nanostructures 66) in the source/drain recesses 92. The liner layers 98A may be grown by a similar process as that described for FIG. 15, except the epitaxial growth process may be performed for a longer duration. In some embodiments, the second nanostructures 66 and the fins 62 are exposed to the semiconductor-containing precursor, the blocker-containing precursor, and the dopant-containing precursor (if present) for a duration in the range of 5 second to 20 minutes. As such, the liner layers 98A are initially grown on the fins 62 and the second nanostructures 66 in the source/drain recesses 92 (as shown by FIG. 27), but then growth is continued until the portions of the liner layers 98A in each source/drain recess 92 merge (as shown by FIG. 28). Performing the epitaxial growth process for a long duration allows the portions of the liner layers 98A in each source/drain recess 92 to merge, but also causes the liner layers 98A to be formed to a large thickness.

In FIG. 29, the liner layers 98A in the n-type region 50N are thinned to reduce their thickness. The liner layers 98A may be thinned by a dry etch using a chlorine-containing etchant such as hydrogen chloride (HCl) gas, chlorine (Cl_2) gas, or the like. The etching may be anisotropic. After the liner layers 98A are thinned, they have the thicknesses T_1 , T_2 , T_3 (previously described).

In FIG. 30, the main layers 98B in the n-type region 50N are epitaxially grown on the liner layers 98A. The main layers 98B may be grown by a similar process as that described for FIG. 16.

In FIG. 31, the finishing layers 98C in the n-type region 50N are epitaxially grown on the main layers 98B. The finishing layers 98C may be grown by a similar process as that described for FIG. 17.

FIGS. 32A-32C are views of nano-FETs, in accordance with some other embodiments. This embodiment is similar to that described for FIGS. 25A-25C, except the spacer layer 94 is patterned (see FIGS. 11A-11C) so that bottom spacers 152 are also formed at the bottoms of the source/drain recesses 92 (see FIGS. 11A-11C). As such, the bottom spacers 152 separate the bottom surfaces of the epitaxial

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source/drain regions 98 from the fins 62. Forming the bottom spacers 152 along the bottom surfaces of the epitaxial source/drain regions 98 helps reduce the leakage from the epitaxial source/drain regions 98 to the substrate 50. In some embodiments, the bottom spacers 152 along the bottom surfaces of the epitaxial source/drain regions 98 have a thickness T_4 in the range of 1 nm to 30 nm. Forming the bottom spacers 152 along the bottom surfaces of the epitaxial source/drain regions 98 with a thickness T_4 that is less than 1 nm may not sufficiently reduce leakage from the epitaxial source/drain regions 98. Forming the bottom spacers 152 along the bottom surfaces of the epitaxial source/drain regions 98 with a thickness T_4 that is greater than 30 nm may cause the resistance of the epitaxial source/drain regions 98 to be excessively high, decreasing device performance.

FIG. 33 is a view of nano-FETs, in accordance with some other embodiments. This embodiment is similar to that described for FIG. 26, except the spacer layer 94 is patterned (see FIGS. 11A-11C) so that bottom spacers 152 are also formed at the bottoms of the source/drain recesses 92 (see FIGS. 11A-11C), in a similar manner as described for FIGS. 32A-32C.

FIGS. 34A-34C are views of nano-FETs, in accordance with some other embodiments. This embodiment is similar to that described for FIGS. 25A-25C, except the liner layers 98A on the fins 62 are formed to a greater thickness than other liner layers 98A. Specifically, the liner layers 98A_F grown from the fins 62 are thicker than the liner layers 98A_N grown from the second nanostructures 66. Growing the liner layers 98A_F on the fins 62 to a greater thickness helps reduce the leakage from the epitaxial source/drain regions 98 to the substrate 50. The liner layers 98A_F are grown so that the top surfaces of the liner layers 98A_F are disposed above the top surfaces of the fins 62 and below the bottom surfaces of the second nanostructures 66. In some embodiments, the distance D_1 between the top surfaces of the liner layers 98A_F and the bottom surfaces of the second nanostructures 66 is less than 3 nm. Forming the liner layers 98A_F so that the distance D_1 is greater than 3 nm may cause the resistance of the epitaxial source/drain regions 98 to be excessively high, decreasing device performance.

The liner layers 98A_F may be grown to a greater thickness than the liner layers 98A_N by growing the liner layers 98A with an epitaxial growth process that has a large bottom-up growth rate. In some embodiments, bottom-up epitaxial growth is performed by using a chlorine-containing precursor during growth of the liner layers 98A (described for FIG. 15). The chlorine-containing precursor may be chlorine (Cl_2) gas, hydrogen chloride (HCl), or the like. Additionally or alternatively, bottom-up epitaxial growth may be performed by growing the liner layers 98A at a high temperature, such as a temperature in the range of 600° C. to 800° C. The epitaxial growth is performed until the top surfaces of the liner layers 98A_F are disposed a desired distance D_1 below the bottom surfaces of the second nanostructures 66.

Embodiments may achieve advantages. Forming the liner layers 98A in the n-type region 50N of a semiconductor material that is composed of a blocker element helps reduce out-diffusion of dopants from the main layers 98B into the channel region 68 in subsequent processing. In an experiment where the n-type dopant was arsenic, the dopant concentration 8 nm beyond the sidewalls of the second nanostructures 66 was decreased up to 66% for channel regions 68 under a tensile strain of 0% to 2%. In another experiment where the n-type dopant was phosphorous, the dopant concentration 8 nm beneath the sidewalls of the

second nanostructures 66 was decreased up to 80%. In some embodiments, the dopant concentration in the channel regions 68 was as low as $5 \times 10^{17} \text{ cm}^{-3}$. By reducing the diffusion of dopants into the channel regions 68, channel mobility may be increased. Device performance may thus be increased.

In an embodiment, a device includes: a first nanostructure; a source/drain region adjoining a first channel region of the first nanostructure, the source/drain region including: a main layer; and a first liner layer between the main layer and the first nanostructure, a carbon concentration of the first liner layer being greater than a carbon concentration of the main layer; an inter-layer dielectric on the source/drain region; and a contact extending through the inter-layer dielectric, the contact connected to the main layer, the contact spaced apart from the first liner layer. In some embodiments, the device further includes: a second nanostructure, the source/drain region adjoining a second channel region of the second nanostructure, where the source/drain region further includes: a second liner layer between the main layer and the second nanostructure, the main layer having a first width between the first liner layer and the second liner layer, the contact having a second width, the second width less than the first width. In some embodiments of the device, the source/drain region further includes: a finishing layer on the main layer, the carbon concentration of the first liner layer being greater than a carbon concentration of the finishing layer. In some embodiments of the device, the carbon concentration of the first liner layer is in a range of 0.1 at % to 2 at %, and the carbon concentration of the main layer is zero. In some embodiments of the device, the carbon concentration of the first liner layer is in a range of 0.1 at % to 2 at %, and the carbon concentration of the main layer is in a range of 0 at % to 2 at %. In some embodiments, the device further includes: a second nanostructure, the source/drain region adjoining a second channel region of the second nanostructure, the first liner layer between the main layer and the second nanostructure; a gate structure wrapped around the first channel region of the first nanostructure and around the second channel region of the second nanostructure; and a spacer between the gate structure and the source/drain region, the first liner layer between the main layer and the spacer. In some embodiments, the device further includes: a second nanostructure, the source/drain region adjoining a second channel region of the second nanostructure; a gate structure wrapped around the first channel region of the first nanostructure and around the second channel region of the second nanostructure; and a spacer between the gate structure and the source/drain region, the main layer contacting the spacer, where the source/drain region further includes: a second liner layer between the main layer and the second nanostructure, the second liner layer separated from the first liner layer.

In an embodiment, a device includes: a nanostructure; and a source/drain region adjoining a channel region of the nanostructure, the source/drain region including: a first epitaxial layer on a sidewall of the nanostructure, the first epitaxial layer including a carbon-containing semiconductor material and an n-type dopant; a second epitaxial layer on the first epitaxial layer, the second epitaxial layer including a carbon-free semiconductor material and the n-type dopant; and a third epitaxial layer on the second epitaxial layer, the third epitaxial layer including the carbon-free semiconductor material and the n-type dopant. In some embodiments, the device further includes: an inter-layer dielectric on the source/drain region; and a contact extending through the inter-layer dielectric to contact the second epitaxial layer of

the source/drain region, the contact spaced apart from the first epitaxial layer of the source/drain region. In some embodiments of the device, the carbon-containing semiconductor material is silicon carbide, the carbon-free semiconductor material is silicon, and the n-type dopant is phosphorous or arsenic. In some embodiments, the device further includes: a fin extending from a substrate, where the first epitaxial layer of the source/drain region is on a top surface of the fin. In some embodiments, the device further includes: a fin extending from a substrate; and a spacer between the fin and the first epitaxial layer of the source/drain region.

In an embodiment, a method includes: etching a source/drain recess in a nanostructure, a sidewall of the nanostructure exposed to the source/drain recess; growing a first epitaxial layer in the source/drain recess by exposing the sidewall of the nanostructure to a semiconductor-containing precursor, a blocker-containing precursor, and a dopant-containing precursor; growing a second epitaxial layer in the source/drain recess by exposing the first epitaxial layer to the semiconductor-containing precursor and the dopant-containing precursor, the first epitaxial layer not exposed to the blocker-containing precursor when growing the second epitaxial layer; depositing a dielectric layer on the second epitaxial layer; and forming a contact through the dielectric layer, the contact connected to the second epitaxial layer, the contact spaced apart from the first epitaxial layer. In some embodiments of the method, the semiconductor-containing precursor is a silane and the blocker-containing precursor is trimethylsilane. In some embodiments of the method, the sidewall of the nanostructure is exposed to the semiconductor-containing precursor, the blocker-containing precursor, and the dopant-containing precursor at a temperature in a range of 600° C. to 700° C. and at a pressure in a range of 1 Torr to 760 Torr. In some embodiments of the method, the first epitaxial layer is exposed to the semiconductor-containing precursor and the dopant-containing precursor at a temperature in a range of 400° C. to 800° C. and at a pressure in a range of 1 Torr to 760 Torr. In some embodiments of the method, the nanostructure is under a tensile strain after etching the source/drain recess, and the dopant-containing precursor is an arsenic-containing precursor or a phosphorous-containing precursor. In some embodiments of the method, the sidewall of the nanostructure is also exposed to a chlorine-containing precursor when growing the first epitaxial layer. In some embodiments of the method, the nanostructure is disposed over a spacer, a sidewall of the spacer is exposed to the source/drain recess, and growth of the first epitaxial layer is stopped before the first epitaxial layer covers the sidewall of the spacer. In some embodiments of the method, the nanostructure is disposed over a spacer, a sidewall of the spacer is exposed to the source/drain recess, and growth of the first epitaxial layer is performed until the first epitaxial layer covers the sidewall of the spacer.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

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What is claimed is:

1. A method comprising:

etching a source/drain recess in a nanostructure and a fin,
a sidewall of the nanostructure exposed to the source/
drain recess, a top surface of the fin exposed to the
source/drain recess;

growing a first epitaxial layer and a second epitaxial layer
in the source/drain recess, wherein the first epitaxial
layer is grown on the sidewall of the nanostructure by
exposing the sidewall of the nanostructure to a semi-
conductor-containing precursor, a blocker-containing
precursor, and a dopant-containing precursor, wherein
the second epitaxial layer is grown to cover an entirety
of the top surface of the fin by exposing the top surface
of the fin to the semiconductor-containing precursor,
the blocker-containing precursor, and the dopant-con-
taining precursor, wherein the second epitaxial layer is
separated from the first epitaxial layer, wherein the
second epitaxial layer is thicker than the first epitaxial
layer;

growing a third epitaxial layer in the source/drain recess
by exposing the first epitaxial layer and the second
epitaxial layer to the semiconductor-containing precursor
and the dopant-containing precursor, the first epitaxial
layer and the second epitaxial layer not exposed to the
blocker-containing precursor when growing the
third epitaxial layer;

depositing a dielectric layer on the third epitaxial layer;
and

forming a contact through the dielectric layer, the contact
connected to the third epitaxial layer, the contact
spaced apart from the first epitaxial layer.

2. The method of claim 1, wherein the semiconductor-
containing precursor is a silane and the blocker-containing
precursor is trimethylsilane.

3. The method of claim 1, wherein the sidewall of the
nanostructure is exposed to the semiconductor-containing
precursor, the blocker-containing precursor, and the dopant-
containing precursor at a temperature in a range of 600° C.
to 700° C. and at a pressure in a range of 1 Torr to 760 Torr.

4. The method of claim 1, wherein the first epitaxial layer
is exposed to the semiconductor-containing precursor and
the dopant-containing precursor at a temperature in a range
of 400° C. to 800° C. and at a pressure in a range of 1 Torr
to 760 Torr.

5. The method of claim 1, wherein the nanostructure is
under a tensile strain after etching the source/drain recess,
and the dopant-containing precursor is an arsenic-containing
precursor or a phosphorous-containing precursor.

6. The method of claim 1, wherein the sidewall of the
nanostructure is also exposed to a chlorine-containing pre-
cursor when growing the first epitaxial layer.

7. The method of claim 1, wherein the nanostructure is
disposed over a spacer, a sidewall of the spacer is exposed
to the source/drain recess, and growth of the first epitaxial
layer is stopped before the first epitaxial layer covers the
sidewall of the spacer.

8. The method of claim 1, wherein the nanostructure is
disposed over a spacer, a sidewall of the spacer is exposed
to the source/drain recess, and growth of the first epitaxial
layer is performed until the first epitaxial layer covers the
sidewall of the spacer.

9. A method comprising:

etching a source/drain recess in a nanostructure and a fin,
the nanostructure disposed over the fin, a sidewall of
the nanostructure and a top surface of the fin exposed
to the source/drain recess;

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forming a source/drain region in the source/drain recess,
wherein forming the source/drain region comprises:

growing a first epitaxial layer and a second epitaxial
layer, the first epitaxial layer grown on the sidewall
of the nanostructure, the second epitaxial layer
grown to cover an entirety of the top surface of the
fin, the second epitaxial layer being thicker than the
first epitaxial layer, the first epitaxial layer and the
second epitaxial layer comprising a carbon-contain-
ing semiconductor material and an n-type dopant;
and

growing a third epitaxial layer on the first epitaxial
layer and the second epitaxial layer, the third epitaxial
layer comprising a carbon-free semiconductor
material and the n-type dopant;

depositing a dielectric layer on the third epitaxial layer;
and

forming a contact through the dielectric layer, the contact
connected to the source/drain region.

10. The method of claim 9, wherein forming the source/
drain region further comprises:

growing a fourth epitaxial layer on the third epitaxial
layer, the fourth epitaxial layer comprising the carbon-
free semiconductor material and the n-type dopant.

11. The method of claim 9, wherein the contact is spaced
apart from the first epitaxial layer.

12. The method of claim 9, wherein the carbon-containing
semiconductor material is silicon carbide, the carbon-free
semiconductor material is silicon, and the n-type dopant is
phosphorous or arsenic.

13. The method of claim 9, wherein the nanostructure is
under a tensile strain after etching the source/drain recess.

14. A method comprising:

forming a first spacer and a second spacer under, respec-
tively, a first nanostructure and a second nanostructure;
forming a source/drain region between the first nanostruc-
ture and the second nanostructure, wherein forming the
source/drain region comprises:

epitaxially growing a first liner layer, a second liner
layer, and a third liner layer, the first liner layer
grown on a sidewall of the first nanostructure, the
second liner layer grown on a sidewall of the second
nanostructure, the first liner layer separated from the
second liner layer, and the third liner layer extending
between and contacting both the first spacer and the
second spacer; and

epitaxially growing a main layer on the first liner layer
and the second liner layer, the main layer having a
first width between the first liner layer and the
second liner layer in a cross-sectional view, wherein
a carbon concentration of the first liner layer and the
second liner layer is greater than a carbon concen-
tration of the main layer;

depositing an inter-layer dielectric on the source/drain
region; and

forming a contact extending through the inter-layer
dielectric, the contact connected to the source/drain
region, the contact having a second width in the cross-
sectional view, the second width being less than the first
width.

15. The method of claim 14, wherein forming the source/
drain region further comprises:

epitaxially growing a finishing layer on the main layer,
wherein the carbon concentration of the first liner layer
and the second liner layer is greater than a carbon
concentration of the finishing layer.

16. The method of claim **14**, wherein the contact is connected to the main layer and spaced apart from the first liner layer and the second liner layer.

17. The method of claim **14**, wherein the main layer is grown on a sidewall of the first spacer, and the method further includes:

forming a gate structure around the first nanostructure, the first spacer disposed between the gate structure and the source/drain region.

18. The method of claim **14**, wherein the first liner layer is grown on a sidewall of the first spacer, and the method further includes:

forming a gate structure around the first nanostructure, the first spacer disposed between the gate structure and the source/drain region.

19. The method of claim **14**, wherein the carbon concentration of the main layer is zero.

20. The method of claim **14**, wherein the carbon concentration of the main layer is non-zero.

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